



**BEYKENT UNIVERSITY
FACULTY OF ENGINEERING ARCHITECTURE
MECHANICAL ENGINEERING**

REPORT 1

Yusuf Yıldızlar

2103040020

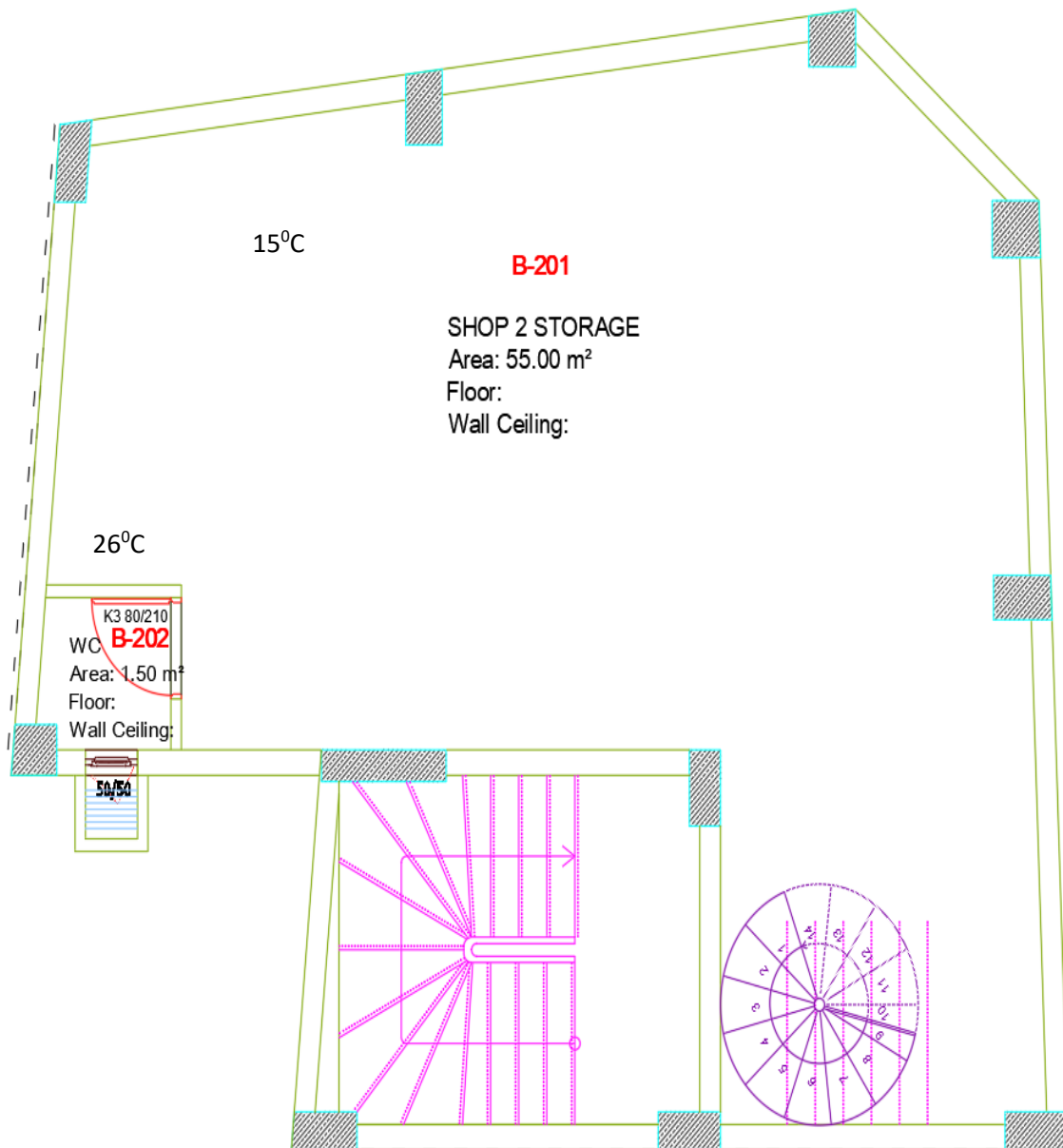
DESIGN OF THERMAL SYSTEM

ADVISOR: Dr Öğr. Üyesi MUAMMER KANLI

İSTANBUL, 2025

Information About the Structure	
City:	Istanbul
District:	Esenyurt
Neighborhood:	Atatürk Mahallesi
Street:	394. Sok-396.Sok
Sheet No:	23B-3A
Island No:	884
Parcel No:	3
Construction Nevi:	B.A.K
Total Construction Area:	584.14m²
No of Floors:	6
The Greatest Height	3.00
The Largest Span	3.00 m
Building Class	3B B.A.K
Purpose of Use	Residential + Commercial

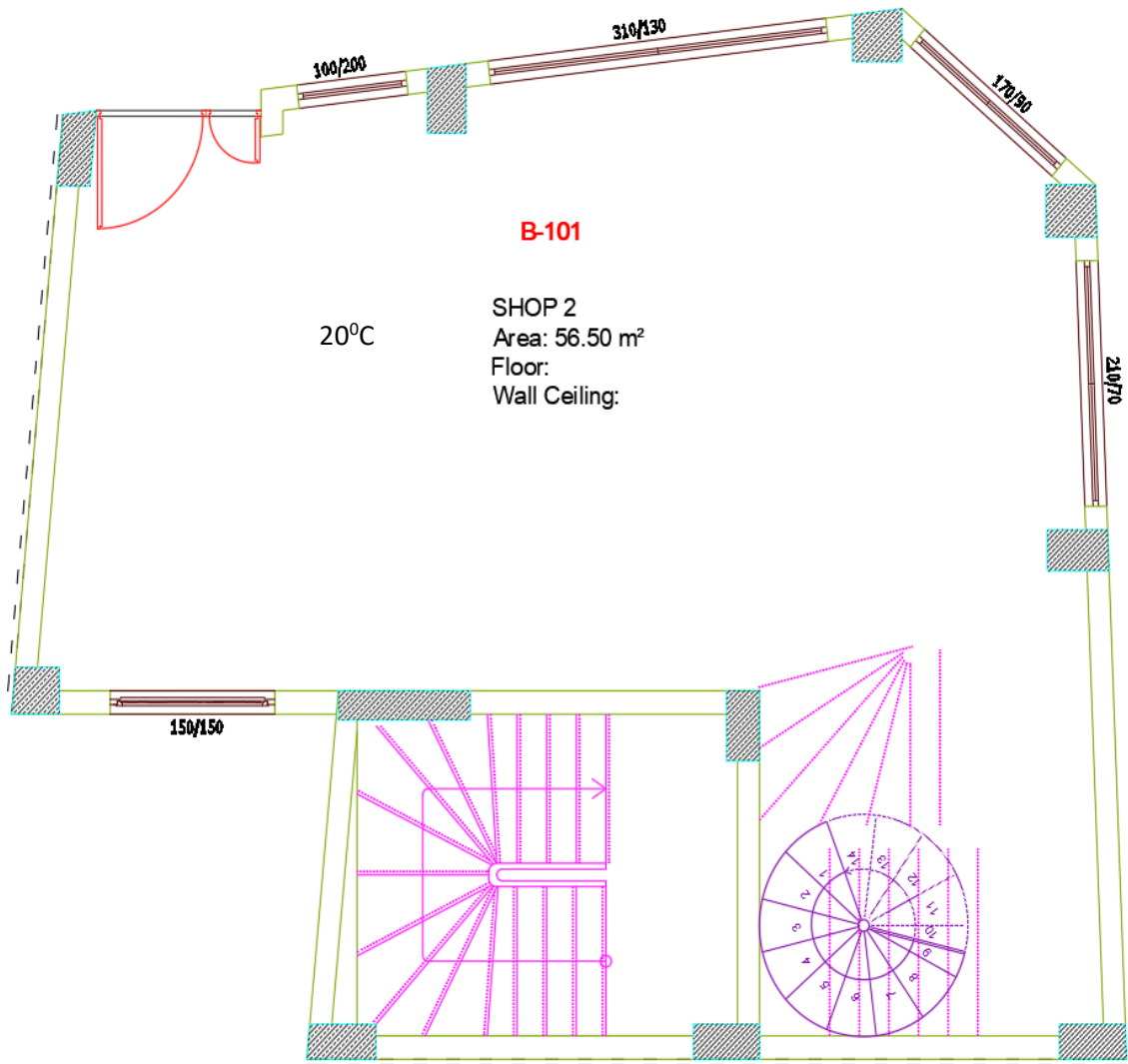
Total Construction Area	
Second Basement Floor:	75.00m²
First Basement Floor:	75.00m²
Ground Floor:	75.00m²
First Floor:	96.07m²
Second Floor:	96.07m²
Third Floor:	167.00m²



2nd Basement Floor Plan

Scale: 1:50

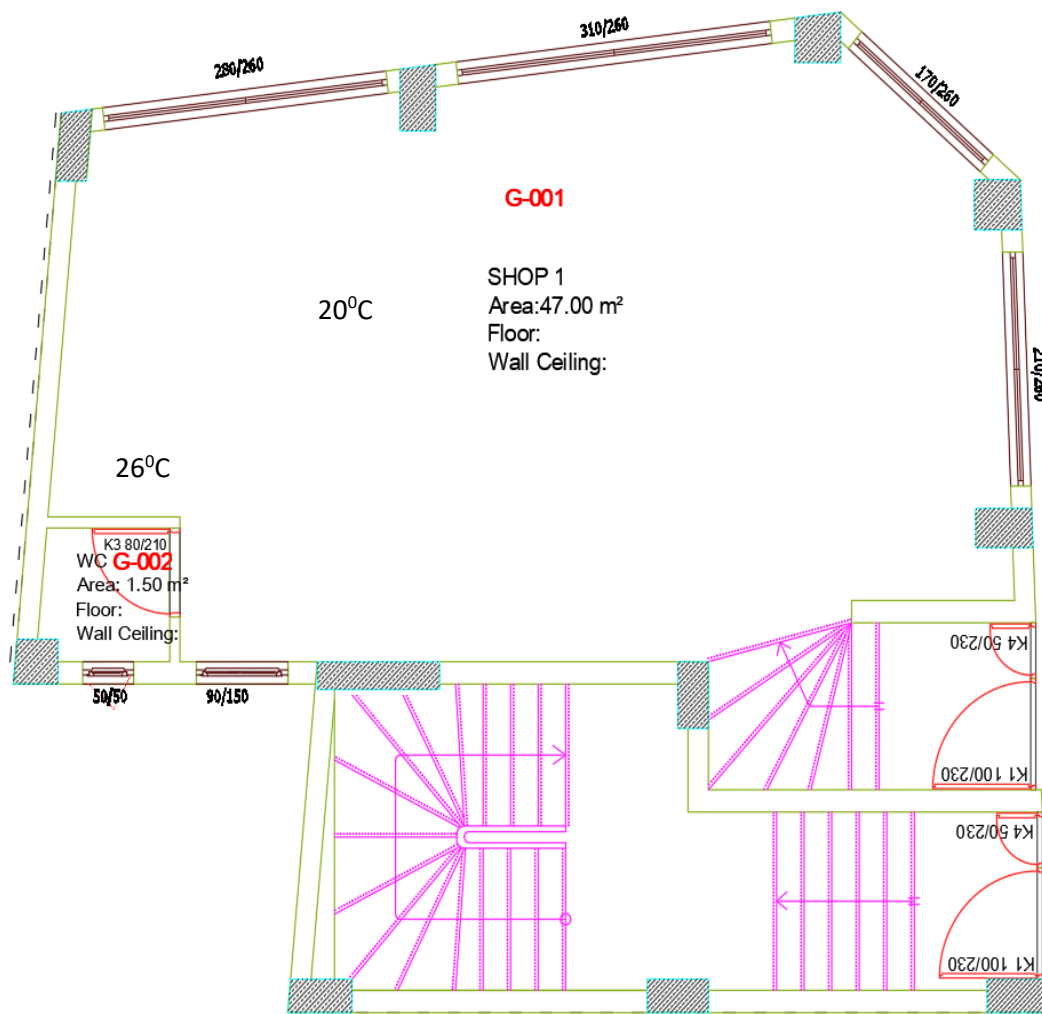
Area: 75.00 m²



1st Basement Floor Plan

Scale: 1:50

Area: 75.00m²

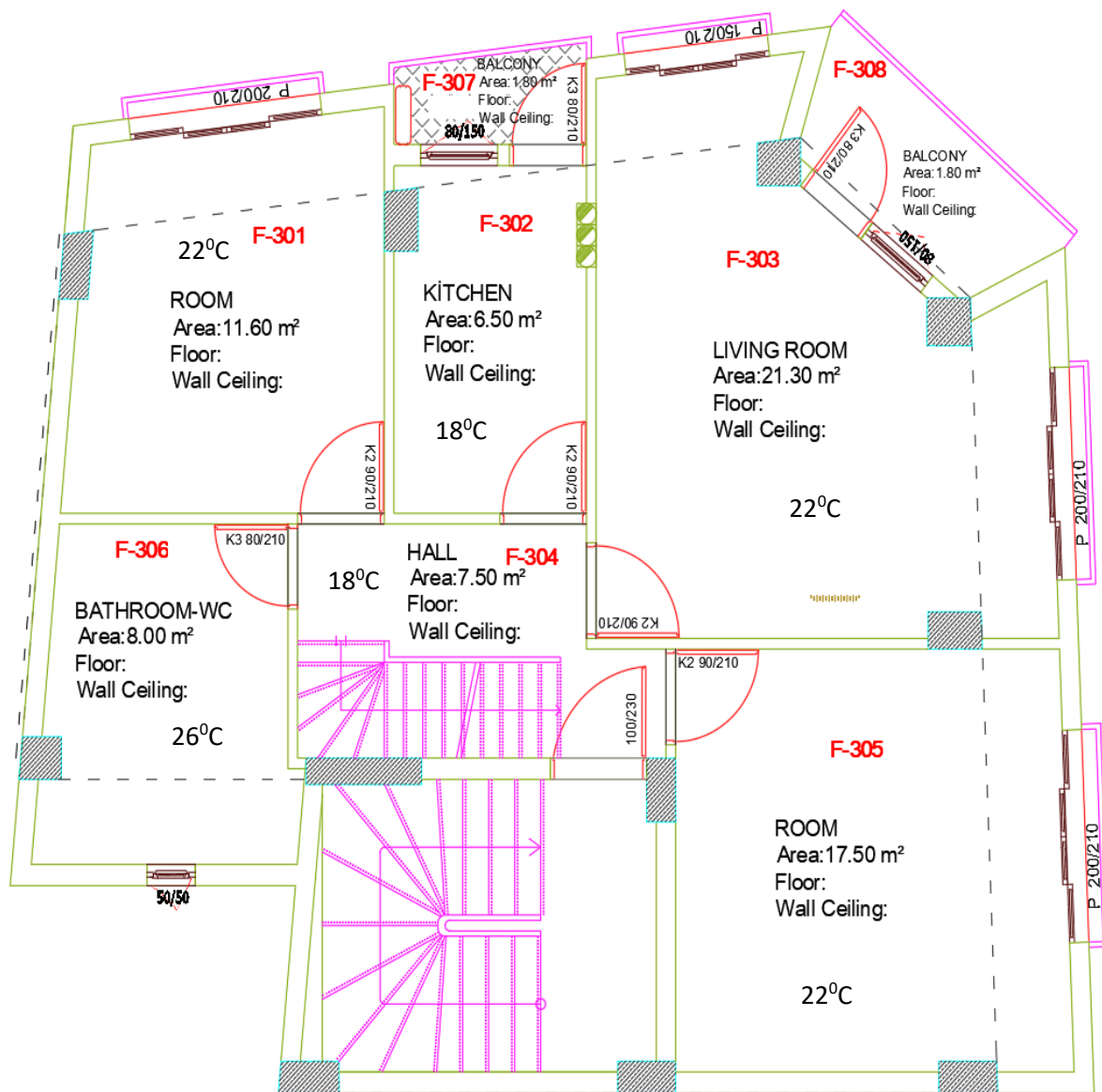


Ground Floor Plan

Scale: 1:50

Area: 75.00m²

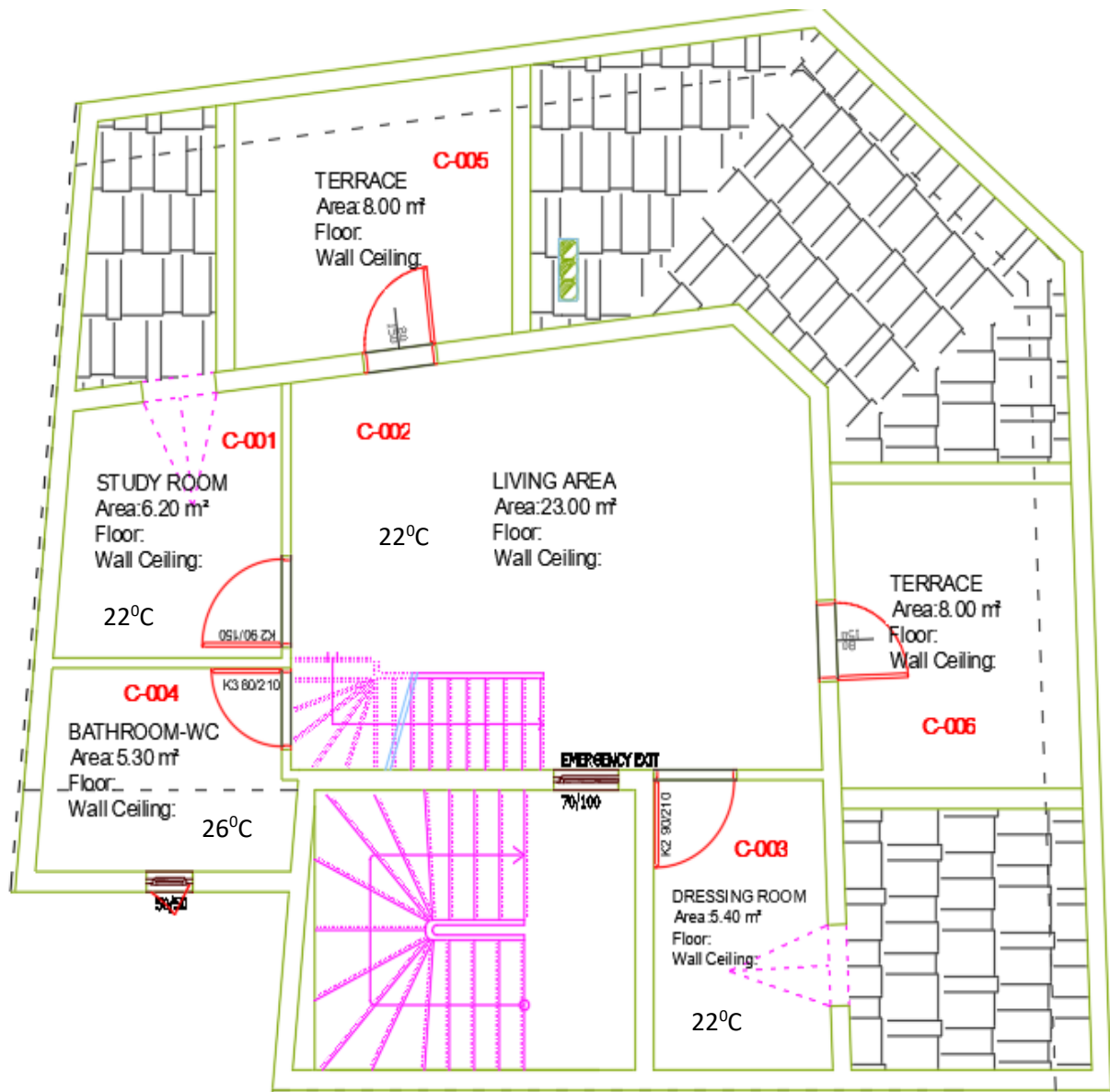
Area:96.07m²



Third Floor Plan

Scale: 1:50

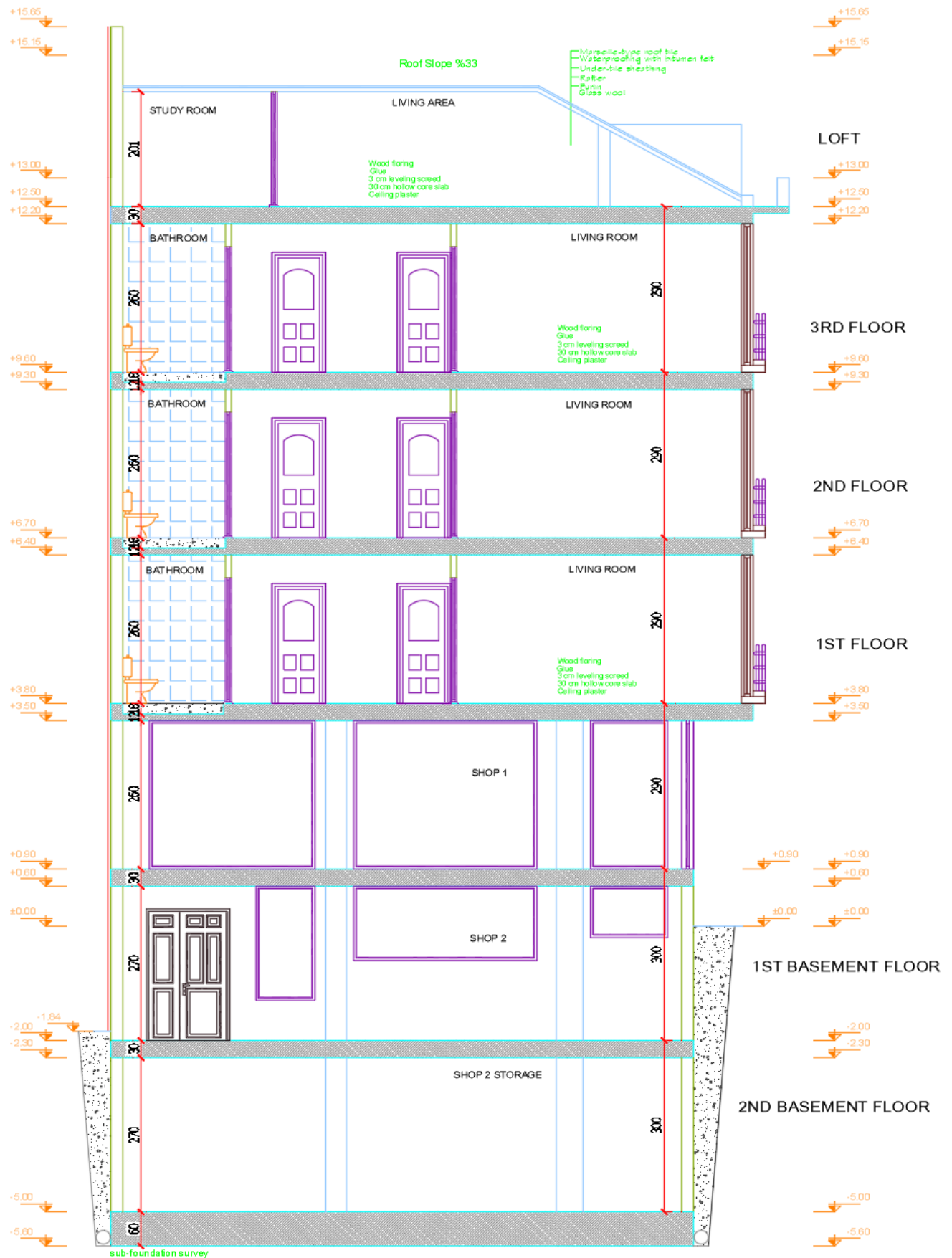
Area:96.07m²



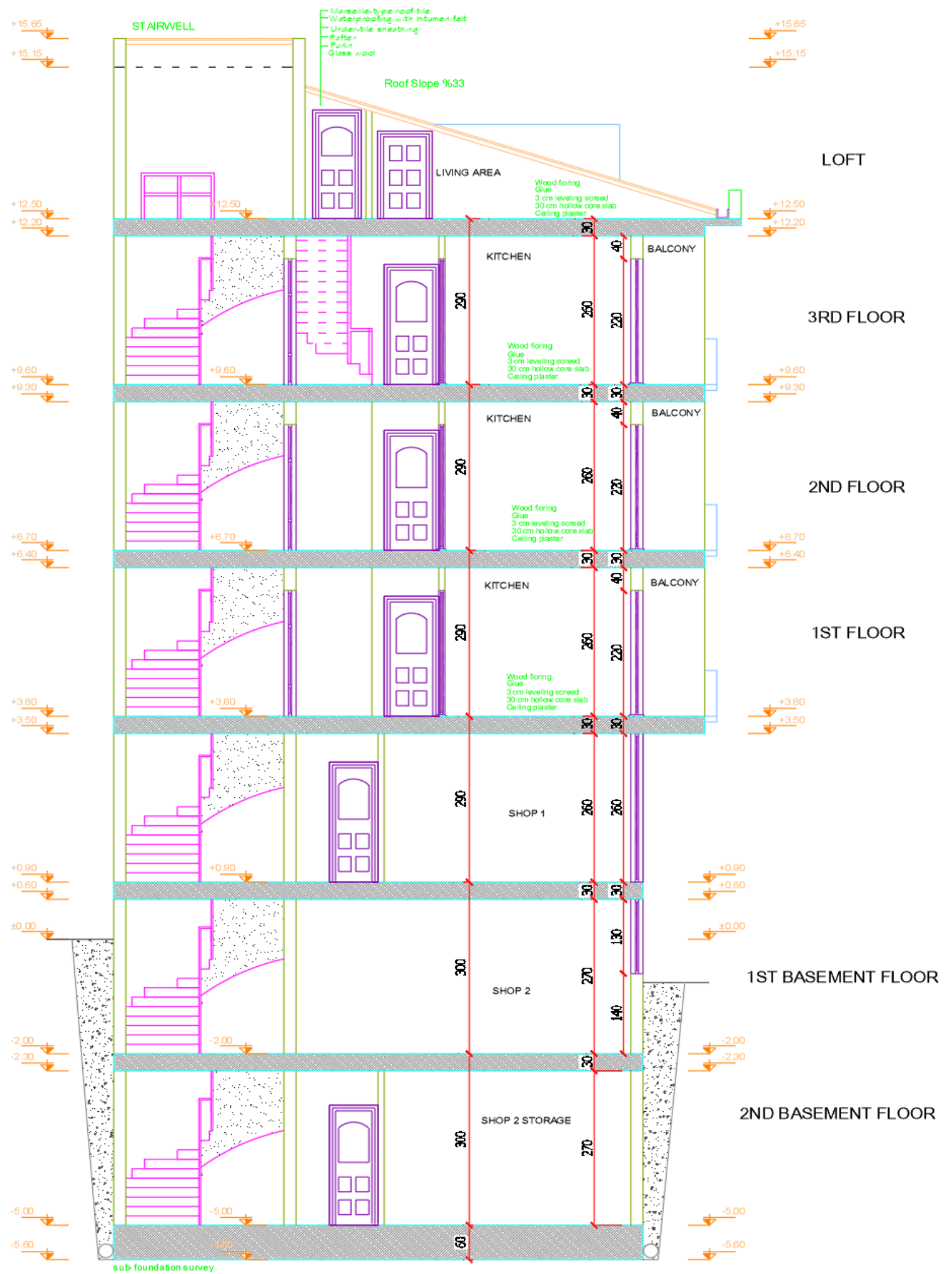
Ceiling Floor Plan

Scale: 1:50

Area: 167.00 m²



A-A Section Scale: 1:50



B-B Section Scale: 1:50



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MECHANICAL ENGINEERING**

REPORT 2

Yusuf Yıldızlar

2103040020

DESIGN OF THERMAL SYSTEM

ADVISOR: Dr Öğr. Üyesi MUAMMER KANLI

İSTANBUL, 2025

Heat Transfer Coefficient "U"				Adjacent Wall
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R (m^2K/W)
	Thermal Convective Resistance $R_i(\frac{m^2K}{W})$			0.13
1	Eartquake Joint	5	0.025	2
2	Exterior Plastering	3	1.4	0.021
3	Adhesive	0		0
4	Airbrick	19.5	0.5	0.39
5	Interior Plastering	2	0.87	0.023
6	Interior Paint	0		0
	Thermal Convective Resistance $R_e(\frac{m^2K}{W})$			0.13
			ΣR	2.694
			U (AW)	0.371 W/m^2K

Heat Transfer Coefficient "U"				External Wall
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R (m^2K/W)
	Thermal Convective Resistance $R_i(\frac{m^2K}{W})$			0.13
1	Strafor	5	0.04	1.25
2	Adhesive	0		0
3	Exterior Plastering	3	1.4	0.02
4	Airbrick	19.5	0.5	0.39
5	Interior Plastering	2	0.87	0.023
6	Interior Paint	0		0
	Thermal Convective Resistance $R_e(\frac{m^2K}{W})$			0.04
			ΣR	2.06
			U (EW)	0.49 W/m^2K

Heat Transfer Coefficient "U"				Ceramic Internal Wall
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R ($m^2 K/W$)
	Thermal Convective Resistance $R_i(\frac{m^2 K}{W})$			0.13
1	Ceramic	0.8	1.2	0.006
2	Resistance	0		0
3	Interior Plaster	2	0.87	0.023
4	Airbrick	8.5	0.5	0.17
5	Interior Plaster	2	0.87	0.023
6	Interior Paint	0		0
	Thermal Convective Resistance $R_e(\frac{m^2 K}{W})$			0.13
			ΣR	0.482
			U (IW-C)	2.074 $W/m^2 K$

Heat Transfer Coefficient "U"				Internal Wall 1
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R ($m^2 K/W$)
	Thermal Convective Resistance $R_i(\frac{m^2 K}{W})$			0.13
1	Interior Paint	0		0
2	Interior Plastering	2	0.87	0.023
3	Airbrick	8.5	0.5	0.17
4	Interior Plastering	2	0.87	0.023
5	Interior Paint	0		0
	Thermal Convective Resistance $R_e(\frac{m^2 K}{W})$			0.13
			ΣR	0.476
			U (IW-1)	2.10 $W/m^2 K$

Heat Transfer Coefficient "U"				Internal Wall 2
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R (m^2K/W)
	Thermal Convective Resistance $R_i(\frac{m^2K}{W})$			0.13
1	Interio Piant	0		0
2	Interior Plastering	2	0.87	0.023
3	Airbrick	19.5	0.5	0.17
4	Interior Plastering	2	0.87	0.023
5	Interior Paint	0		0
	Thermal Convective Resistance $R_e(\frac{m^2K}{W})$			0.13
			ΣR	0.696
			U (IW-1)	1.44 W/m^2K

Heat Transfer Coefficient "U"				Floor 1
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R (m^2K/W)
	Thermal Convective Resistance $R_i(\frac{m^2K}{W})$			0.17
1	Wooden	3	0.2	0.15
2	Adhesive	0		0
3	Screed	5	1.4	0.036
4	Reinforced Concrete	12	2.1	0.057
5	Hollow Block	30	0.04	7.5
6	Foam Board	4	0.04	1
7	Heraklit	2	0.14	0.14
8	Interior Plaster	2	0.87	0.023
	Thermal Convective Resistance $R_e(\frac{m^2K}{W})$			0.17
			ΣR	9.246
			U (IW-1)	0,108 W/m^2K

Heat Transfer Coefficient "U"				Floor 2
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R (m^2K/W)
	Thermal Convective Resistance $R_i(\frac{m^2K}{W})$			0.17
1	Ceramic	0.8	1.2	0.006
2	Adhesive	0		0
3	Screed	5	1.4	0.036
4	Reinforced Concrete	12	2.1	0.057
5	Hollow Block	30	0.04	7.5
6	Foam Board	4	0.04	1
7	Heraklit	2	0.14	0.14
8	Interior Plaster	2	0.87	0.023
	Thermal Convective Resistance $R_e(\frac{m^2K}{W})$			0.17
			ΣR	1.602
			U (IW-1)	0,11 W/m^2K

Heat Transfer Coefficient "U"				Floor 3 (bathroom)
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R (m^2K/W)
	Thermal Convective Resistance $R_i(\frac{m^2K}{W})$			0.17
1	Ceramic	0.8	1.2	0.006
2	Adhesive	0		0
3	Screed	18	1.4	0.129
4	Reinforced Concrete	12	2.1	0.057
5	Foam Board	4	0.04	1
6	Heraklit	2	0.14	0.14
7	Interior Plaster	2	0.87	0.023
	Thermal Convective Resistance $R_e(\frac{m^2K}{W})$			0.17
			ΣR	1.695
			U (IW-1)	0,59 W/m^2K

Heat Transfer Coefficient "U"				Roof
No	Material Type	Thick Ness d (cm)	Heat Transfer Coefficient λ (W/mK)	Thermal Resistance R (m^2K/W)
	Thermal Convective Resistance $R_i(\frac{m^2K}{W})$			0.13
1	Gypsum Board	0.0125	0.25	0.05
2	Stone Wool	0.05	0.04	1.250
3	Reinforced Concrete Slab	0.15	2.10	0.0714
4	Leveling Concrete	0.05	1.40	0.0357
5	Waterproofing	0.0040	0.23	0.0174
6	Thermal Insulation	0.0800	0.035	2.2857
7	Gravel	0.050	0.36	0.1389
	Thermal Convective Resistance $R_e(\frac{m^2K}{W})$			0.04
			ΣR	3.9791
			U (IW-1)	0,251 W/m^2K

(1) <https://argevetasarim.com/malzemelerin-ve-gazlarin-isil-iletkenligi/>

CALCULATION FORMULAS

FORMULA = $1/U = (1/\alpha_i) + (d_1/\lambda_1) + \dots (d_n/\lambda_n) + (1/\alpha_0)$

Exterior Wall (U-Value) = $1 / (0.13 + (0.02 / 0.87) + (0.20 / 0.21) + (0.02 / 1.40) + 0.04)$

Interior Wall (U-Value) = $1 / (0.13 + (0.02 / 0.87) + (0.10 / 0.21) + (0.02 / 0.87) + 0.13)$

Ceiling (Insulation in Roof Space) = $1 / (0.13 + (0.0125 / 0.25) + (0.10 / 0.04) + 0.04)$

Floor (Insulation on Ground) = $1 / (0.17 + (0.03 / 2.10) + (0.05 / 1.40) + (0.15 / 2.10) + (0.05 / 0.04) + 0.04)$

Floor (No Insulation) = $1 / (0.17 + (0.03 / 2.10) + (0.05 / 1.40) + (0.15 / 2.10) + 0.04)$

Floor - 2 (No Insulation) = $1 / (0.17 + (0.03 / 2.10) + (0.05 / 1.40) + (0.15 / 2.10) + 0.04)$

Roof (7-Layer) = $1 / (0.13 + (0.0125 / 0.25) + (0.05 / 0.04) + (0.15 / 2.10) + (0.05 / 1.40) + (0.004 / 0.23) + (0.08 / 0.035) + (0.05 / 0.36) + 0.04)$

L : The opening length (peripheral) of windows (l) are read from the layout appearances of the building

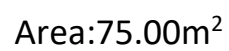
and then calculated.

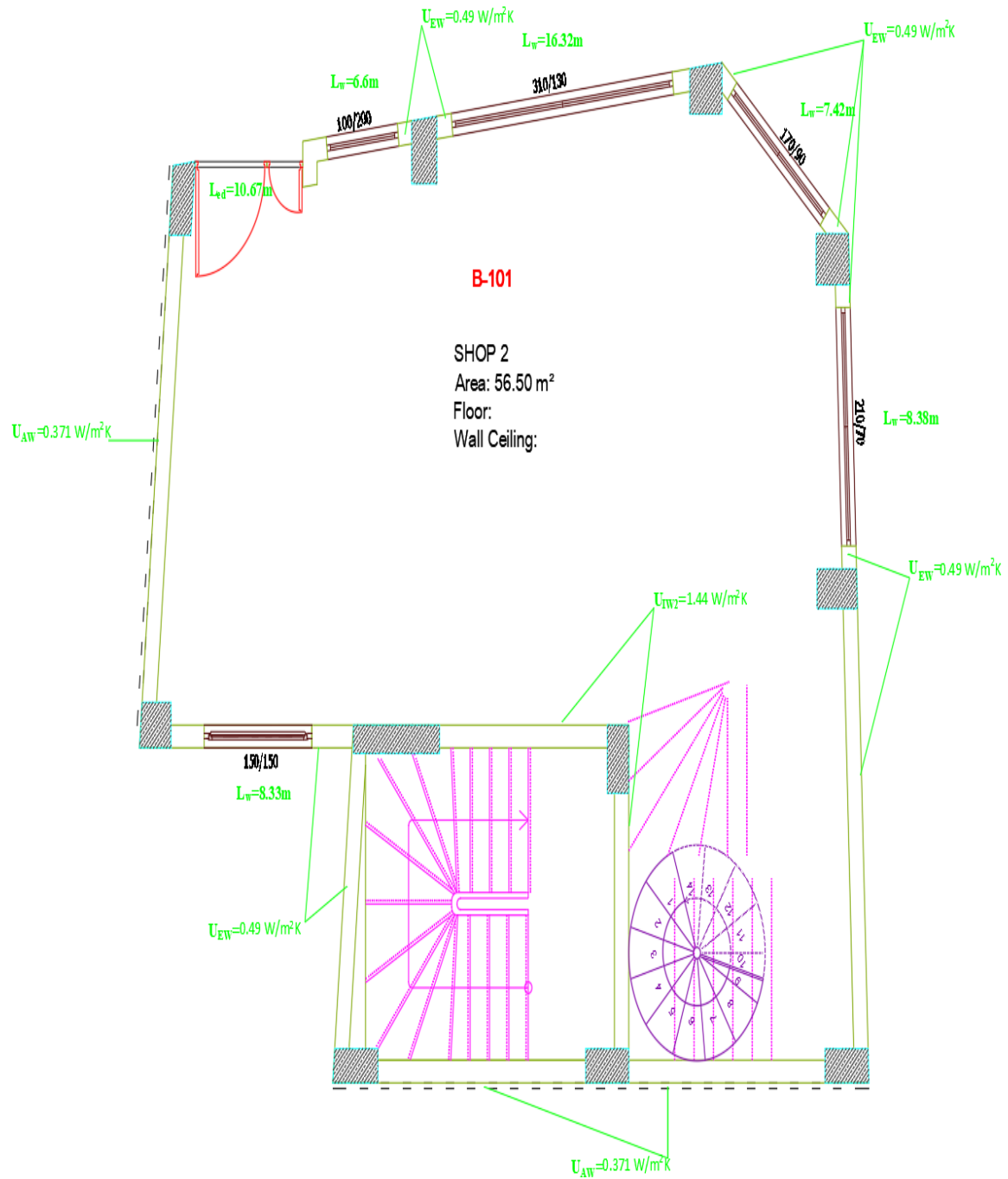
Lw for F102–Kitchen

80/150 :

80/150 ; The value in the denominator gives us the vertical length of the opened part of the window. To find the width, we measure the width of only the opened part of the window in the drawing program. F102-We have a 60cm wide glass for the kitchen room. Therefore: $L = w \times A$
 $(1.5 \times 0.8) \times 3.70$

The w value was taken from page 21 of the 3rd week's PDF.



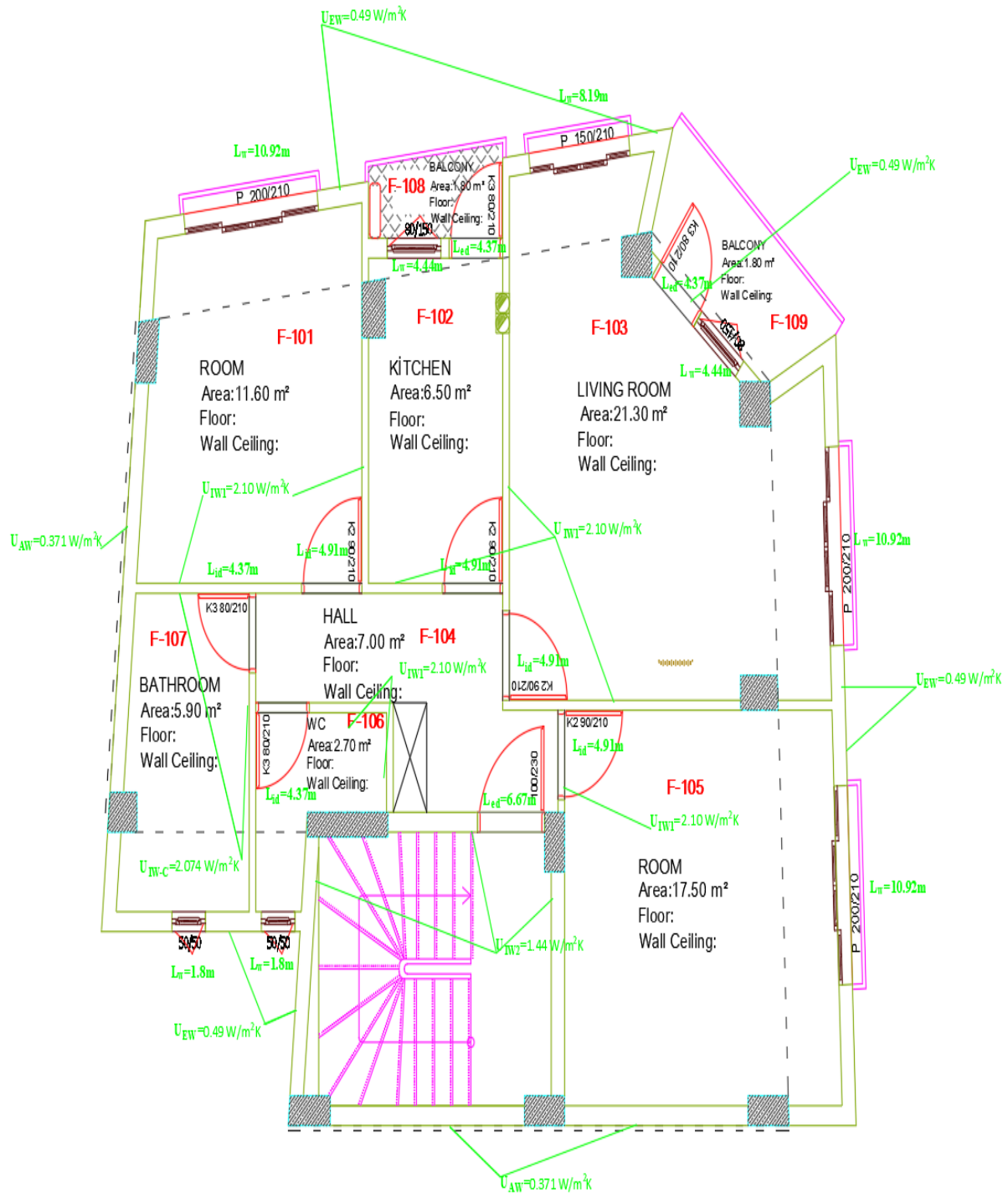


1st Basement Floor Plan

Scale: 1:50

Area: 75.00m²

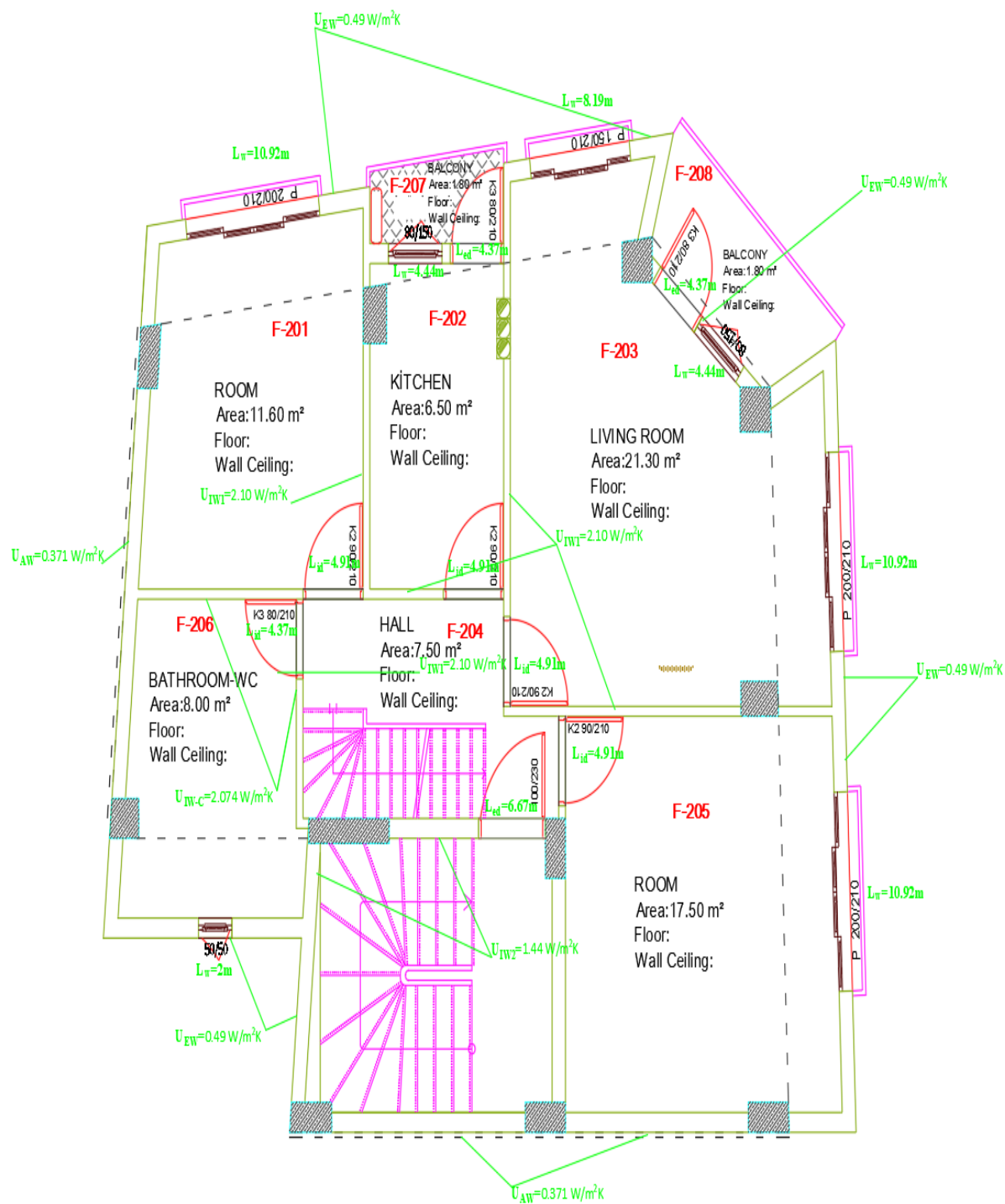
Area:75.00m²



First and Second Floor Plans

Scale: 1:50

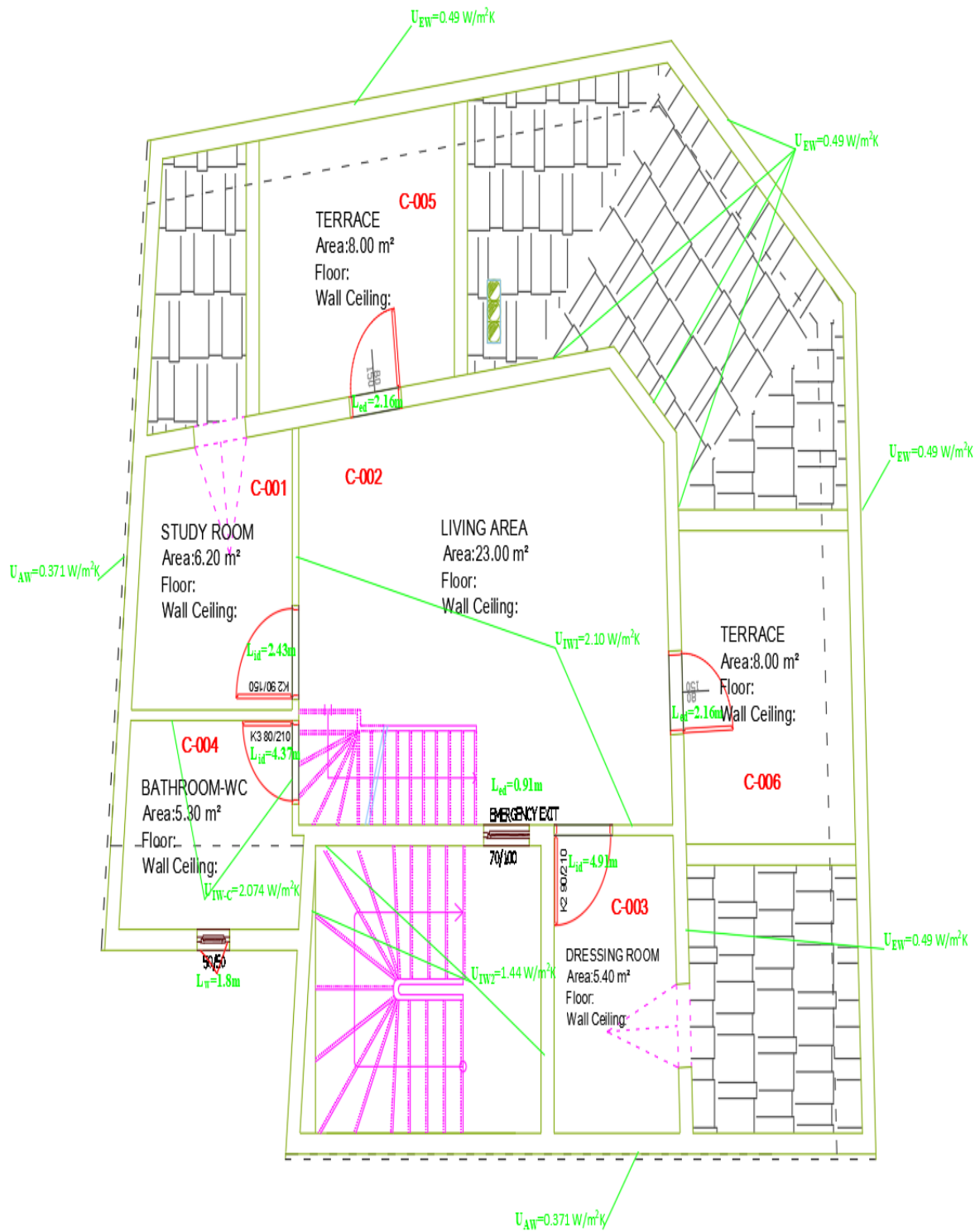
Area: 96.07 m²



Third Floor Plan

Scale: 1:50

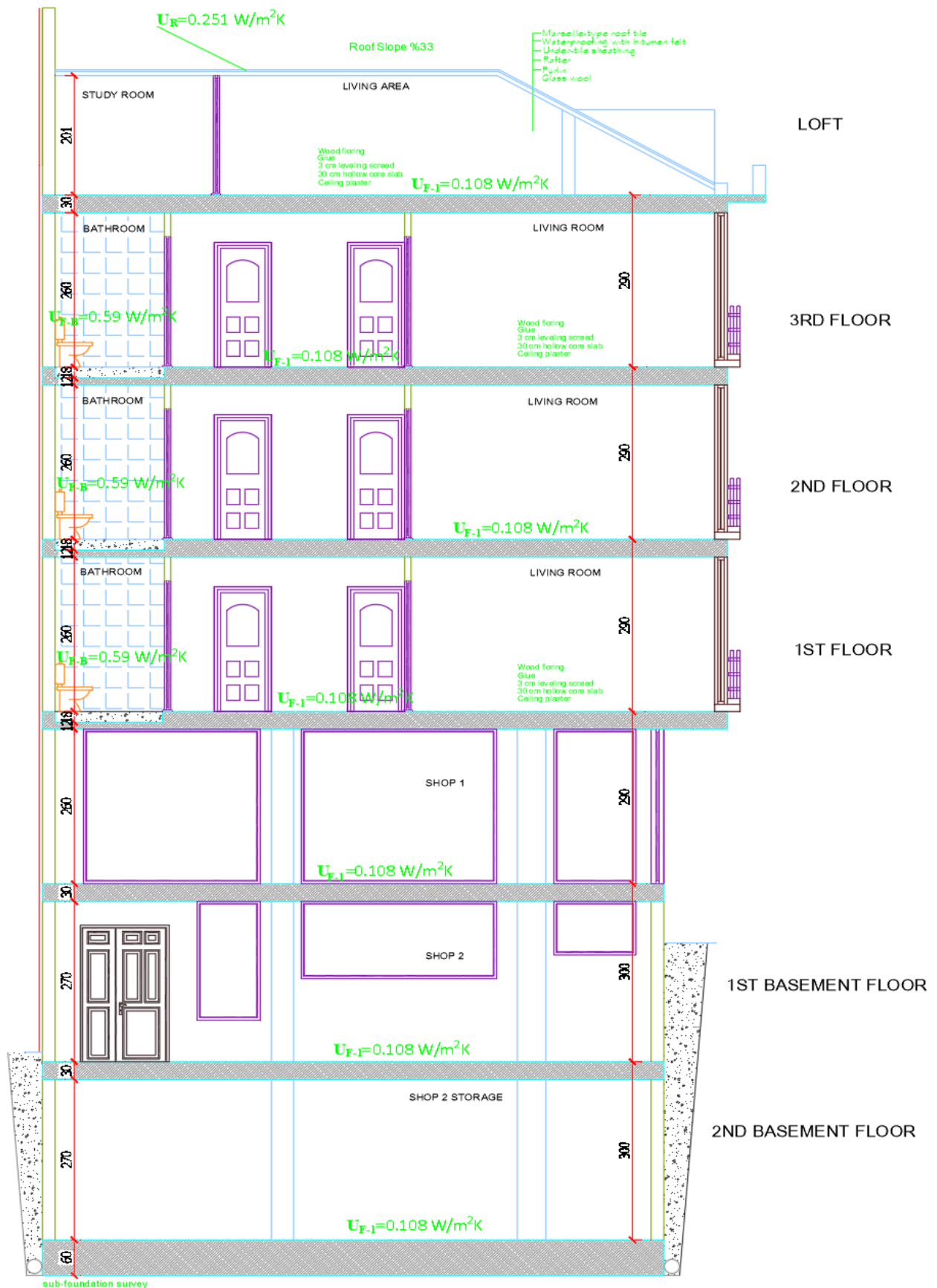
Area: 96.07 m²



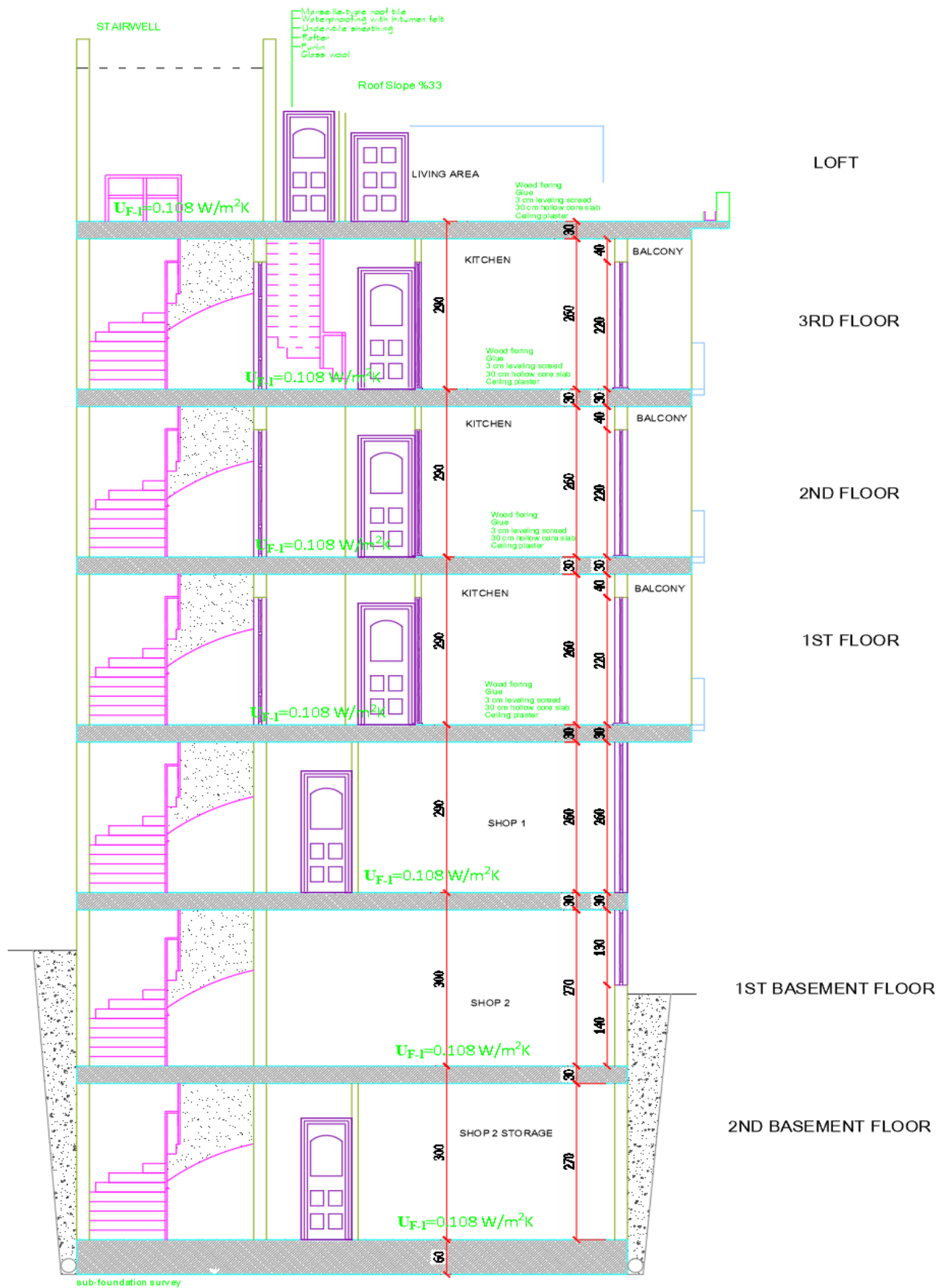
Ceiling Floor Plan

Scale: 1:50

Area: 167.00 m²



A-A Section Scale: 1:50



B-B Section Scale: 1:50



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MECHANICAL ENGINEERING**

REPORT 3

Yusuf Yıldızlar

2103040020

DESIGN OF THERMAL SYSTEM

ADVISOR: Dr Öğr. Üyesi MUAMMER KANLI

İSTANBUL, 2025

Introduction

The purpose of this report is to calculate the detailed heat loss and total heating energy requirements for the residential and commercial building located in Esenyurt, Istanbul (Block: 884, Parcel: 3). This study constitutes the third phase of the thermal system design project, building upon the architectural analysis (Report 1) and the thermal transmittance calculations (Report 2).

The heat loss calculations are performed individually for each room on every floor, ranging from the 2nd Basement Floor to the Loft Floor. The objective is to determine the specific heat load (Q_h) required to maintain desired indoor comfort conditions against external weather conditions and adjacent unheated spaces.

Methodology and Design Parameters

The calculations in this report are based on the following standards and parameters:

Formulation: The nominal heat loss (Q_0) is calculated using the standard transmission formula:

$$Q_0 = A \cdot U \cdot \Delta T$$

Where A is the net area of the component, U is the overall heat transfer coefficient (calculated in Report 2), and ΔT is the temperature difference.

Design Temperatures:

External Design Temperature (T_{ext}): Based on the local climate data for the project location, the outdoor temperature is taken as +4°C.

Soil Temperature: For basement walls and floors in contact with the ground, the soil temperature is assumed to be +11°C.

Internal Design Temperatures (T_i): Determined by room function (e.g., 22°C for Living Rooms/Bedrooms, 26°C for Bathrooms, 18°C for Kitchens/Halls/Stairwells).

Increments (Z): The nominal heat loss is adjusted by a total increment factor (Z), which includes:

Operation Regime (Z_D): Taken as %7 for continuous residential heating.

Orientation (Z_H): Applied based on the direction of external walls (e.g., +5% for North).

Storey Height (Z_W): Applied as 0% for this building height.

Infiltration / Leakage (Q_s): Heat loss due to air leakage through window and door gaps is calculated for rooms with external openings, using the calculated leakage lengths (l) from Report 2 and the infiltration formula:

$$Q_s = \frac{1}{3.6} \cdot \sum (a \cdot l) \cdot R \cdot H \cdot \Delta T \cdot Z_e$$

The results presented in the following "Heat Loss Calculation Charts" will serve as the foundational data for sizing the heating plant (boiler) and selecting the appropriate radiators for each room in the subsequent stages of the mechanical design.

Parameter	Value	Unit	Note / Source
Location	Istanbul / Esenyurt	-	2nd Climate Zone
External Design Temperature(T_{ext})	+4	°C	Project Assumption
Soil Temperature (T_{soil})	+11	°C	For unheated basement walls/floors
Indoor Temp. (Living Room/Bedroom)	+22	°C	TS 2164 / DIN 4701
Indoor Temp. (Kitchen/Hall/Stairwell)	+18	°C	TS 2164 / DIN 4701
Indoor Temp. (Bathroom / WC)	+26	°C	TS 2164 / DIN 4701

Component Code	Description	U-Value (W/m ² K)	Source (Report 2)
U_{EW}	External Wall (Standard)	0,490	
U_{AW}	Adjacent Wall (Soil Contact)	0,371	
$U_{IW\ 1}$	Internal Wall (Standard 8.5cm)	2,100	
$U_{IW\ 2}$	Internal Wall (Thick/Stair 19.5cm)	1,440	
$U_{IW\ C}$	Internal Wall (Bathroom/WC)	2,074	
U_{Floor}	Floor / Ceiling (Standard)	0,108	
$U_{Floor\ B}$	Floor / Ceiling (Bathroom)	0,590	
U_{Roof}	Roof (Loft Ceiling)	0,251	
U_{Win}	Window (PVC Double Glazed)	2,600	Project Standard
$U_{Win\ S}$	Window (Small/Single Glazed)	5,000	For small internal windows
U_{Door}	Internal Door	2,300	Standard Assumption

2nd Basement Floor Calculations

[illegible]

HEAT LOSS CALCULATION CHART																	Page	2
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	B-2
																	Date	12.11.2025
Component			Area Calculation						Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total			
					A ₀			A	U	ΔT	Q ₀	Z _D	Z _W	ZH	Z	Q _h =Q _i +Q ₀		
		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%	W		
B-202										T _i =26°C								
Floor 3 (Bath)	Fl	58,00	1,03	1,23	1,27	1	0,00	1,27	0,590	15	11,24							
Floor 1 (Ceiling)	Ce	58,00	1,03	1,23	1,27	1	0,00	1,27	0,108	6	0,82							
Internal Door (ID)	W	-	0,80	2,10	1,68	1	0,00	1,68	2,300	11	42,50							
Int. Wall (IW-C)	W	8,50	1,03	2,70	2,78	1	1,68	1,10	2,074	11	25,10							
Int. Wall (IW-C)	S	8,50	1,23	2,70	3,32	1	0,00	3,32	2,074	11	75,74							
Window (SW)	N	-	0,50	0,50	0,25	1	0,00	0,25	5,800	16	23,20							
Int. Wall (IW-C)	N	8,50	1,23	2,70	3,32	1	0,25	3,07	2,074	16	101,87							
Int. Wall (IW-C)	E	8,50	1,03	2,70	2,78	1	0,00	2,78	2,074	16	92,25							
TOTAL (Nominal Heat Loss)											372,73	7%	0%	0%	1,07			
Q (Heat Loss w/ Increments)																	398,82	
Q _s (Leakage Heat Loss)																	0	
TOTAL HEAT NEED (Q _h)																	398,82	

1st Basement Floor Calculations

HEAT LOSS CALCULATION CHART																	Page	3	
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	B-1	
																	Date	12.11.2025	
Component			Area Calculation							Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total				
					A ₀			A	U	ΔT	Q ₀	Z ₀	Z _w	ZH	Z	Q _h =Q _i +Q ₀			
		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%	W			
B-101										T _i =20°C									
Floor 1 (Downstairs)	Fl	58,00	-	-	56,50	1	0,00	56,50	0,108	5	30,51								
Floor 1 (Ceiling)	Ce	58,00	-	-	56,50	1	0,00	56,50	0,108	0	0,00								
Adjacent Wall (AW)	N	29,50	4,66	2,70	12,58	1	0,00	12,58	0,371	9	42,00								
Ext. Door (150/230)	W	-	1,50	2,30	3,45	1	0,00	3,45	2,600	16	143,52								
Window (100/200)	W	-	1,00	2,00	2,00	1	0,00	2,00	2,600	16	83,20								
Window (310/130)	W	-	3,10	1,30	4,03	1	0,00	4,03	2,600	16	167,65								
External Wall (EW)	W	29,50	6,70	2,70	18,09	1	9,48	8,61	0,490	16	67,50								
Window (170/90)	S	-	1,70	0,90	1,53	1	0,00	1,53	2,600	16	63,65								
Window (210/70)	S	-	2,10	0,70	1,47	1	0,00	1,47	2,600	16	61,15								
External Wall (EW)	S	29,50	4,91	2,70	13,26	1	3,00	10,26	0,490	16	80,44								
Window (150/150)	E	-	1,50	1,50	2,25	1	0,00	2,25	2,600	16	93,60								
External Wall (EW)	E	29,50	2,96	2,70	7,99	1	2,25	5,74	0,490	16	45,00								
Internal Wall 2 (IW-2)	E	23,50	8,43	2,70	22,76	1	0,00	22,76	1,440	10	327,74								
TOTAL (Nominal Heat Loss)												1205,97	7%	0%	0%	1,07			
Q (Heat Loss w/ Increments)																	1290,39		
Q _s (Leakage Heat Loss)																	950,04		
TOTAL HEAT NEED (Q _h)																	2240,43		
Total L = 57,60m																			

Ground Floor Calculations

HEAT LOSS CALCULATION CHART																Page	4			
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																Floor	G-0			
																Date	12.11.2025			
Component			Area Calculation						Heat Loss			Increments				Total Heat Need				
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total					
					A ₀			A	U	ΔT	Q ₀	Z _D	Z _W	Z _H	Z	Q _h =Q _i +Q ₀				
		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%	W				
G-001										T _i =20°C										
Floor 1 (Downstairs)	Fl	58,00	-	-	47,00	1	0,00	47,00	0,108	0	0,00									
Floor 1 (Ceiling @ 22°C)	Ce	58,00	-	-	32,90	1	0,00	32,90	0,108	-2	-7,11									
Floor 1 (Ceiling @ 18°C)	Ce	58,00	-	-	13,50	1	0,00	13,50	0,108	2	2,92									
Window (280/260)	N	-	2,80	2,60	7,28	1	0,00	7,28	2,600	16	302,85									
External Wall (EW)	N	29,50	3,46	3,00	10,38	1	7,28	3,10	0,490	16	24,30									
Window (310/260)	W	-	3,10	2,60	8,06	1	0,00	8,06	2,600	16	335,30									
External Wall (EW)	W	29,50	4,49	3,00	13,47	1	8,06	5,41	0,490	16	42,41									
Window (170/260)	S	-	1,70	2,60	4,42	1	0,00	4,42	2,600	16	183,87									
Window (210/260)	S	-	2,10	2,60	5,46	1	0,00	5,46	2,600	16	227,14									
External Wall (EW)	S	29,50	4,64	3,00	13,92	1	9,88	4,04	0,490	16	31,67									
Internal Door (ID)	E	-	0,80	2,10	1,68	1	0,00	1,68	2,300	-6	-23,18									
Int. Wall (IW-C)	E	8,50	1,20	3,00	3,60	1	1,68	1,92	2,074	-6	-23,89									
Window (90/150)	E	-	0,90	1,50	1,35	1	0,00	1,35	2,600	16	56,16									
External Wall (EW)	E	29,50	0,90	3,00	2,70	1	1,35	1,35	0,490	16	10,58									
Internal Wall 2 (IW-2)	E	23,50	8,19	3,00	24,57	1	0,00	24,57	1,440	10	353,81									
TOTAL (Nominal Heat Loss)											1516,83			7%	0%	5%	1,12			
Q (Heat Loss w/ Increments)																	1698,85			
Q _s (Leakage Heat Loss)																	1303,46			
TOTAL HEAT NEED (Q _h)																	3002,31			
Total L = 78,90m																				

HEAT LOSS CALCULATION CHART																	Page	5
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	G-0
																	Date	12.11.2025
Component			Area Calculation						Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total			
					A ₀			A	U	ΔT	Q ₀	Z _D	Z _W	ZH	Z		Q _h =Q _i +Q ₀	
		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%		W	
G-002										T _i =26°C								
Floor 3 (Bath Down)	Fl	58,00	-	-	1,50	1	0,00	1,50	0,590	0	0,00							
Floor 3 (Bath Up)	Ce	58,00	-	-	1,50	1	0,00	1,50	0,590	0	0,00							
Internal Door (ID)	W	-	0,80	2,10	1,68	1	0,00	1,68	2,300	6	23,18							
Int. Wall (IW-C)	W	8,50	1,25	3,00	3,75	1	1,68	2,07	2,074	6	25,76							
Int. Wall (IW-C)	N	8,50	1,20	3,00	3,60	1	0,00	3,60	2,074	6	44,80							
Internal Wall 2 (IW-2)	E	23,50	1,25	3,00	3,75	1	0,00	3,75	1,440	16	86,40							
Window (50/50)	S	-	0,50	0,50	0,25	1	0,00	0,25	2,600	22	14,30							
External Wall (EW)	S	29,50	1,20	3,00	3,60	1	0,25	3,35	0,490	22	36,11							
TOTAL (Nominal Heat Loss)												230,55	7%	0%	-5%	1,02		
Q (Heat Loss w/ Increments)																	235,17	
Q _s (Leakage Heat Loss)																	34,06	
TOTAL HEAT NEED (Q _h)																	269,23	
Total L = 1,8m																		

1st Floor Calculations

[illegible]

HEAT LOSS CALCULATION CHART																	Page	9
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	F-1
																	Date	12.11.2025
Component			Area Calculation						Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total			
					A ₀		A	U	ΔT	Q ₀	Z _D	Z _W	ZH	Z	Q _h =Q _i +Q ₀			
		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%	W		
F-104									T _i =18°C									
Floor 1 (Down)	Fl	58,00	-	-	7,00	1	0,00	7,00	0,108	-2	-1,51							
Floor 1 (Up)	Ce	58,00	-	-	7,00	1	0,00	7,00	0,108	0	0,00							
Internal Door (K2 90/210)	N	-	0,90	2,10	1,89	1	0,00	1,89	2,300	0	0,00							
Internal Wall 1 (IW-1)	N	8,50	2,00	2,90	5,80	1	1,89	3,91	2,100	0	0,00							
Internal Door (K2 90/210)	E	-	0,90	2,10	1,89	1	0,00	1,89	2,300	-4	-17,39							
Internal Wall 1 (IW-1)	E	8,50	3,99	2,90	11,57	1	1,89	9,68	2,100	-4	-81,31							
Internal Door (K2 90/210)	E	-	0,90	2,10	1,89	1	0,00	1,89	2,300	-4	-17,39							
Internal Wall 1 (IW-1)	E	8,50	3,95	2,90	11,46	1	1,89	9,57	2,100	-4	-80,39							
Internal Door (ID 100/230)	S	-	1,00	2,30	2,30	1	0,00	2,30	2,300	8	42,32							
Internal Wall 2 (IW-2)	S	23,50	3,02	2,90	8,76	1	2,30	6,46	1,440	8	74,42							
Internal Door (K3 80/210)	W	-	0,80	2,10	1,68	1	0,00	1,68	2,300	-8	-30,91							
Int. Wall (IW-C)	W	8,50	1,10	2,90	3,19	1	1,68	1,51	2,074	-8	-25,05							
Internal Door (K3 80/210)	W	-	0,80	2,10	1,68	1	0,00	1,68	2,300	-8	-30,91							
Internal Wall 1 (IW-1)	W	8,50	2,20	2,90	6,38	1	1,68	4,70	2,100	-8	-78,96							
TOTAL (Nominal Heat Loss)											-247,09	7%	0%	0%	1,07			
Q (Heat Loss w/ Increments)																-264,38		
Q _s (Leakage Heat Loss)																0		
TOTAL HEAT NEED (Q _h)																-264,38		

2nd Floor Calculations

HEAT LOSS CALCULATION CHART																	Page	13
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	F-2
																	Date	12.11.2025
Component			Area Calculation						Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total			
					A ₀			A	U	ΔT	Q ₀	Z _D	Z _W	ZH	Z		Q _h =Q _i +Q ₀	
					m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%		W	
F-201		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	T _i =22°C								
Floor 1 (Down)	Fl	58,00	-	-	11,60	1	0,00	11,60	0,108	0	0,00							
Floor 1 (Up)	Ce	58,00	-	-	11,60	1	0,00	11,60	0,108	0	0,00							
Adjacent Wall (AW)	W	29,50	6,15	2,90	17,84	1	0,00	17,84	0,371	11	72,81							
Window (P 200/210)	N	-	2,00	2,10	4,20	1	0,00	4,20	2,600	18	196,56							
External Wall (EW)	N	29,50	3,51	2,90	10,18	1	4,20	5,98	0,490	18	52,74							
Internal Door (K2 90/210)	E	-	0,90	2,10	1,89	1	0,00	1,89	2,300	4	17,39							
Internal Wall 1 (IW-1)	E	8,50	3,94	2,90	11,43	1	1,89	9,54	2,100	4	80,14							
Internal Door (K3 80/210)	S	-	0,80	2,10	1,68	1	0,00	1,68	2,300	-4	-15,46							
Int. Wall (IW-C)	S	8,50	3,34	2,90	9,69	1	1,68	8,01	2,074	-4	-66,45							
TOTAL (Nominal Heat Loss)											337,73	7%	0%	5%	1,12			
Q (Heat Loss w/ Increments)																378,25		
Q _s (Leakage Heat Loss)																168,76		
TOTAL HEAT NEED (Q _h)																547,01		
Total L = 10,92m																		

HEAT LOSS CALCULATION CHART																	Page	16
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	F-2
																	Date	12.11.2025
Component			Area Calculation						Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total			
					A ₀		A	U	ΔT	Q ₀	Z ₀	Z _w	ZH	Z	Q _h =Q _i +Q ₀			
		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%	W		
F-204										T _i =18°C								
Floor 1 (Down)	Fl	58,00	-	-	7,00	1	0,00	7,00	0,108	0	0,00							
Floor 1 (Up)	Ce	58,00	-	-	7,00	1	0,00	7,00	0,108	0	0,00							
Internal Door (K2 90/210)	N	-	0,90	2,10	1,89	1	0,00	1,89	2,300	0	0,00							
Internal Wall 1 (IW-1)	N	8,50	2,00	2,90	5,80	1	1,89	3,91	2,100	0	0,00							
Internal Door (K2 90/210)	E	-	0,90	2,10	1,89	1	0,00	1,89	2,300	-4	-17,39							
Internal Wall 1 (IW-1)	E	8,50	3,99	2,90	11,57	1	1,89	9,68	2,100	-4	-81,31							
Internal Door (K2 90/210)	E	-	0,90	2,10	1,89	1	0,00	1,89	2,300	-4	-17,39							
Internal Wall 1 (IW-1)	E	8,50	3,95	2,90	11,46	1	1,89	9,57	2,100	-4	-80,39							
Internal Door (ID 100/230)	S	-	1,00	2,30	2,30	1	0,00	2,30	2,300	8	42,32							
Internal Wall 2 (IW-2)	S	23,50	3,02	2,90	8,76	1	2,30	6,46	1,440	8	74,42							
Internal Door (K3 80/210)	W	-	0,80	2,10	1,68	1	0,00	1,68	2,300	-8	-30,91							
Int. Wall (IW-C)	W	8,50	1,10	2,90	3,19	1	1,68	1,51	2,074	-8	-25,05							
Internal Door (K3 80/210)	W	-	0,80	2,10	1,68	1	0,00	1,68	2,300	-8	-30,91							
Internal Wall 1 (IW-1)	W	8,50	2,20	2,90	6,38	1	1,68	4,70	2,100	-8	-78,96							
TOTAL (Nominal Heat Loss)											-245,57	7%	0%	0%	1,07			
Q (Heat Loss w/ Increments)																-262,76		
Q _s (Leakage Heat Loss)																0		
TOTAL HEAT NEED (Q _h)																-262,76		

3rd Floor Calculations

[illegible]

Loft Floor Calculations

HEAT LOSS CALCULATION CHART																	Page	26
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	C-0
																	Date	12.11.2025
Component			Area Calculation						Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total			
					A ₀			A	U	ΔT	Q ₀	Z _D	Z _W	ZH	Z		Q _h =Q _i +Q ₀	
					m²	pc	m²	m²	W/m²K	K	W	%	%	%	1+%		W	
					T _i =22°C													
C-001		cm	m	m														
Floor 1 (Down)	Fl	58,00	-	-	6,20	1	0,00	6,20	0,108	0	0,00							
Roof	Ce	-	-	-	6,20	1	0,00	6,20	0,251	18	28,01							
Adjacent Wall (AW)	W	29,50	2,49	2,01	5,00	1	0,00	5,00	0,371	18	33,39							
Internal Door (ID)	N	-	0,90	1,98	1,78	1	0,00	1,78	2,300	13	53,22							
External Wall (EW)	N	29,50	2,43	2,01	4,88	1	1,78	3,10	0,490	13	19,75							
Internal Wall 1 (IW-1)	E	8,50	3,34	2,01	6,71	1	0,00	6,71	2,100	0	0,00							
Internal Door (K3 80/210)	S	-	0,80	1,98	1,58	1	0,00	1,58	2,300	-4	-14,54							
Int. Wall (IW-C)	S	8,50	2,43	2,01	4,88	1	1,58	3,30	2,074	-4	-27,38							
TOTAL (Nominal Heat Loss)											92,46	7%	0%	5%	1,12			
Q (Heat Loss w/ Increments)																103,55		
Q _s (Leakage Heat Loss)																0		
TOTAL HEAT NEED (Q _h)																103,55		

HEAT LOSS CALCULATION CHART																	Page	29
BUILDING'S NAME:Residential + Commercial Building (Esenyurt, Block: 884, Parcel: 3)																	Floor	C-0
																	Date	12.11.2025
Component			Area Calculation						Heat Loss			Increments				Total Heat Need		
Label	Direction	Thickness	Length	Height or Width	Total Area	Quantity	Omitted Area	Accounted Area	Heat Transfer Coefficient	Temperature Difference	Nominal Heat Loss	Combined	Storey Height	Direction	Total			
					A ₀			A	U	ΔT	Q ₀	Z _D	Z _W	ZH	Z	Q _h =Q _i +Q ₀		
		cm	m	m	m ²	pc	m ²	m ²	W/m ² K	K	W	%	%	%	1+%	W		
C-004										T _i =26°C								
Floor 3 (Bath F-306)	Fl	58,00	-	-	4,00	1	0,00	4,00	0,590	0	0,00							
Floor 1 (F-304)	Fl	58,00	-	-	1,30	1	0,00	1,30	0,108	8	1,12							
Roof	Ce	-	-	-	5,59	1	0,00	5,59	0,251	22	30,87							
Internal Door (K3 80/198)	N	-	0,80	1,98	1,58	1	0,00	1,58	2,300	4	14,54							
Int. Wall (IW-C) (C-001)	N	8,50	2,43	2,01	4,88	1	1,58	3,30	2,074	4	27,38							
Internal Wall 1 (IW-1)	E	8,50	2,70	2,01	5,43	1	0,00	5,43	2,100	4	45,61							
Internal Wall 2 (IW-2)	S	23,50	2,96	2,01	5,95	1	0,00	5,95	1,440	16	137,09							
Internal Wall 2 (IW-2)	W	23,50	3,20	2,01	6,43	1	0,00	6,43	1,440	16	148,15							
TOTAL (Nominal Heat Loss)											404,75	7%	0%	0%	1,07			
Q (Heat Loss w/ Increments)																433,08		
Q _s (Leakage Heat Loss)																0		
TOTAL HEAT NEED (Q _h)																433,08		

Total Heat Loss Summary

Total Heat Loss Chart			
Floor	Room No	Room Heat Load (Q _h)[W]	
2nd Basement	B-201	161,27	
	B-202	398,82	
1st Basement	B-101	2240,43	
Ground	G-001	3002,31	
	G-002	269,23	
1st Floor	F-101	549,82	
	F-102	141,49	
	F-103	895,50	
	F-104	-264,38	
	F-105	1337,15	
	F-106	67,65	
	F-107	437,67	
2nd Floor	F-201	547,01	
	F-202	143,00	
	F-203	890,34	
	F-204	-262,76	
	F-205	1326,87	
	F-206	54,01	
	F-207	437,67	
3rd Floor	F-301	462,19	
	F-302	166,31	
	F-303	937,80	
	F-304	-349,54	
	F-305	1315,70	
	F-306	697,73	
Loft	C-001	103,55	
	C-002	283,23	
	C-003	104,66	
	C-004	433,08	
Total Building Load		16527,81	W
Total Capacity		16,53	kW

Conclusion

In this report, the heat loss calculations for the building located in Esenyurt have been completed in accordance with TS 825 standards. The calculations considered transmission losses (Q_0), necessary increments (Z), and infiltration losses (Q_s).

The total heat load of the building is calculated as 16,53 kW. This value represents the peak heating demand required at the design temperature of +4°C. This total capacity will be used as the basis for selecting the central heating boiler and pump sizing in the next stage of the project.



**BEYKENT UNIVERSITY
FACULTY OF ENGINEERING ARCHITECTURE
MECHANICAL ENGINEERING**

REPORT 4

Yusuf Yıldızlar

2103040020

DESIGN OF THERMAL SYSTEM

ADVISOR: Dr Öğr. Üyesi MUAMMER KANLI

İSTANBUL, 2025

Introduction

This report presents the fourth phase of the thermal system design project for the residential and commercial building located in Esenyurt, Istanbul (Block: 884, Parcel: 3). Following the architectural analysis (Report 1), thermal transmittance calculations (Report 2), and detailed heat loss calculations (Report 3), this study focuses on the selection, sizing, and placement of heating terminal units (radiators).

The primary objective of this report is to determine the appropriate radiator dimensions and types required to satisfy the specific heat load (Q_h) of each room, ensuring thermal comfort under design conditions.

Methodology and Design Parameters:

The radiator selection process is based on the following standards and project guidelines:

Heating System Regime: A central heating system with a standard water temperature regime of 90/70°C is assumed for the calculations.

Radiator Types:

Panel Radiators (Type P100): Selected for standard living spaces (Living rooms, Bedrooms, Kitchens, Shops) with a standard height of 500 mm.

Towel Radiators: Selected for wet volumes such as Bathrooms and WCs to provide both heating and towel drying functions.

Capacity Correction: Since the standard radiator heat outputs are typically given for a 20°C indoor temperature, the actual heat output for each room is calculated using "Capacity Factors" (F_k). These factors correct the radiator performance based on the specific design temperature of the room (e.g., 15°C, 18°C, 22°C, or 26°C).

Sizing Strategy: The required radiator length is calculated by dividing the room's heat loss by the corrected unit efficiency (W/m). The calculated lengths are rounded up to the nearest commercially available size (multiples of 10 cm), with a minimum applied length of 40 cm for practical and aesthetic reasons.

Placement: Radiators are positioned primarily under windows or on external walls to counteract heat loss and prevent cold drafts, in accordance with the architectural constraints.

The results of this study, detailed in the "Radiator Selection Charts," provide the Total Installed Capacity of the building, which will be used to select the central heating boiler in the final stage of the design.

Calculation Methodology

To demonstrate the sizing process applied throughout this project, the detailed calculation steps for a sample room (Room F-105 - Bedroom) are presented below. The same methodology was applied to all panels and towel radiators in the building.

- **Room:** F-105 (1st Floor Bedroom)
- **Design Indoor Temperature** (T_{in}): 22°C
- **Calculated Heat Loss** (Q_{loss}): 1337.15 W (From Report 3)
- **Selected Radiator Type:** Panel Radiator P100 (Height: 500 mm)
- **Catalog Heat Output** ($T_{cat} = 20^{\circ}C$): 2429 W/m

Step 1: Correcting the Unit Efficiency (Q_u)

Since the room temperature (22°C) differs from the standard catalog temperature (20°C), a capacity factor (F_k) is used to adjust the efficiency.

- From the "Capacity Factor vs Room Temperature" chart, for 22°C $> F_k = 1.05$.

$$Q_u = \frac{Q_{catalog}}{F_k} = \frac{2429}{1.05} = 2313.33 \text{ W/m}$$

Step 2: Calculating the Theoretical Length ($L_{theoretical}$)

The required radiator length is found by dividing the total heat loss by the corrected unit efficiency.

$$L_{theoretical} = \frac{Q_{loss}}{Q_u} = \frac{1337.15}{2313.33} = 0.578 \text{ m}$$

Step 3: Selection and Rounding ($L_{selection}$)

Radiators are manufactured in multiples of 10 cm. Therefore, the theoretical length is rounded up to the nearest standard size.

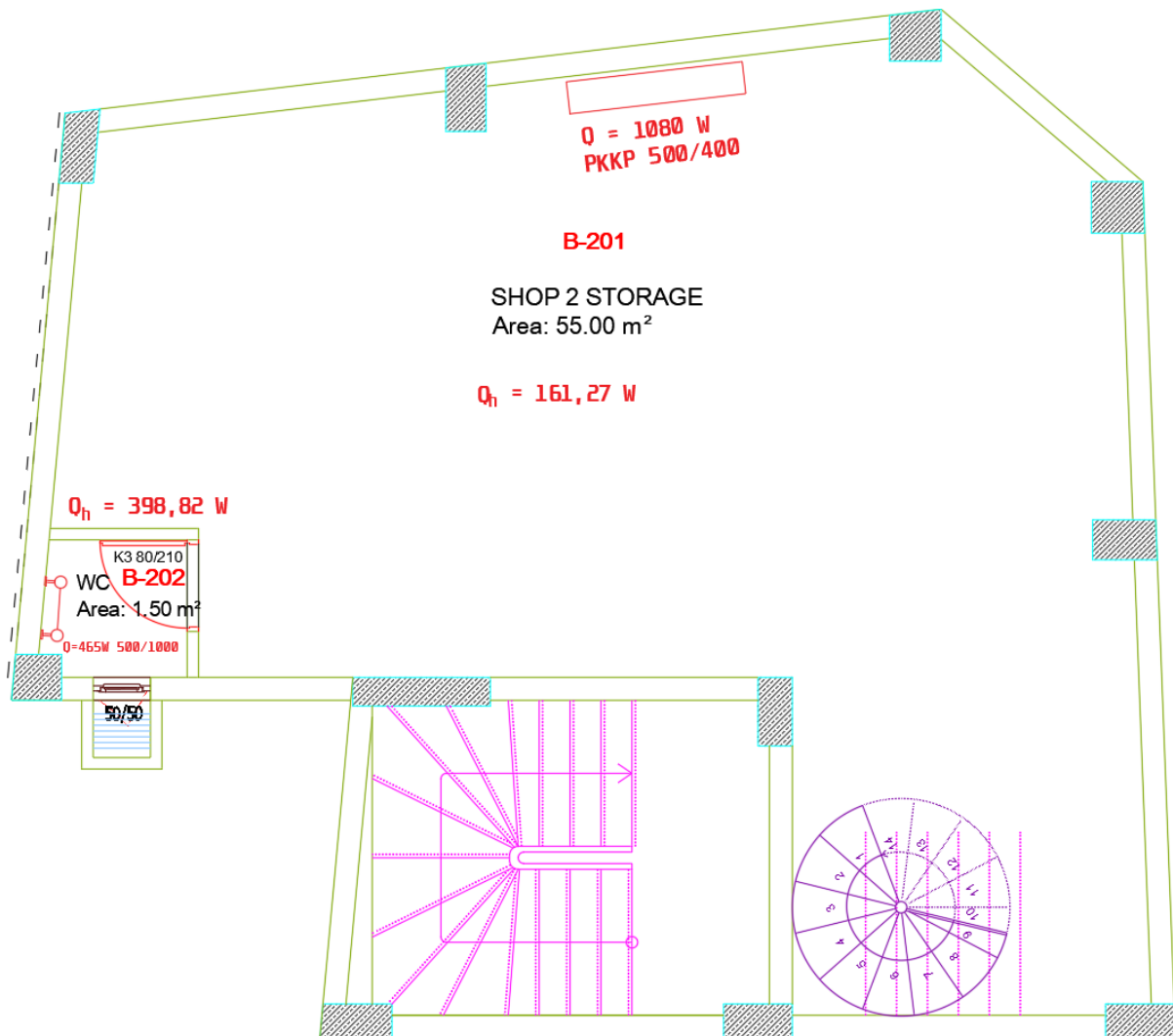
$$L_{theoretical} = 0.578 \text{ m} > L_{selection} = \mathbf{0.6 \text{ m}} \text{ (Selected Length)}$$

Step 4: Calculating the Actual Installed Capacity ($Q_{installed}$)

The final heating capacity provided to the room is calculated using the selected length.

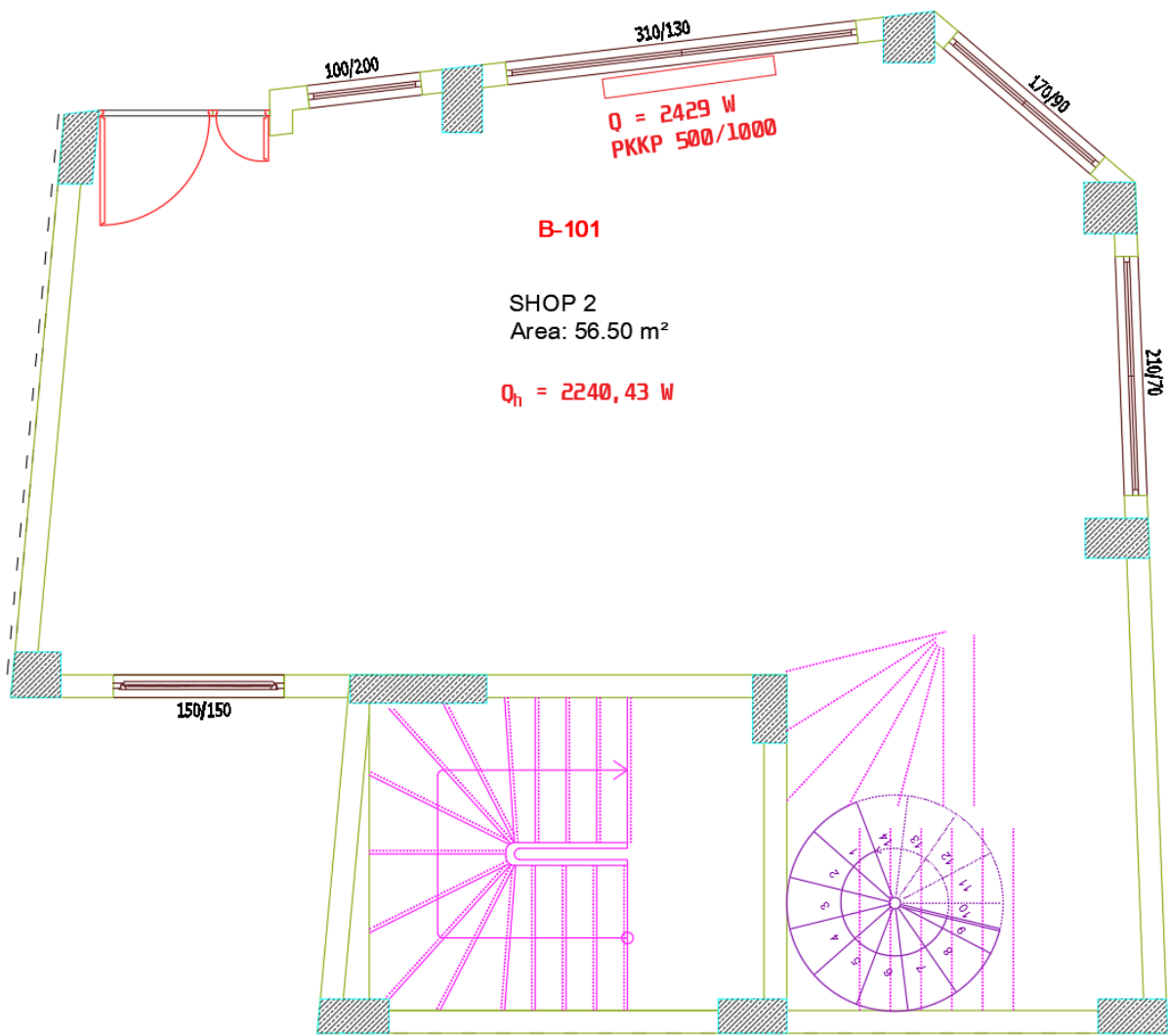
$$Q_{installed} = L_{selection} \times Q_u = 0.6 \times 2313.33 = \mathbf{1388 \text{ W}}$$

Result: Since the installed capacity (1388 W) is greater than the heat loss (1337.15 W), the selected radiator is sufficient for this room.



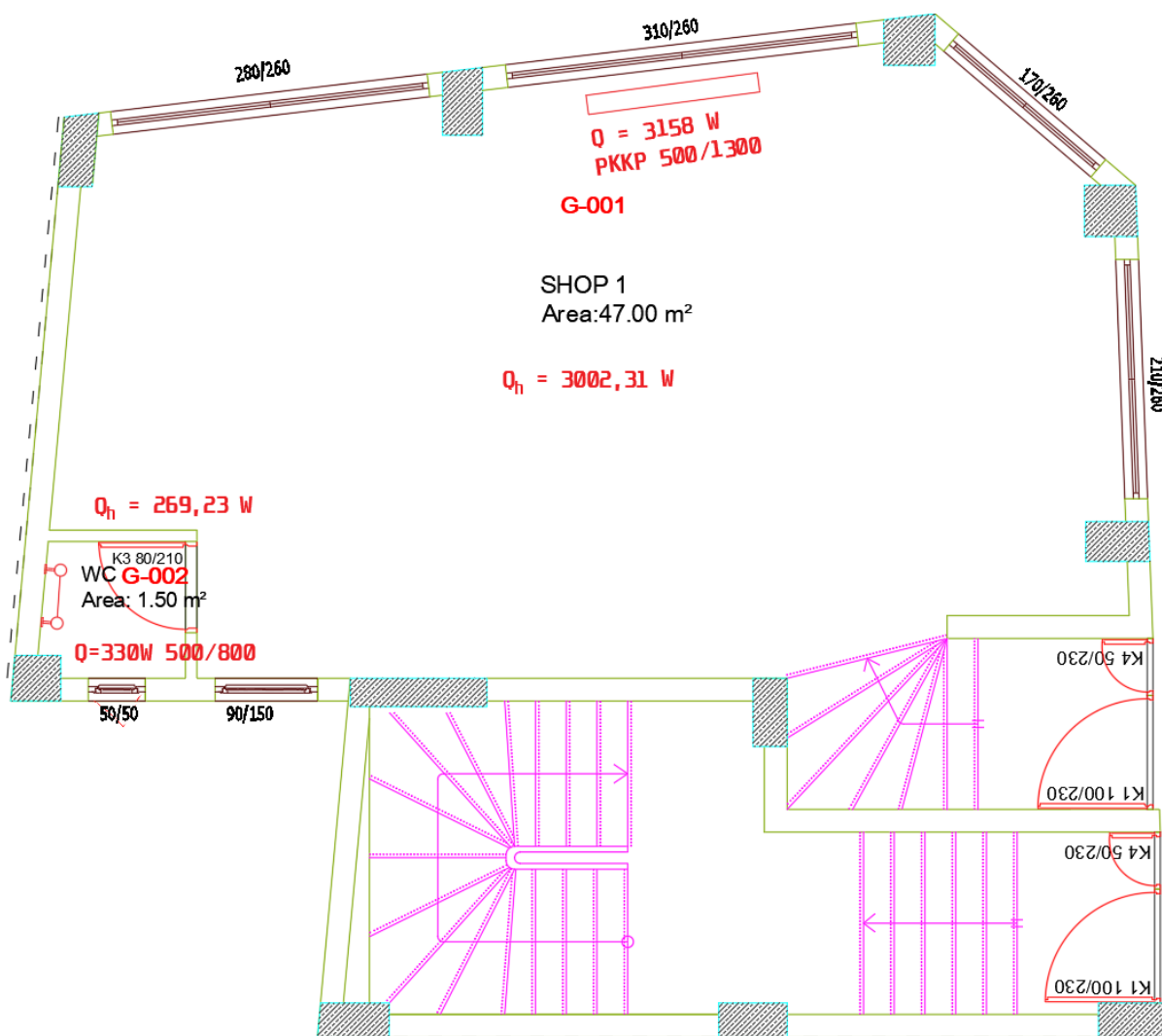
2nd Basement Floor (B-2) Radiator Placement Plan

Scale: 1/50



1st Basement Floor (B-1) Radiator Placement Plan

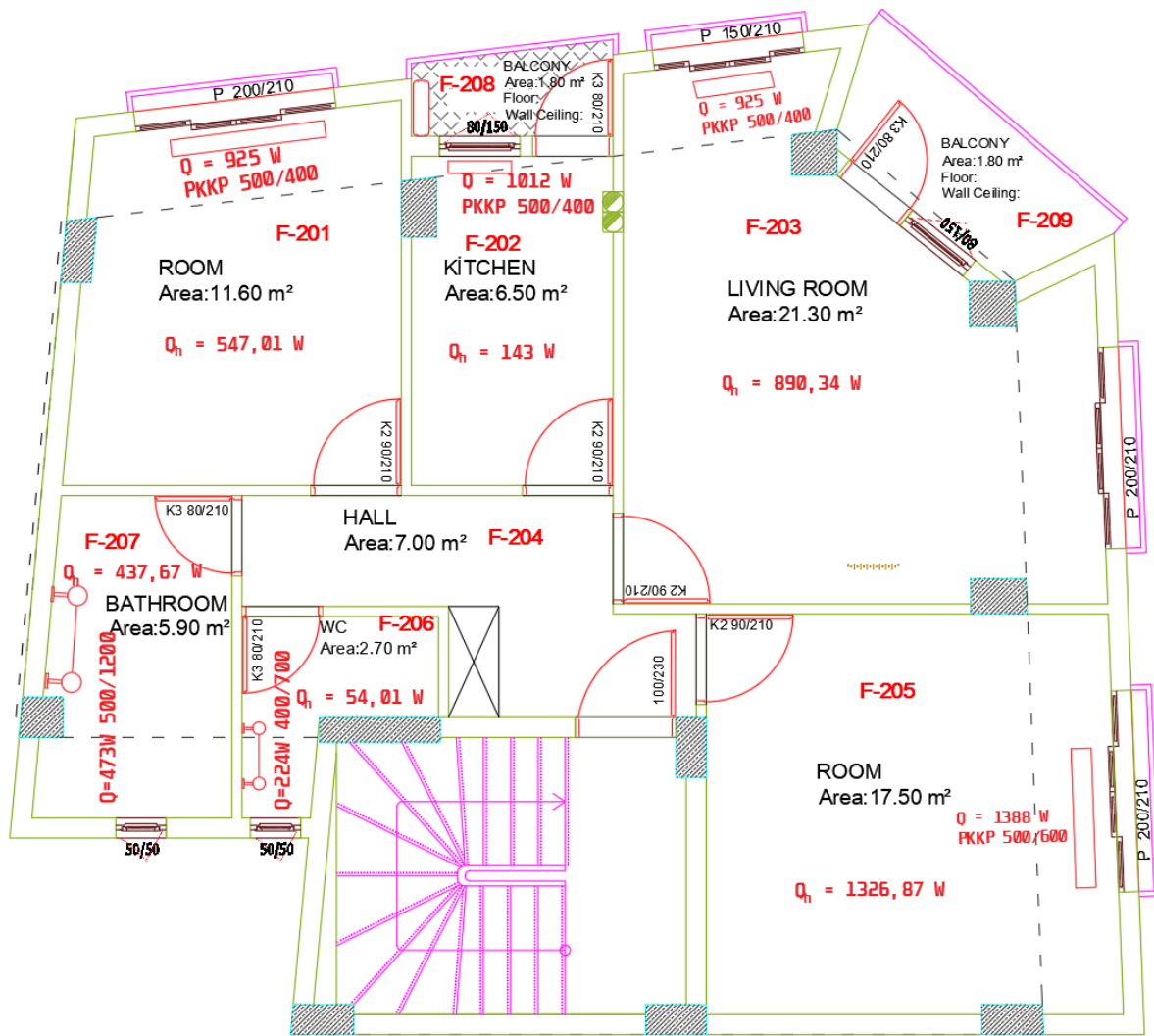
Scale: 1/50



Ground Floor (G-0) Radiator Placement Plan

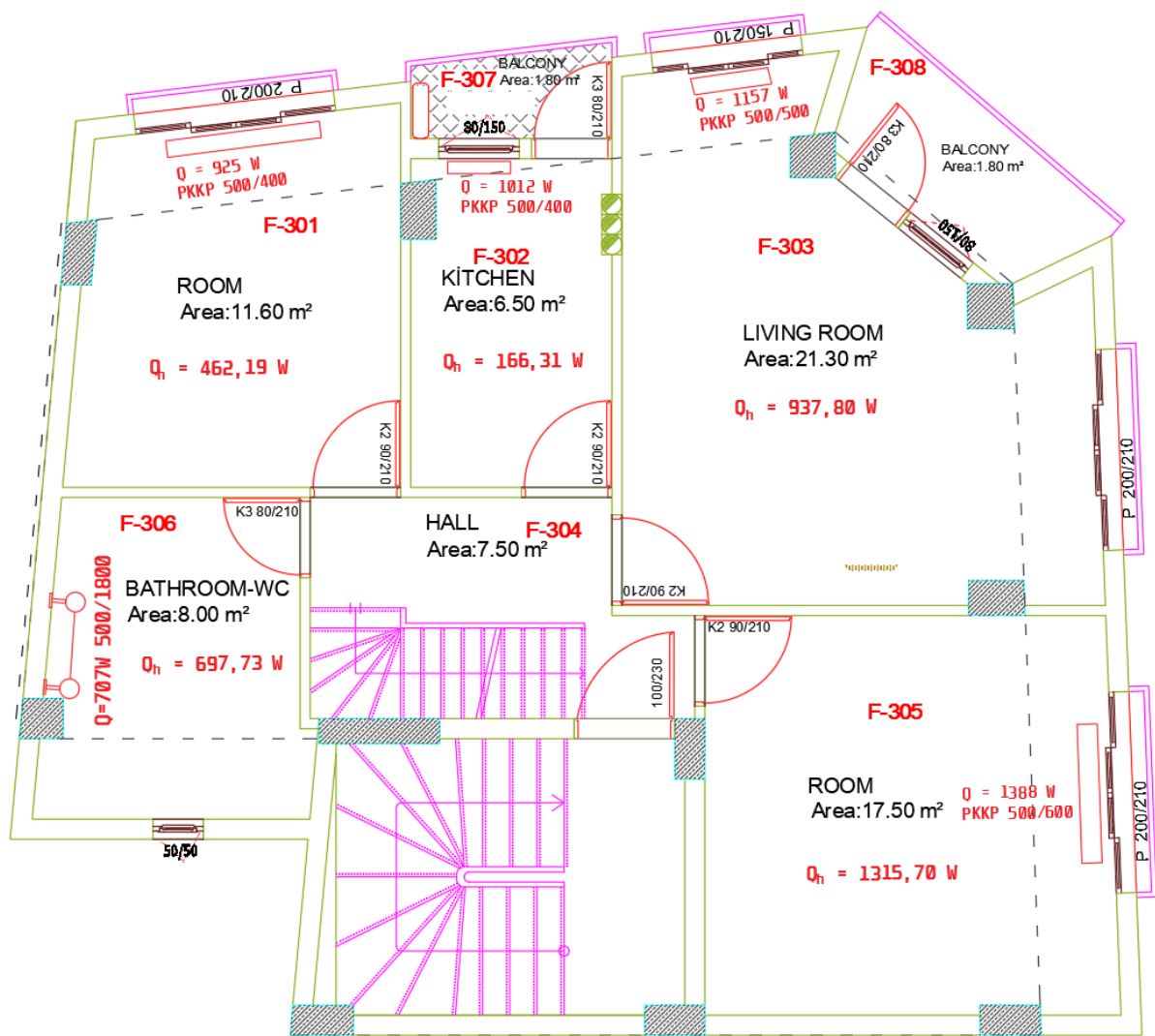
Scale: 1/50

Scale: 1/50



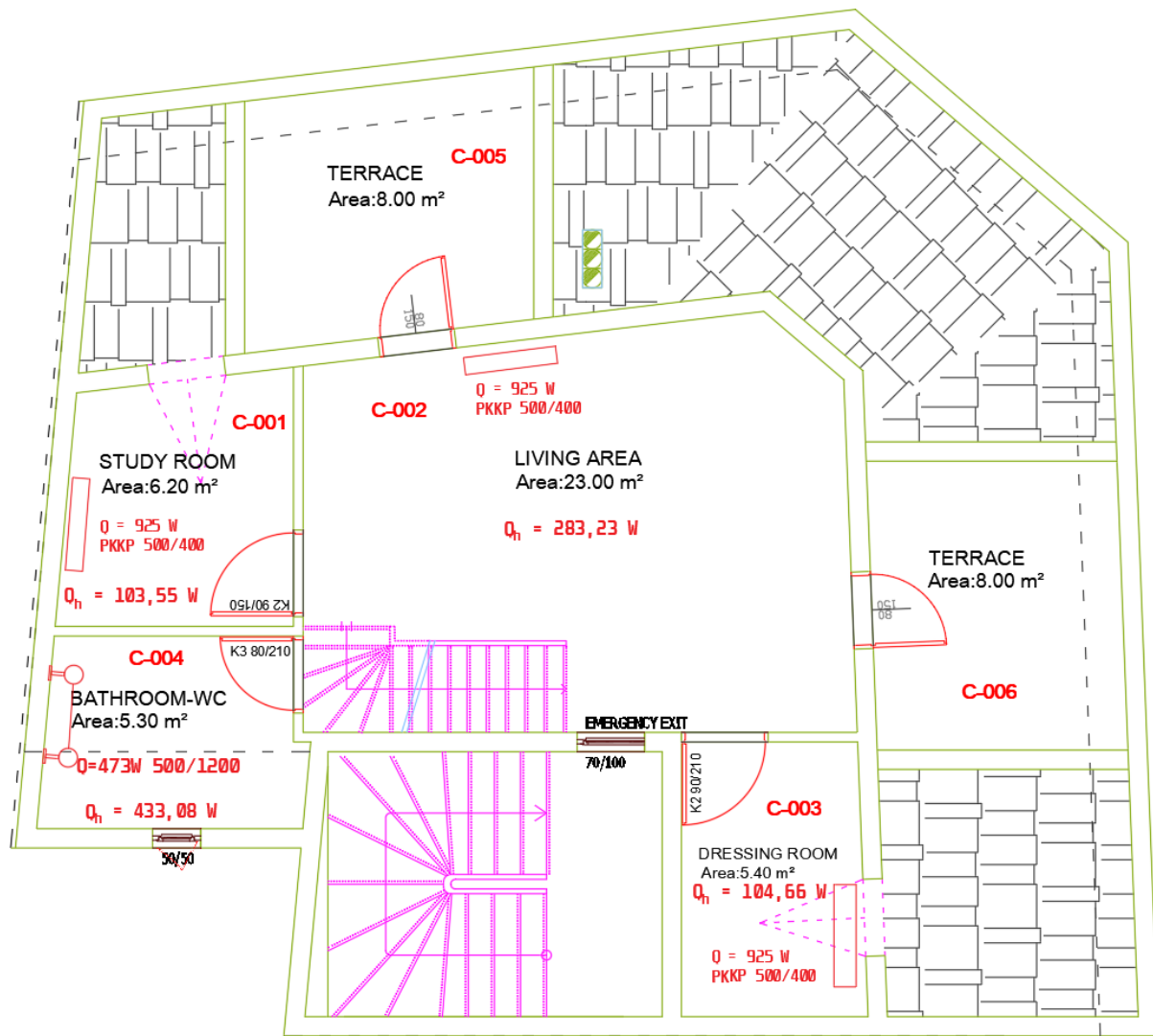
2nd Floor (F-2) Radiator Placement Plan

Scale: 1/50



3rd Floor (F-3) Radiator Placement Plan

Scale: 1/50



Loft Floor (C-0) Radiator Placement Plan

Scale: 1/50

Conclusion & Boiler Selection

In this study, the selection, sizing, and placement of heating terminal units (radiators) were performed for the residential and commercial building located in Esenyurt, Istanbul. The study is based on the architectural analysis (Report 1) and the detailed heat loss calculations (Report 3).

Radiator Selection Criteria: The following parameters were applied during the selection process:

- **Radiator Type:** Panel Radiators (Type P100 / PKKP, Height: 500 mm) were selected for standard living spaces (Living rooms, Bedrooms, Kitchens, Shops). Towel Radiators (Havlupan) were selected for wet volumes (Bathrooms and WCs).
- **Design Parameters:** A central heating system with a 90/70°C water regime was assumed.
- **Sizing Methodology:** The required radiator lengths were calculated by correcting the standard heat outputs using Capacity Factors (F_k) based on the design room temperatures (15°C – 26°C).
- **Constraints:** Calculated lengths were rounded up to the nearest commercially available size (multiples of 10 cm). A minimum radiator length of 40 cm was applied for all rooms to ensuring practical installation and aesthetic integrity.

Total Installed Capacity Analysis: The sum of the heating capacities of all selected radiators in the building is calculated as follows:

$$Q_{Total\ Installed} = 25\ 793\ W \approx 26\ kW$$

Evaluation: While the calculated transmission and infiltration heat loss of the building was 16.53 kW (as determined in Report 3), the Total Installed Radiator Capacity is 26 kW. This difference is due to:

1. Applying a minimum radiator size of 40 cm (900-1000 W) even for small rooms with very low heat demand.
2. Rounding up theoretical lengths to the nearest upper standard size.
3. Safety margins ensuring rapid heating (shorter regime time).

Final Boiler Selection: To ensure sufficient heating performance under all conditions and peak demands, the Central Heating Boiler capacity will be selected based on the total installed capacity of 26 kW.

References

1. **TS 825:** "Thermal Insulation Requirements for Buildings", Turkish Standards Institution.
2. **Course Notes:** Dr. Muammer Kanlı, "Design of Thermal Systems", Week 5 (Radiators), Week 6 (Regulations), and Week 7 (Heater Calculation), Beykent University, 2022.
3. **Architectural Project:** "Residential + Commercial Building (Block: 884, Parcel: 3)", Esenyurt/Istanbul.
4. **Product Catalogues:** Standard heating capacity tables for Panel Radiators (Type PKKP/22) and Towel Radiators



**BEYKENT UNIVERSITY
FACULTY OF ENGINEERING ARCHITECTURE
MECHANICAL ENGINEERING**

REPORT 5

Yusuf Yıldızlar

2103040020

DESIGN OF THERMAL SYSTEM

ADVISOR: Dr Öğr. Üyesi MUAMMER KANLI

İSTANBUL, 2025

Introduction

In this study, the thermal system of a multi-storey building was designed and analysed.

The aim of the project is to determine the critical heating circuit and to calculate the pressure losses along this circuit in order to ensure proper system operation.

The building consists of two basement floors, a ground floor, three typical floors, and a loft floor. Heating loads for each floor were previously calculated, and the total system heat demand was obtained accordingly. Based on these results, the heating distribution network was designed using a two-pipe vertical riser system.

Special attention was given to the identification of the critical circuit, which represents the longest and most hydraulically demanding flow path from the boiler to the most remote radiator and back to the boiler.

Design Assumptions

1. The heating system is assumed to operate as a closed-loop hot water system.
2. The temperature difference between supply and return lines is taken as 20 °C.
3. Water properties are assumed constant within the operating temperature range.
4. Minor losses are calculated using standard loss coefficients.
5. The critical circuit is defined as the longest hydraulic path with the highest pressure loss.

Description of the Heating System

General System Layout:

The heating system is designed as a two-pipe hot water heating system with vertical risers.

The boiler is located on the second basement floor, and the supply and return pipes rise vertically to serve all floors.

Each floor is connected to the main riser through a single supply and a single return branch.

Within each floor, radiators are connected to the floor distribution pipes; however, floor-level horizontal piping and radiator connections are not included in the riser loss calculations, except for the critical floor.

Heating Loads:

The heating loads of the building floors were determined previously and are summarised below:

Loft Floor: 3248.6 W

3rd Floor: 5188.5 W

2nd Floor: 4947.2 W

1st Floor: 4947.2 W

Ground floor: 3487.7 W

1st Basement: 2429 W

2nd Basement: 1544.6 W

CRITICAL CIRCUIT DEFINITION

Identification of the Critical Circuit:

The critical circuit is defined as the flow path with the maximum total pressure loss in the system.

In this project, the critical circuit starts from the boiler located on the second basement floor, rises through the vertical riser, reaches the loft floor, and supplies the most remote radiator (study room radiator) before returning back to the boiler through the return line.

This circuit was selected because it represents the longest flow path and includes the highest number of fittings and local losses.

Pipe Diameter Selection:

Pipe diameters along the critical circuit were selected based on the calculated heat loads and allowable flow velocities using the Pipe Diameter Calculation Chart (PDCC).

The selected nominal pipe diameters vary along the riser, decreasing as the carried heat load reduces towards the upper floors. The corresponding inch values were used in the local loss calculations.

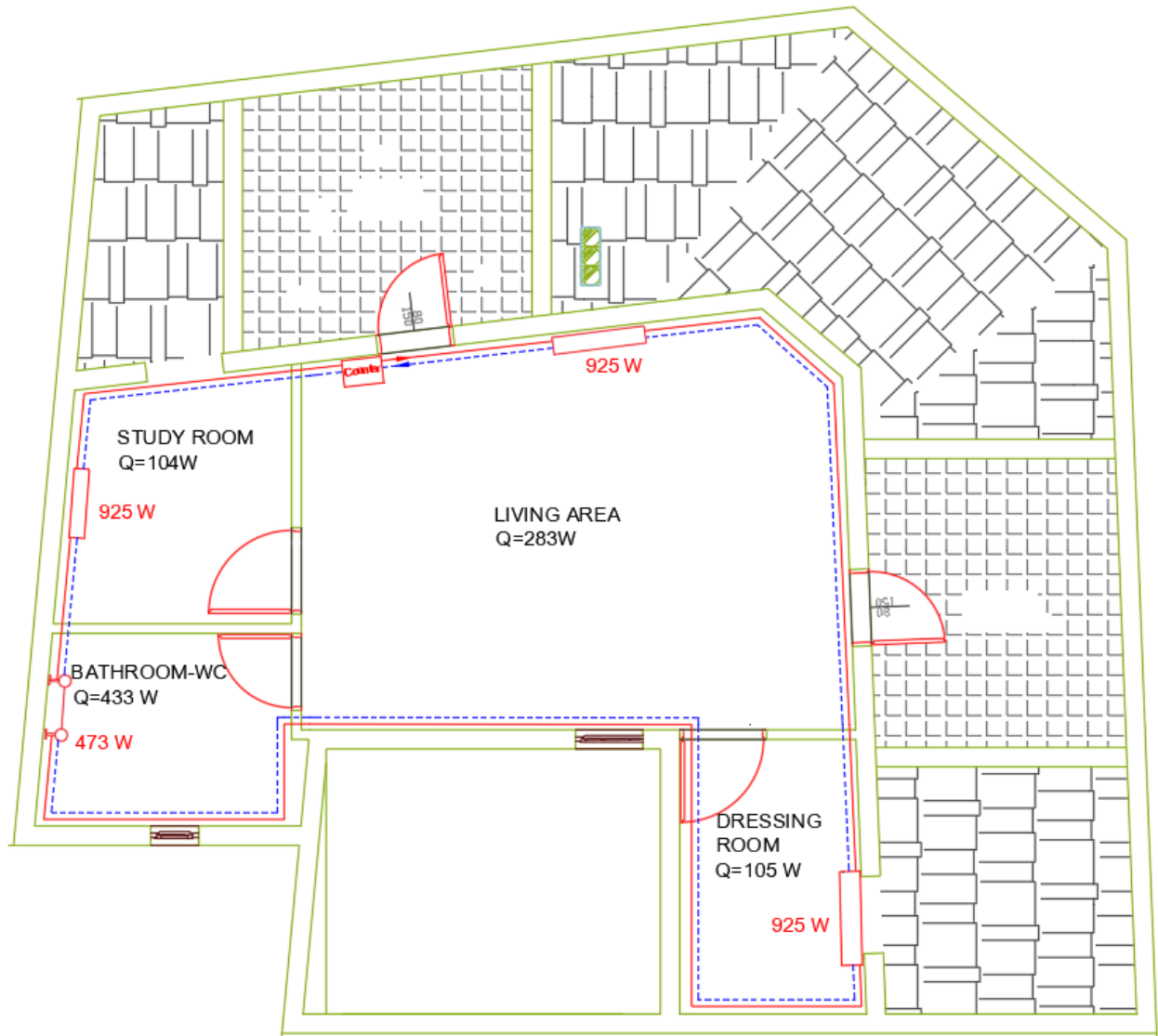


Figure 1 shows the piping layout of the critical floor and the critical circuit extending from the vertical riser to the most remote radiator.

LOCAL PRESSURE LOSSES (ζ CALCULATION)

General Approach:

Local pressure losses were calculated using the loss coefficient (ζ) method.

Equivalent pipe length values were not used in order to avoid double counting of losses.

The ζ values were selected according to the pipe diameter corresponding to each section of the critical circuit.

Assumptions for Local Loss Calculations:

The following assumptions were made:

- Only the vertical riser is considered for intermediate floors.
- Floor-level horizontal piping and radiators are excluded from the riser loss calculations.
- Each intermediate floor includes:
 - 1 T-separation (supply)
 - 1 T-combination (return)
- The loft floor (critical floor) additionally includes:
 - All local losses from the riser outlet to the critical radiator.
- Elbows are assumed to be standard 90° elbows.
- Ogee (long-radius) elbows are not considered unless explicitly shown in the drawings.
- Valves are assumed to be fully open.

STRUCTURE OF THE LOCAL LOSS TABLE

The local loss table is organised into eight rows, representing different sections of the critical circuit:

1. Boiler room
2. Second basement floor (riser start)
3. First basement floor
4. Ground floor
5. First floor
6. Second floor
7. Third floor
8. Loft floor (critical floor)

For each row:

- The corresponding pipe diameter is specified.
- The number of local loss elements (elbows, T-separations, T-combinations, valves) is listed.
- The total local loss contribution is calculated by multiplying the quantity of each element by its ζ value.

SPECIAL TREATMENT OF THE LOFT FLOOR

The loft floor is treated separately because it contains the critical radiator.

In addition to the standard riser losses (one separation and one combination), the loft floor includes:

- Local losses from internal floor piping up to the critical radiator
- Additional T-separations and T-combinations within the floor
- Elbows along the critical flow path
- One radiator valve for the critical radiator

This approach ensures that all local losses along the critical circuit are fully accounted for without double counting.

Riser Diagram

The riser diagram was prepared to illustrate the vertical layout of the heating system and to represent the main system components in a two-dimensional schematic form. The riser diagram includes the boiler, circulation pump, supply and return riser pipes, and the floor connections of the heating system.

In the riser diagram, the supply and return lines are shown separately in order to clearly distinguish the flow directions. The supply line represents the upward flow of hot water from the circulation pump to the upper floors, while the return line indicates the downward flow of cooled water back to the boiler. Vertical dimensions are shown schematically, and horizontal floor connections are represented without scale, as commonly applied in heating system design.

Each floor connection is indicated on the riser diagram to demonstrate the distribution of the heating system to individual floors. Pipe sections between consecutive junctions were numbered systematically to enable pipe diameter calculations and pressure loss analysis. The prepared riser diagram served as the basis for determining the critical circuit of the system and for performing subsequent pipe diameter and circulation pump calculations.

Riser Diagram of the Heating System

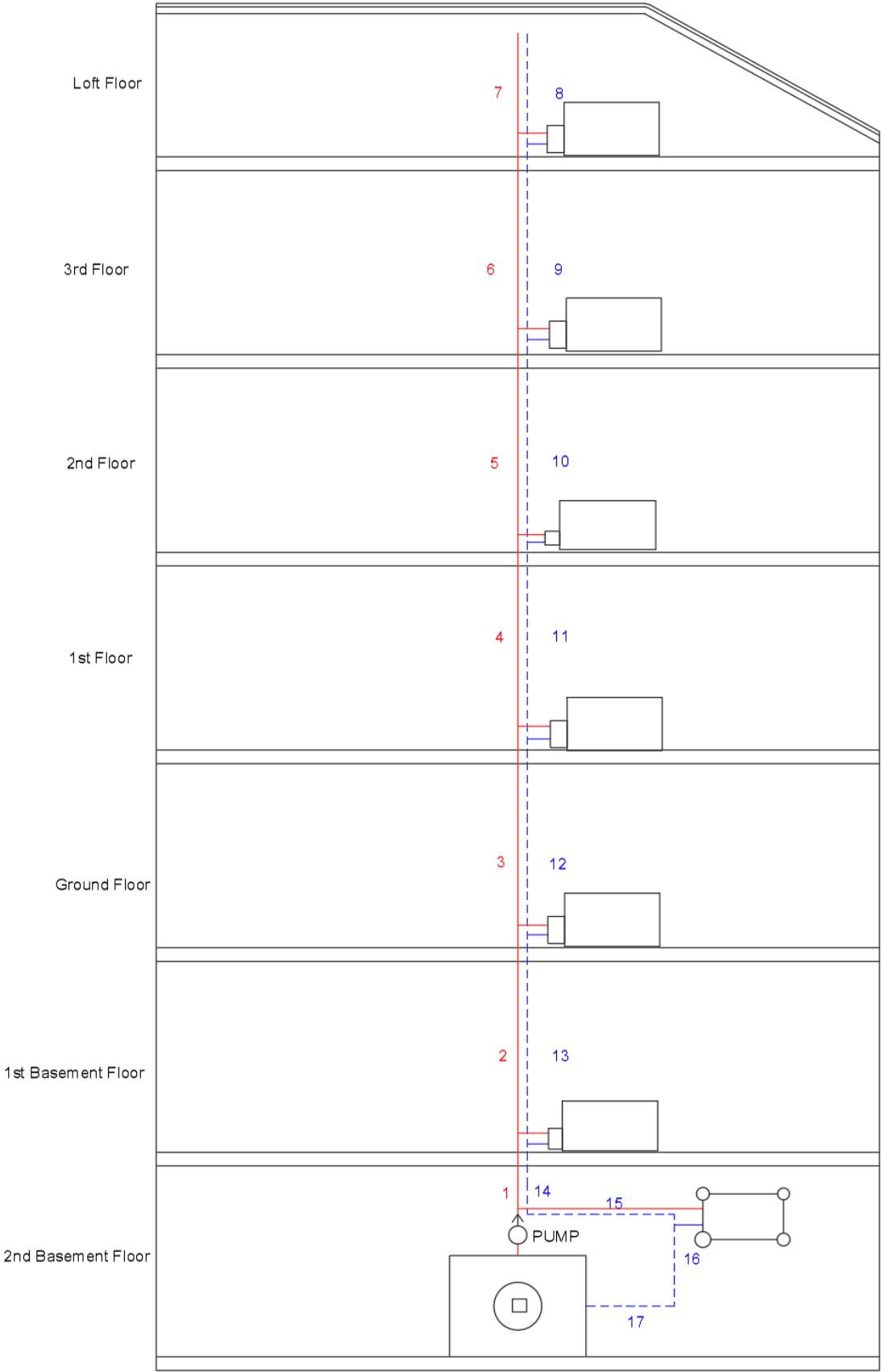


Figure 2: Riser diagram showing the supply and return lines with pipe numbering.

Determination of the Critical Circuit

The critical circuit of the heating system was determined in order to perform pipe diameter calculations and circulation pump selection. The critical circuit is defined as the flow path between the circulation pump and the most remote radiator located on the uppermost floor, which represents the highest pressure loss in the system.

Based on the prepared riser diagram, the critical circuit starts from the circulation pump and continues along the supply line through pipes numbered **1, 2, 3, 4, 5, 6, and 7** up to the uppermost floor. After reaching the most remote radiator, the flow returns to the circulation pump through the return line consisting of pipes numbered **8, 9, 10, 11, 12, 13, 14, 15, 16, and 17**.

Pipe diameter calculations and pressure loss analyses were carried out only for the pipes located on this critical circuit, while the remaining pipe sections were excluded from the calculations.

The selected critical circuit represents the hydraulically most unfavorable path, combining the maximum pipe length, elevation difference, and number of fittings.

LOCAL PRESSURE LOSS CALCULATION (ζ VALUES)

Local pressure losses along the critical circuit were calculated using the loss coefficient (ζ) method.

This method allows the determination of pressure losses caused by fittings, valves, and flow direction changes along the piping system.

The calculations were carried out only for the critical circuit, which extends from the boiler to the most remote radiator located on the loft floor and back to the boiler. In order to avoid double counting, equivalent pipe length values were not used, and all local losses were evaluated directly by their corresponding ζ coefficients.

The local loss table was prepared by dividing the critical circuit into several sections, including the boiler room, vertical riser segments for each floor, and the loft floor containing the critical radiator. Each row of the table represents a specific section of the critical circuit, and the pipe diameter corresponding to that section was used to select the appropriate ζ values.

For intermediate floors, only the vertical riser was considered, and one T-separation (supply) and one T-combination (return) were included for each floor. Floor-level horizontal piping and radiator connections were excluded from these rows.

The loft floor was treated separately, and all local losses from the riser outlet to the critical radiator were included in the calculation.

PIPE DIAMETER CALCULATION CHART															Page	1	
BUILDING'S NAME: Apartment Building															Floor	-	
Room's				Radiator's						Hardware's							
a	b	c	d	e	f	g	h	i	k	l	m	n	o	p	r	s	t
Pipes	Heat Value	Heat Value for ΔT =20°	Pipe Length	Depending on Approx D _{pipe}						Depending on Revised D _{pipe}						Difference	
				d	V	R	LR	Σζ	Z	d	V	R	LR	Σζ	Z	LR	Z
No	W	W	m	mm	m/s	Pa/m	Pa		Pa		m/s	Pa/m	Pa		Pa	Pa	Pa
1	25792,8	25792,8	3	40	0,28	23,537	70,611	1,5	58,8								
2	24248,2	24248,2	3	32	0,3	32,363	97,089	1,5	67,5								
3	21819,2	21819,2	2,9	32	0,28	27,46	79,634	1,5	58,8								
4	18331,5	18331,5	2,9	32	0,24	21,575	62,568	1,5	43,2								
5	13384,3	13384,3	2,9	25	0,3	44,132	127,98	1,5	67,5								
6	8437,1	8437,1	2,9	25	0,2	21,575	62,568	1,5	30								
7	3248,6	3248,6	2,1	15	0,22	49,035	102,97	1,5	36,3								
8	3248,6	3248,6	2,1	15	0,22	49,035	102,97	1	24,2								
9	8437,1	8437,1	2,9	25	0,2	21,575	62,568	1	20								
10	13384,3	13384,3	2,9	25	0,3	44,132	127,98	1	45								
11	18331,5	18331,5	2,9	32	0,24	21,575	62,568	1	28,8								
12	21819,2	21819,2	2,9	32	0,28	27,46	79,634	1	39,2								
13	24248,2	24248,2	3	32	0,3	32,363	97,089	1	45								
14	24248,2	24248,2	0,1	32	0,3	32,363	3,2363	1	45								
15	24248,2	24248,2	2,15	32	0,3	32,363	69,58	1	45								
16	25792,8	25792,8	1,35	40	0,28	23,537	31,775	1	39,2								
17	25792,8	25792,8	1,3	40	0,28	23,537	30,598	1	39,2								

Table 1 shows the calculation of local pressure loss coefficients (ζ) for each section of the critical circuit.

Local Pressure Loss Calculation (ζ Values) for the Critical Circuit

In addition to frictional pressure losses along the pipes, local pressure losses caused by fittings and components were also taken into account. These losses occur due to flow disturbances at elbows, valves, T-connections, and radiator connections, and are represented by dimensionless loss coefficients (ζ).

For the critical circuit, local loss coefficients were determined based on the actual piping layout and the components encountered along the flow path. The system was divided into eight segments, including the boiler section, vertical risers, intermediate floors, and the critical floor where the most hydraulically unfavorable radiator is located.

For each segment, the number and type of fittings (such as elbows, T-separations, T-combinations, column valves, radiator valves, and other relevant components) were identified. Corresponding ζ values were selected from standard reference tables according to the pipe diameters used in each segment. The total local loss coefficient ($\Sigma\zeta$) for each segment was calculated by summing the individual loss coefficients.

The resulting local pressure losses were then computed using the calculated flow velocities and were later combined with frictional losses to determine the total pressure loss of the critical circuit.

Local pressure losses were evaluated by identifying all fittings and components along the critical circuit. The total loss coefficient (ζ) was obtained by summing the individual contributions for each segment.

CALCULATION CHART FOR ζ VALUES																					
BUILDING'S NAME: Apartment Building																		Page	2		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Part No	Pipe Diameter	Boiler or Radiator		Trousers Leg	S Part	Double Band (Wide)	Double Band (Narrow)	T Combination	T Seperation	T Counter Flow	T Passage (Seperation)	T Entrance (Seperation)	Pipe Diameters		Elbow 90°	Gate Valve	Column Valve (Straight)	Column Valve (Inclined)	Radiator Valve (Straight)	Radiator Valve (Corner)	TOTAL
		3.0	0.5										1 1/4"	1 1/2"							
1	1 1/2"	3											1 1/4"	1 1/2"		0.6	7.0				3.6
2	1 1/2"														3			2.5			5.5
3	1 1/4"							1	1.5												2.5
4	1 1/4"							1	1.5												2.5
5	1 1/4"							1	1.5												2.5
6	1"							1	1.5												2.5
7	1"							1	1.5												2.5
8	1 1/2"							5	7.5						11.9					3	27.4
																					49

Table 2 shows the calculation of total local loss coefficients ($\Sigma\zeta$) and the corresponding local pressure losses for each segment of the critical circuit.

After determining the frictional pressure losses along the pipes and the local pressure losses caused by fittings and components, the total pressure loss of the critical circuit was calculated. This total pressure loss represents the minimum pressure rise that must be provided by the circulation pump to ensure sufficient water flow through the most hydraulically unfavorable circuit of the system.

The total pressure loss consists of three main components:

- (1) frictional pressure losses along straight pipe sections,
- (2) local pressure losses due to fittings, valves, and flow disturbances, and
- (3) static pressure difference caused by the elevation between the boiler and the highest radiator.

The frictional losses (LR) and local losses (Z) were obtained from the Pipe Diameter Calculation Chart and the Local Loss Calculation Table, respectively. The static pressure difference was calculated based on the total vertical height between the boiler located on the second basement floor and the critical radiator located on the loft floor.

The static pressure difference was calculated for completeness; however, it is not used in the circulation pump selection since the system operates as a closed-loop heating system.

Pipe No	LR (Pa)	Z (Pa)	ΔH (Pa)	Total ΔP (Pa)
1	70,611	58,8		129,411
2	97,089	67,5		164,589
3	79,634	58,8		138,434
4	62,568	43,2		105,768
5	127,98	67,5		195,48
6	62,568	30		92,568
7	102,97	36,3		139,27
8	102,97	24,2		127,17
9	62,568	20		82,568
10	127,98	45		172,98
11	62,568	28,8		91,368
12	79,634	39,2		118,834
13	97,089	45		142,089
14	3,2363	45		48,2363
15	69,58	45		114,58
16	31,775	39,2		70,975
17	30,598	39,2		69,798
TOTAL	1271,418	732,7	193257	195261,1183

The total pressure loss due to friction and local losses was obtained by summing the values listed in Table 3.

The static pressure difference is calculated based on the vertical elevation between the boiler located on the second basement floor and the critical radiator located on the loft floor.

$$\Delta H = \rho g h$$

$$\Delta H = 1000 \times 9.81 \times 19.7 = 193,257 \text{ Pa}$$

$$\Delta P_{total} = \sum (LR + Z) + \Delta H$$

$$\Delta P_{total} = 195261.1183 \text{ Pa}$$

PUMP SELECTION

Based on the total pressure loss calculated for the critical circuit, the required pump head was determined.

From the previous calculations, the total pressure loss of the critical circuit was found to be approximately 2 kPa. Therefore, the circulation pump must be capable of overcoming at least this pressure difference to ensure adequate water circulation throughout the heating system.

In practical applications, a safety margin is usually added to account for uncertainties and operating conditions. Accordingly, a pump with a nominal head slightly higher than the calculated value is recommended.

The selected pump should provide the required head at the design flow rate corresponding to the total heating load of the building.

Although the total pressure loss including static head is calculated as 195 kPa, in a closed-loop heating system the static pressure is balanced. Therefore, the circulation pump is required to overcome mainly the frictional and local losses, which are approximately 2 kPa.

Pump Flow Rate and Required Head

The circulation pump flow rate was determined based on a temperature difference of 20 °C between the supply and return lines, in accordance with the design assumptions of the heating system. The total heating load of the building was used to obtain the required volumetric flow rate to ensure proper heat transfer throughout the system.

The required pump head was evaluated considering the pressure losses along the critical circuit. These losses include frictional pressure losses in the pipes as well as local losses caused by fittings, valves, and other hydraulic components. Although the static pressure difference between the boiler and the highest radiator was calculated, it does not contribute to the pump head requirement since the system operates as a closed-loop heating system.

Therefore, the circulation pump was selected to overcome the total frictional and local pressure losses of the critical circuit, ensuring reliable and efficient operation of the heating system.

Since the heating system operates as a closed loop, the static pressure does not affect the pump selection, and the circulation pump is sized based on frictional and local losses only.

$$\dot{V}_p = \frac{3.6 Q_k}{\rho c_p (T_g - T_d)}$$

$$\Delta T = 20^\circ C \quad Q = 25792.8 W \quad \rho = 1000 kg/m^3$$

$$\dot{V}_p = \frac{92854.08}{83740} = 1.109 m^3/h$$

$$H_p = \frac{\sum(LR) + \sum(Z)}{11.1} + \text{Boiler loss} \quad \{mmSS\}$$

$$H_p = \frac{2004.118}{1.11} = 180.55 mmSS$$

$$H_p = 181 mmSS \text{ (excluding boiler loss)}$$

With the determination of the pump head and the critical circuit analysis, the hydraulic design of the heating system has been completed.



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REPORT 6

Yusuf Yıldızlar

2103040020

DESIGN OF THERMAL SYSTEM

ADVISOR: Dr Öğr. Üyesi MUAMMER KANLI

İSTANBUL, 2025

Introduction

In the previous stages of this project, the architectural properties of the building, heat transfer coefficients of the building envelope, detailed heat loss calculations, and the hydraulic design of the heating system were completed. Based on these analyses, the total heating demand of the building and the required boiler capacity were determined.

The aim of this report is to complete the thermal system design by performing the chimney (shaft) sizing and expansion tank sizing calculations for the selected boiler. Proper chimney design is essential to ensure safe and efficient removal of combustion gases, while correct expansion tank sizing is required to compensate for the volumetric expansion of water in the closed-loop heating system and to maintain stable operating pressure.

In this report, the boiler capacity obtained from previous calculations is first converted into appropriate units and used for chimney sizing according to standard design charts. Subsequently, the total water volume of the heating system is estimated, and the expansion tank volume is calculated based on thermal expansion, static pressure, and maximum operating pressure of the system. The results of this report finalize the mechanical design of the heating system and ensure safe, reliable, and efficient system operation.

Chimney (Shaft) Sizing

The chimney sizing calculation is performed to ensure safe and efficient discharge of combustion gases produced by the boiler. The chimney cross-section is determined based on the boiler capacity and the effective chimney height using standard chimney sizing charts. Proper chimney design is essential for maintaining sufficient draft and preventing backflow of flue gases.

The boiler capacity obtained from previous calculations is **26 kW**. For chimney sizing calculations, this value is converted into kilocalories per hour as follows: $26 \text{ kW} \times 860 = \mathbf{22,360 \text{ kcal/h}}$.

The effective chimney height is taken as **20 m** based on the vertical distance between the boiler room and the chimney outlet.

For an effective chimney height of **20 m** and a boiler capacity of **22,360 kcal/h**, the chimney diameter is selected as 26 cm (equivalent rectangular section **20×27 cm**) according to the chimney sizing chart.

Expansion Tank Sizing

The expansion tank is designed to compensate for the volumetric expansion of water in the closed-loop heating system caused by temperature increase during operation. Proper sizing of the expansion tank is necessary to prevent excessive pressure rise in the system and to ensure safe and stable operating conditions. In this study, the expansion tank volume is calculated based on the total water volume of the system, the thermal expansion coefficient of water, and the static and maximum operating pressures.

The heating system is designed as a closed-loop hot water system; therefore, a **closed (diaphragm type) expansion tank** is selected.

$$\mathbf{V = 250 \text{ L}}$$

Based on the boiler capacity and panel radiator heating system, the total water volume of the system is assumed as 250 liters.

$$\mathbf{e = 0.0354}$$

For the operating temperature range, the volumetric expansion coefficient of water is taken as $e = 0.0354$.

$$P_{st} = \frac{19.9}{10} + 1 = 2.99 \text{ bar}$$

$$P_{max} = 2.99 + 1 = 3.99 \text{ bar}$$

The static height between the boiler and the highest radiator is $H_{st} = 19.9$ m. Therefore, the static pressure is calculated as $P_{st} = H_{st}/10 + 1 = 2.99$ bar, and the maximum operating pressure is taken as $P_{max} = P_{st} + 1 = 3.99$ bar.

$$V_{gen,k} = \frac{V \times e}{1 - \frac{P_{st}}{P_{max}} \times 0.8} = \frac{250 \times 0.0354}{1 - \frac{2.99}{3.99} \times 0.8} = 28.25 \text{ L}$$

$$V_{select} = 1.4 \times 28.25 = 39.55 \text{ L} \approx 40 \text{ L}$$

By adding a 40% safety margin, the selected expansion tank volume becomes $V_{select} = 1.4 \times 28.25 = 39.55$ L, therefore a 40 L closed (diaphragm) expansion tank is selected.

Conclusion

In this report, the chimney and expansion tank sizing calculations of the heating system were completed based on the boiler capacity determined in previous studies. For a boiler capacity of 26 kW and an effective chimney height of 20 m, the chimney diameter was selected as 26 cm according to standard chimney sizing charts. Additionally, the expansion tank volume was calculated considering the total system water volume, thermal expansion of water, and system operating pressures. As a result, a 40 L closed diaphragm expansion tank was selected to ensure safe and stable operation of the heating system.

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MECHANICAL ENGINEERING**

REPORT 7

Yusuf Yıldızlar

2103040020

DESIGN OF THERMAL SYSTEM

ADVISOR: Dr Öğr. Üyesi MUAMMER KANLI

İSTANBUL, 2026

Introduction

Heating systems constitute one of the most important mechanical engineering applications in buildings, as they directly affect thermal comfort, energy consumption, and operating costs. A properly designed heating system must be capable of maintaining the desired indoor thermal conditions under design outdoor conditions while operating safely, efficiently, and economically. In this context, accurate determination of building heat losses and appropriate selection of heat emitters and hydraulic components are essential steps in the design process.

In this study, the heating system design of a multi-zone building is carried out based on steady-state heat loss calculations. The heat losses of individual spaces are determined considering transmission losses through building envelope components and ventilation effects in accordance with relevant standards. The obtained heat loss values are used as a basis for sizing radiators and towel radiators, which serve as the terminal heating units in the system.

Following the initial manual calculations, the heating system is modeled and analyzed using the MTH (Mechanical Thesis Heating) software. The software is employed to verify heat load calculations, perform radiator and towel radiator selections, determine pipe diameters, calculate pressure losses, and evaluate pump requirements. During this process, it is observed that some of the initially selected heat emitters are insufficient to fully meet the calculated heat demands. Therefore, these units are revised by selecting higher-capacity standard models to ensure adequate heating performance and to provide a reasonable safety margin.

The final system design consists of a hot-water-based heating system operating at a 90/70 °C temperature regime, supplied by a natural gas-fired boiler. The hydraulic design, including pipe sizing and pump selection, is carried out to ensure acceptable flow velocities and pressure losses. In addition, the chimney design is evaluated in accordance with DIN 4705 to verify safe flue gas evacuation.

The main objective of this study is to demonstrate a systematic approach to heating system design by integrating theoretical calculations with computer-aided analysis, and to present a technically sound and practically applicable solution for building heating applications.

Methodology

MTH-Based Heating System Modelling:

In order to verify the results obtained from manual calculations and to perform detailed system sizing, the heating system is modeled using the MTH software. The software provides an integrated environment for heat load verification, radiator and towel radiator selection, hydraulic analysis, pump sizing, and chimney evaluation.

The previously calculated space heat losses are defined in the software on a room-by-room basis. Indoor design temperatures assigned during the manual calculation stage are preserved, and the heating system is modeled as a hot-water-based system operating under a 90/70 °C supply/return temperature regime. All calculations are carried out assuming water as the heat transfer fluid.

Radiator and Towel Radiator Selection in MTH:

Radiator and towel radiator selections are performed within the MTH environment using manufacturer-independent catalog data. The software is used to check whether the initially selected heat emitters are capable of meeting the calculated heat losses of the spaces.

During this process, it is observed that some radiators and towel radiators selected through manual calculations are marginally insufficient. In such cases, higher-capacity standard models with equivalent performance characteristics are selected to ensure sufficient heating performance and to introduce an appropriate safety margin. The final selections presented in this study therefore represent the revised radiator and towel radiator capacities determined through MTH analysis.

Hydraulic Design and Pipe Sizing:

The hydraulic design of the heating system is carried out using the pipe sizing module of the MTH software. Pipe diameters are determined based on allowable flow velocity limits and pressure loss criteria. Minimum and maximum flow velocities are defined to ensure reliable operation, prevent excessive noise, and avoid unnecessary pressure losses.

The system is analyzed by defining pipe segments along the main supply and return lines, including branch connections. Tee junctions are modeled to represent flow separation and combination points within the system. Pipe diameters, flow rates, and pressure losses are automatically calculated by the software for each segment.

Pump Sizing:

Pump sizing is performed based on the total system flow rate and calculated pressure losses obtained from the hydraulic analysis. The total pressure loss of the critical circuit is used to determine the required pump head. The hydraulic power requirement is calculated theoretically, and the pump selection is evaluated accordingly.

Due to the relatively low pressure losses of the system, the calculated hydraulic power remains small. In practical applications, however, pumps are selected from standard commercially available models to ensure reliable operation and system stability.

Chimney Design:

The chimney design is evaluated using the DIN 4705 chimney calculation module within the MTH software. The chimney height, flue material, boiler capacity, and fuel type are defined according to the project specifications. The calculation is performed to verify that flue gas evacuation conditions are satisfied and that the selected chimney configuration operates safely under design conditions.

Presentation of MTH Outputs:



Figure 2.1. U-value (K-value) calculation module icon in MTH software indicating access to building envelope thermal analysis.

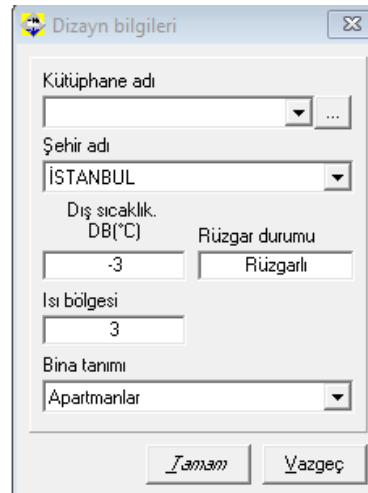
A screenshot of the "Dizayn bilgileri" (Design Information) form in the MTH software. It contains fields for "Kütüphane adı" (Library name), "Şehir adı" (City name) set to "İSTANBUL", "Dış sıcaklık DB(°C)" (Outdoor temperature) set to "-3", "Rüzgar durumu" (Wind condition) set to "Rüzgarlı", "Isı bölgesi" (Climate zone) set to "3", and "Bina tanımı" (Building type) set to "Apartmanlar" (Apartments). There are "İlanam" and "Vazgeç" buttons at the bottom.

Figure 2.2. Selection of project location and design climate data in the U-value (K-value) module of MTH software.

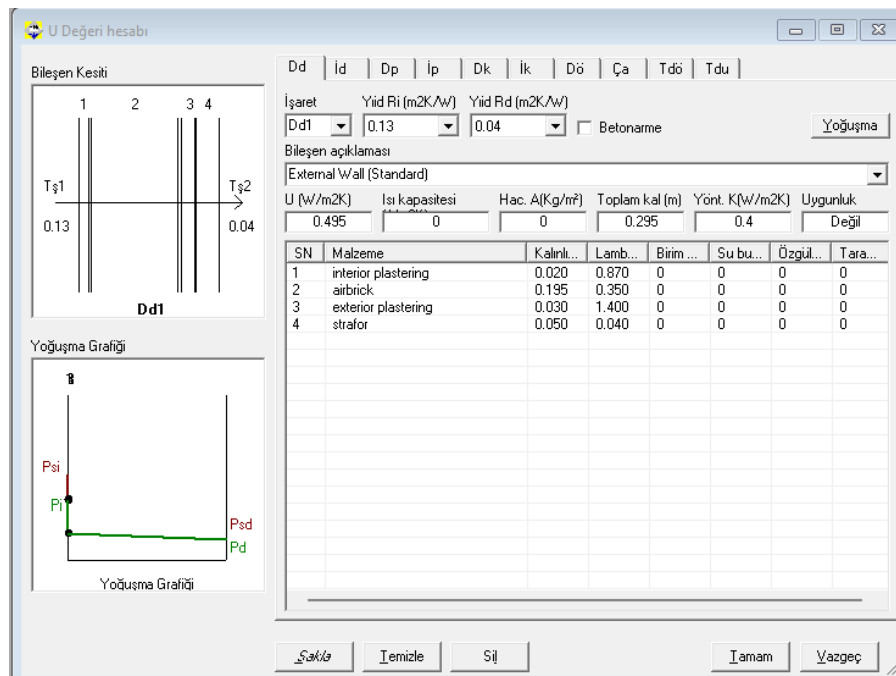
A screenshot of the "U Değeri hesabı" (U-value calculation) form. It shows a cross-section diagram of a wall with four layers, labeled 1 to 4, with temperatures T_{s1} and T_{s2} indicated. Below the diagram is a "Yoğuşma Grafiği" (Condensation graph) showing the temperature profile and condensation points. The main table lists material layers with their properties: SN, Malzeme, Kalınlık, Lamb, Birim, Su bu, Özgül, and Tara. The table includes data for interior plastering, airbrick, exterior plastering, and strafor (foam). The bottom of the form has "Sil", "İlanam", and "Vazgeç" buttons.

Figure 2.3. Calculation of U-value (K-value) for the external wall in the MTH software, including material layers, thicknesses, thermal conductivities, and condensation control.

U Değeri hesabı

Bileşen Kesiti

Yoğuşma Grafiği

İşaret: Yüid Ri (m²K/W) Yüid Rd (m²K/W)

İd2: 0.13 0.13

Bileşen açıklaması: Internal Wall 2

U (W/m²K) Isı kapasitesi Hac. A(Kg/m³) Toplam kal (m)

1.437 0 0 0.235

SN	Malzeme	Kalınlı...	Lamb...	Birim ...	Su bu...	Özgül...	Tara...
1	interior plastering	0.020	0.870	0	0	0	0
2	airbrick	0.195	0.500	0	0	0	0
3	interior plastering	0.020	0.870	0	0	0	0

Sakla Temizle Sil Tamam Vazgeç

Figure 2.4. U-value (K-value) calculation interface for the internal wall in MTH software based on the defined material layers.

U Değeri hesabı

Bileşen Kesiti

İşaret: Malzeme

İp 100x2 3. Ahşap Doğrama (90s)

Tipi: 3.1. Tek Camlı pencere U (Watt/m²K) 2.6

Kanat tipi: Ahşap veya sentetik çerçeveli tek pencere

Cam tipi: Normal cam (Adi cam)

Tesisat tipi: Genel ısıtma a (m³/h.m) 3

Gölge tipi: Pencere açısı (°) 0

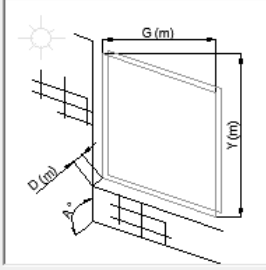
Uz. (m) 2 Yük. (m) 1 Fuga uz. (m) 9.01 Pencere derinliği (m) 0.1

Sakla Temizle Sil Tamam Vazgeç

Figure 2.5. U-value (K-value) calculation interface for the internal window in MTH software.

U Değeri hesabı

Bileşen Kesiti



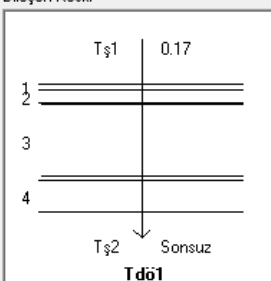
Dd | İd | Dp | İp | Dk | İk | Dö | Ça | Tdö | Tdu
 Kütüphane adı | İşaret | Malzeme
 | 150x230 | 3. Ahşap Doğrama (90s)
 Tipi | U (Watt/m²K)
 3.1. Tek Camlı pencere | 4.4
 Kanat tipi
 Ahşap veya sentetik çerçeveli tek pencere
 Cam tipi
 Normal cam (Adi cam)
 Tesisat tipi | a (m³/h.m)
 Genel ısıtma | 3
 Gölge tipi | Pencere açısı (°)
 | 90
 Uz. (m) | Yük. (m) | Fuga uz. (m) | Pencere derinliği (m)
 1.5 | 2.3 | 10.36 | 0.1

Sakla | İmzile | Sil | Tamam | Vazgeç

Figure 2.6. Calculation of the U-value (K-value) for the external door based on defined door material properties in MTH software.

U Değeri hesabı

Bileşen Kesiti



Dd | İd | Dp | İp | Dk | İk | Dö | Ça | Tdö | Tdu
 İşaret | Yiid Ri (m²K/W) | Yiid Rd (m²K/W)
 Tdö1 | 0.17 | Sonsuz
 Bileşen açıklaması
 U (W/m²K) | Isı kapasitesi | Hac. A(Kg/m³) | Toplam kal (m) | Yönt. K(W/m²K) | Uygunluk
 0.112 | 0 | 0 | 0.2 | 0.35 | Değil

SN	Malzeme	Kalınlı...	Lamb...	Birim ...	Su bu...	Özgütl...	Tara...
1	ceramic tile	0.010	1.300	0	0	0	0
2	cement mortar	0.020	1.400	0	0	0	0
3	concrete	0.120	2.400	0	0	0	0
4	xps	0.050	0.035	0	0	0	0

Sakla | İmzile | Sil | Tamam | Vazgeç

Figure 2.7. Calculation of the U-value (K-value) for the floor in contact with ground based on defined material layers in MTH software.

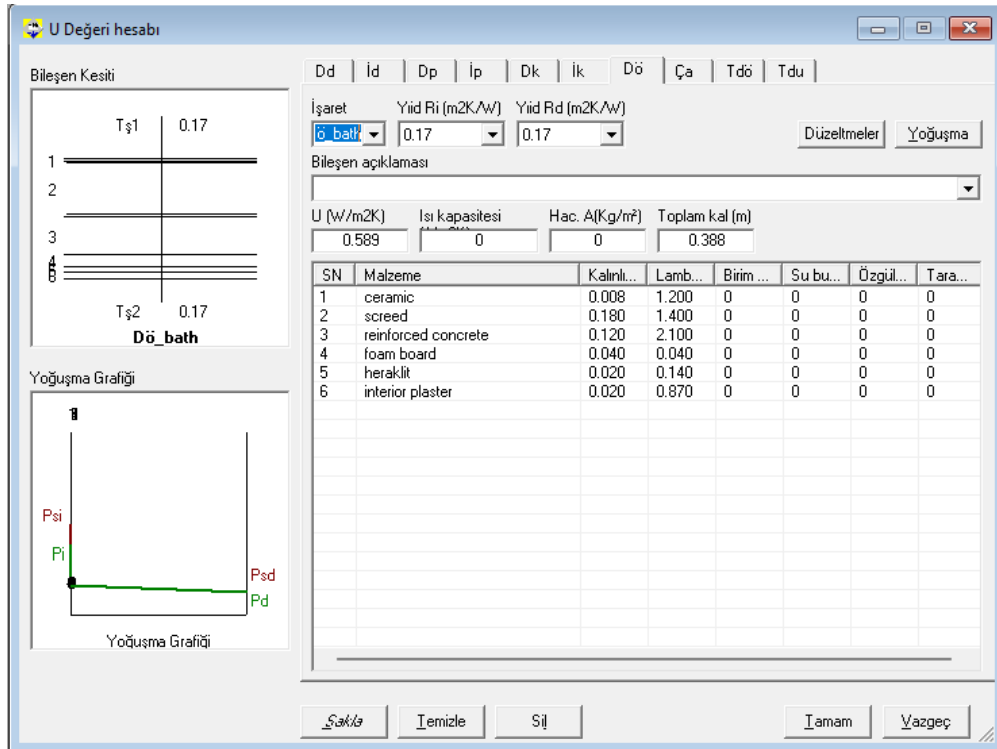


Figure 2.8. Calculation of the U-value (K-value) for the bathroom floor based on defined floor material layers in MTH software.

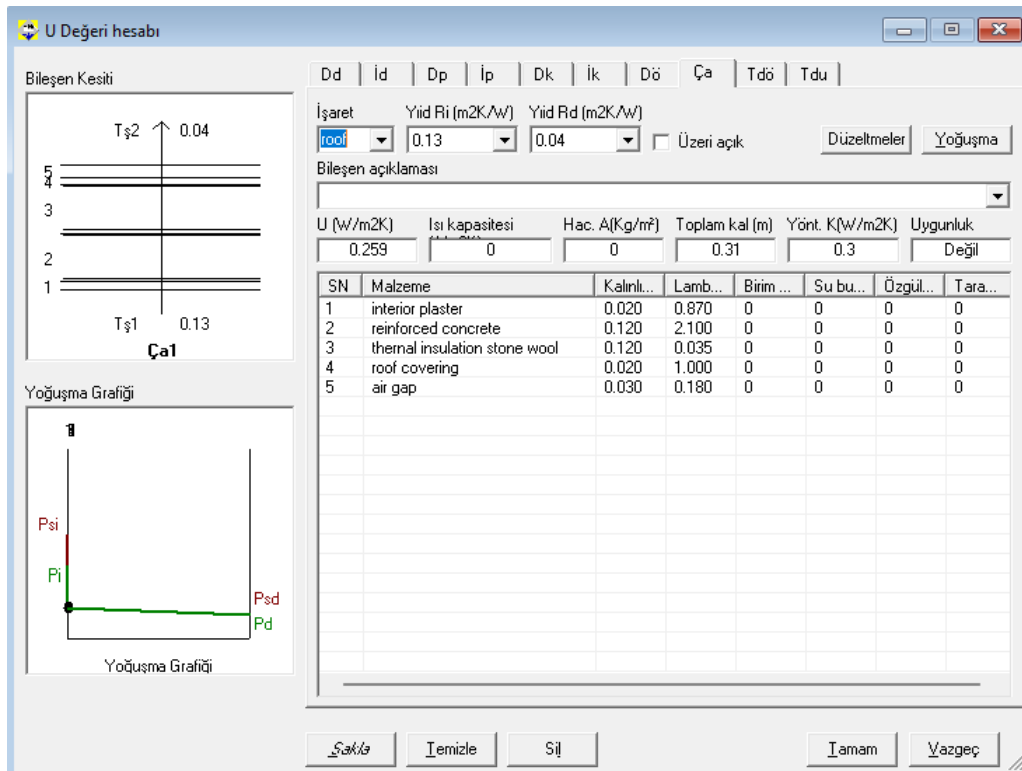


Figure 2.9. Calculation of the U-value (K-value) for the roof based on defined roof material layers in MTH software.

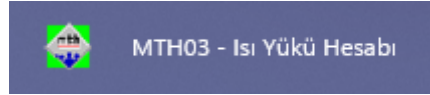


Figure 2.10. Heating load calculation module icon in MTH software indicating the transition to heat load analysis.

Figure 2.11. User interface indicating the transition to the heating load calculation module after completing U-value (K-value) definitions in MTH software.

İşareti	K (K...)	K (W...)
Dd1	0.495	0.495
Dd2	0.371	0.371
İd1	2.101	2.101
İd2	1.437	1.437
W1	2.600	2.600
SW1	5.000	5.000
Tdö1	0.112	0.112
Dö_b...	0.589	0.589
roof	0.259	0.259
Dö_w...	0.108	0.108
ik. an...	0.000	0.000

Figure 2.12. Screen showing the transfer of previously defined U-values (K-values) from the building envelope module to the heating load calculation module in MTH software.

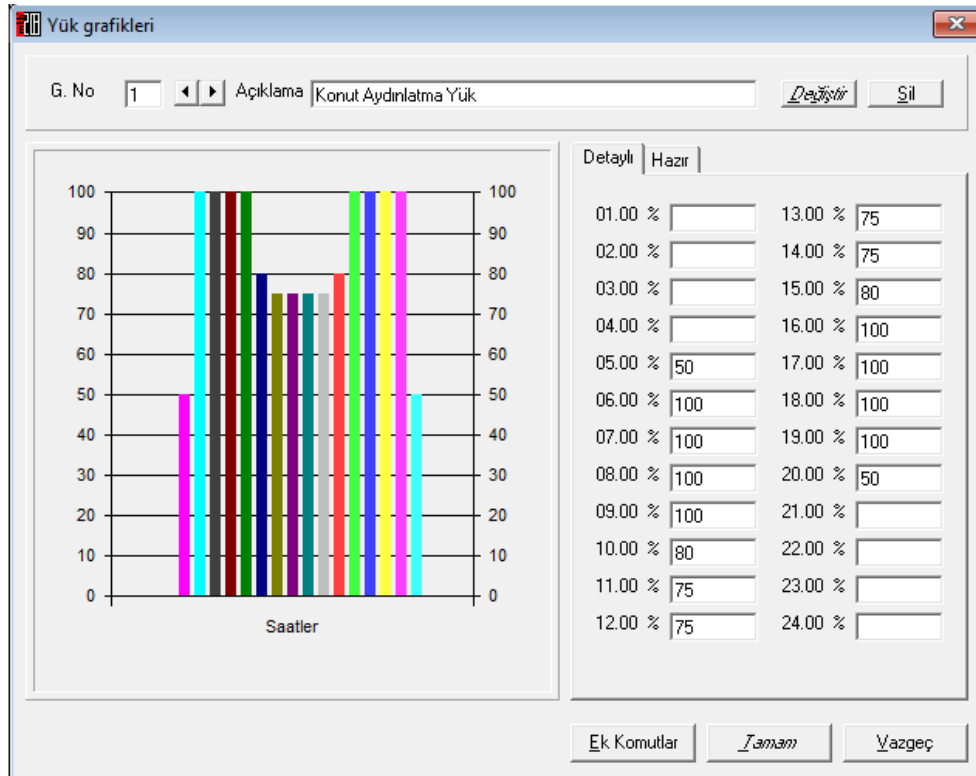


Figure 2.13. Assignment of lighting heat gains using graphical representation in the heating load calculation module of MTH software.

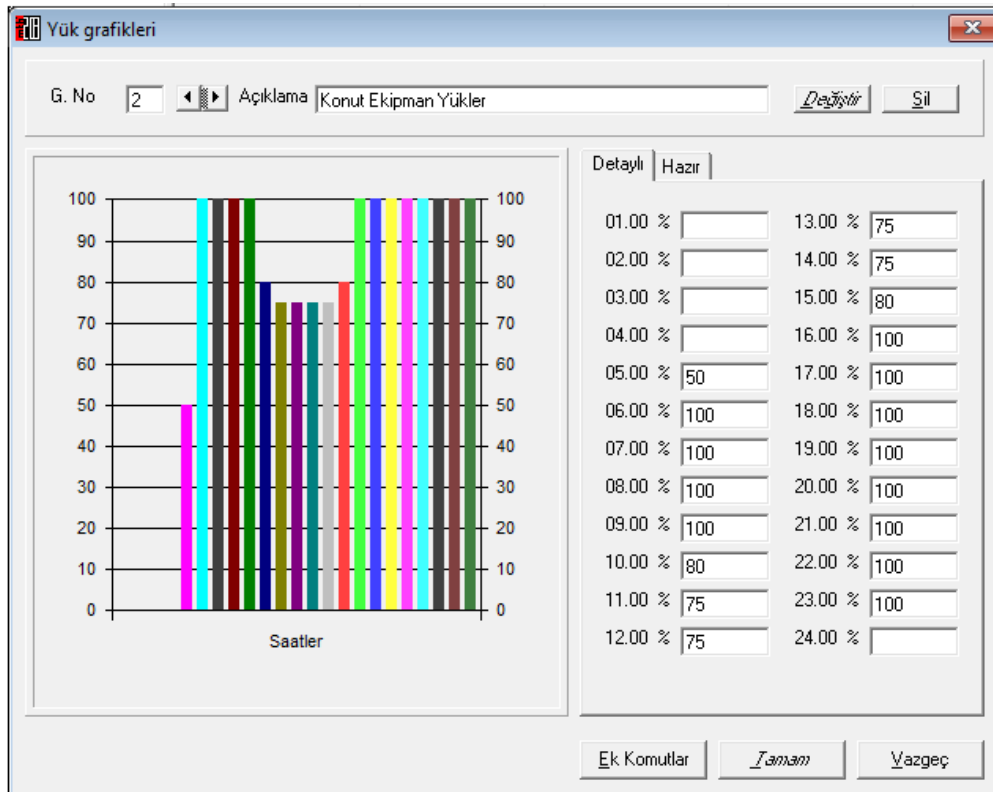


Figure 2.14. Assignment of equipment heat gains using graphical representation in the heating load calculation module of MTH software.



Figure 2.15. Assignment of occupant (human) heat gains using graphical representation in the heating load calculation module of MTH software.

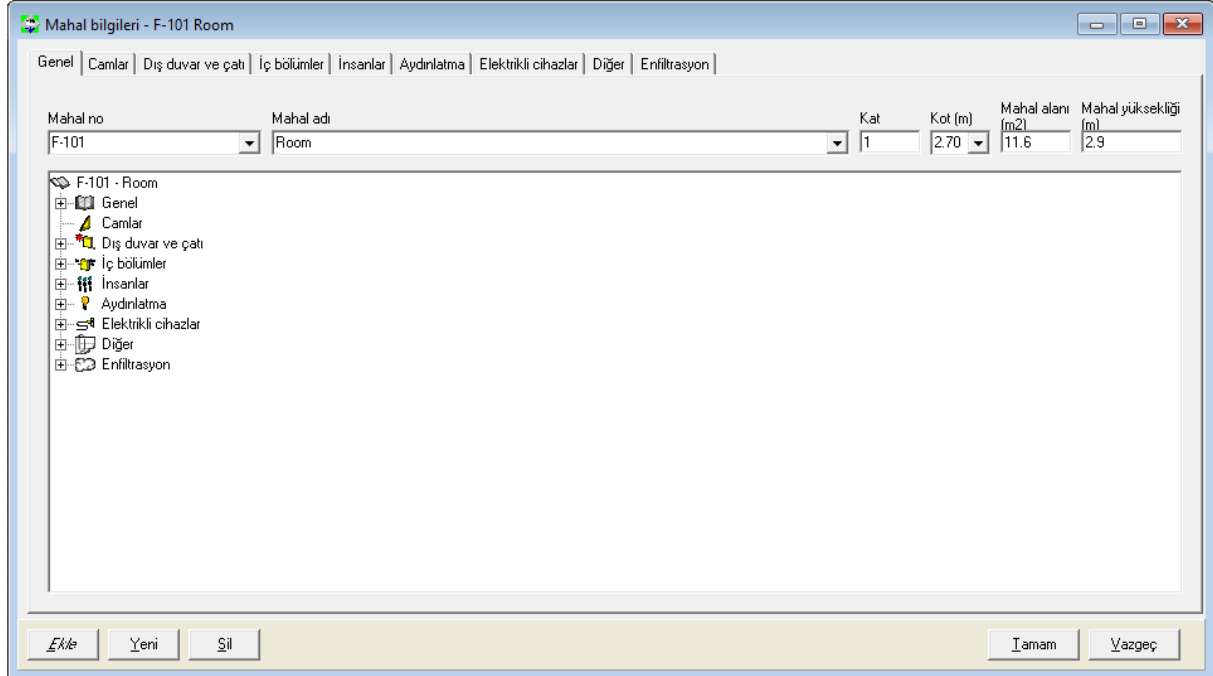


Figure 2.16. Overview of general room information for room F101, including design parameters, in the heating load calculation module of MTH software.

Mahal bilgileri - F-101 Room

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Ağırlık Renk

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Gölge

Hayır

İşaret	Yön	Gen. (m)	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (Wa...	Ağırlık	Renk	Gölge
Dd2	B	0.3	6.15	2.9	1	0	17.84	0.371	Orta	Orta	Hayır
Dd1	K	0.3	3.51	2.9	1	4.2	5.98	0.495	Orta	Orta	Hayır

Vazgeç

Figure 2.17. Assignment of external wall and roof components to room F101 in the heating load calculation module of MTH software.

Mahal bilgileri - F-101 Room

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Yaz dt(°C) Kış dt(°C)

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Ağırlık

Orta

İşaret	Yön	Orta	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (Wa...	Yaz dt...	Kış dt(...	Ağırlık
Dö_wood		0.58	0	0	1	0	11.6	0.108	2	2	Orta
Dö_wood		0.58	0	0	1	0	11.6	0.108	0	0	Orta
İp 200x...	K	0	2	2.1	1	0	4.2	2.6	18	18	Orta
İk 90x210	D	0	0.9	2.1	1	0	1.89	2.3	4	4	Orta
İd1	D	0.13	3.94	2.9	1	1.89	9.54	2.101	4	4	Orta
İK 80x2...	G	0	0.8	2.1	1	0	1.68	2.3	-4	-4	Orta
İd1	G	0.13	3.34	2.9	1	1.68	8.01	2.101	-4	-4	Orta

Vazgeç

Figure 2.18. Assignment of internal wall and partition elements to room F101 in the heating load calculation module of MTH software.

Mahal bilgileri - F-101 Room

Genel | Camlar | Dış duvar ve çatı | İç bölümler | **İnsanlar** | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

Aktivite: DI (Watt/per): GI (Watt/per): Toplam:

m2/İnsan: Toplam insan:

Toplam DI (Watt): Toplam GI (Watt): Toplam (Watt): Grafik adı:

Figure 2.19. Assignment of internal heat gains based on two average occupants in seated and resting condition for room F101 in the heating load calculation module of MTH software.

Mahal bilgileri - F-101 Room

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | **Aydınlatma** | Elektrikli cihazlar | Diğer | Enfiltrasyon

Watt/m2: Toplam (Watt):

☒ Akkor ☐ Flüresan Ballast t.:

Grafik adı:

Figure 2.20. Definition of internal lighting heat gain for room F101 based on one lighting fixture rated at 12 W in the heating load calculation module of MTH software.

Mahal bilgileri - C-002 Living Area

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Ağırlık Renk

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Gölge

İşaret	Yön	Gen. (m)	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (Wa...	Ağırlık	Renk	Gölge
roof		0.31	0	0	1	0	24.26	0.253	Orta	Orta	Hayır
Dd1	K	0.3	4.3	2.01	1	1.58	7.06	0.495	Orta	Orta	Hayır
Dd1	D	0.3	5.2	2.01	1	1.58	8.87	0.495	Orta	Orta	Hayır

Vazgeç

Figure 2.23. Assignment of external wall and roof components to room C-002 in the heating load calculation module of MTH software.

Mahal bilgileri - C-002 Living Area

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Yaz dt(°C) Kış dt(°C)

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Ağırlık

İşaret	Yön	Orta	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (Wa...	Yaz dt...	Kış dt(...	Ağırlık
Do_wood		0.58	0	0	1	0	6.5	0.108	4	4	Orta
Do_wood		0.58	0	0	1	0	6.5	0.108	4	4	Orta
Do_wood		0.58	0	0	1	0	10	0.108	0	0	Orta
Id1	B	0.13	3.34	2.01	1	0	6.71	2.101	0	0	Orta
IK 80x2...	K	0	0.8	1.98	1	0	1.58	2.3	13	13	Orta
IK 80x2...	D	0	0.8	1.98	1	0	1.58	2.3	13	13	Orta
IK 80x2...	G	0	0.8	1.98	1	0	1.58	2.3	-4	-4	Orta
Id1	G	0.13	2.7	2.01	1	1.58	3.85	2.101	-4	-4	Orta
IK 80x2...	G	0	0.9	1.98	1	0	1.78	2.3	0	0	Orta
Id1	G	0.13	4.91	2.01	1	1.78	8.09	2.101	0	0	Orta

Vazgeç

Figure 2.24. Assignment of internal wall and partition elements to room C-002 in the heating load calculation module of MTH software.

Mahal bilgileri - C-002 Living Area

Genel | Camlar | Dış duvar ve çatı | İç bölümler | **İnsanlar** | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

Aktivite: Vasat bir şahıs istirahatte oturmuş

DI (Watt/per): 64 GI (Watt/per): 52 Toplam: 116

m2/İnsan: 23 Toplam insan: 1

Toplam DI (Watt): 64 Toplam GI (Watt): 52 Toplam (Watt): 116

Grafik adı: Konut İnsan Yükleri

İzmirle Vazgeç

Figure 2.25. Definition of occupant heat gain for room C-002 based on one person in the heating load calculation module of MTH software.

Mahal bilgileri - C-002 Living Area

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | **Aydınlatma** | Elektrikli cihazlar | Diğer | Enfiltrasyon

Watt/m2: 1 Toplam (Watt): 24

☒ Akkor ☐ Flüresan Ballast f.: 1.25

Grafik adı: Konut Aydınlatma Yük

İzmirle Vazgeç

Figure 2.26. Definition of internal lighting heat gain for room C-002 based on a lighting load of 24 W in the heating load calculation module of MTH software.

Mahal bilgileri - G-001 Shop 1

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Ağırlık Renk

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Gölge

Hayır

İşaret	Yön	Gen. (m)	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (Wa...)	Ağırlık	Renk	Gölge
Dd1	K	0.3	3.46	2.7	1	7.28	2.06	0.495	Orta	Orta	Hayır
Dd1	B	0.3	4.49	2.7	1	8.06	4.06	0.495	Orta	Orta	Hayır
Dd1	G	0.3	4.64	2.7	1	9.88	2.65	0.495	Orta	Orta	Hayır
Dd1	D	0.3	0.9	2.7	1	1.35	1.08	0.495	Orta	Orta	Hayır

Vazgeç

Figure 2.29. Assignment of external wall and roof components to room G001 in the heating load calculation module of MTH software.

Mahal bilgileri - G-001 Shop 1

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Yaz dt(°C) Kış dt(°C)

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Ağırlık

Orta

İşaret	Yön	Orta	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (Wa...)	Yaz dt...	Kış dt(...)	Ağırlık
Dö_wood		0.58	0	0	1	0	47	0.108	0	0	Orta
Dö_wood		0.58	0	0	1	0	32.9	0.108	-2	-2	Orta
Dö_wood		0.58	0	0	1	0	13.5	0.108	2	2	Orta
İp 280x...	K	0	2.8	2.6	1	0	7.28	2.6	16	16	Orta
İp 310x...	K	0	3.1	2.6	1	0	8.06	2.6	16	16	Orta
İp 170x...	G	0	1.7	2.6	1	0	4.42	2.6	16	16	Orta
İp 210x...	G	0	2.1	2.6	1	0	5.46	2.6	16	16	Orta
İK 80x2...	D	0	0.8	2.1	1	0	1.68	2.3	-6	-6	Orta
İd1	D	0.13	1.2	2.7	1	1.68	1.56	2.101	-6	-6	Orta
İp 90x150	D	0	0.9	1.5	1	0	1.35	2.6	16	16	Orta
İd2	D	0.24	8.19	2.7	1	0	22.11	1.437	10	10	Orta

Vazgeç

Figure 2.30. Definition of internal partitions for room G001 in the heating load calculation module of MTH software, where the floor is modeled as two separate internal floor elements due to adjacency with different neighboring spaces.

Mahal bilgileri - G-001 Shop 1

Genel | Camlar | Dış duvar ve çatı | İç bölümler | **İnsanlar** | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

Aktivite: Orta derecede aktif mağaza çalışanı
 DI (Watt/per): 66
 GI (Watt/per): 110
 Toplam: 176

m2/İnsan: 11.75
 Toplam insan: 4

Toplam DI (Watt): 264
 Toplam GI (Watt): 440
 Toplam (Watt): 704
 Grafik adı: Konut İnsan Yükleri

İmizle Vazgeç

Figure 2.31. Assignment of internal heat gains for room G001 by defining four moderately active retail employees in the heating load calculation module of MTH software.

Mahal bilgileri - G-001 Shop 1

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | **Aydınlatma** | Elektrikli cihazlar | Diğer | Enfiltrasyon

Watt/m2: 1
 Toplam (Watt): 60

☒ Akkor
☐ Flüresan
 Ballast f.: 1.25

Grafik adı: Konut Aydınlatma Yük

İmizle Vazgeç

Figure 2.32. Assignment of lighting heat gain by defining a 60 W lighting load for room G001 in the heating load calculation module of MTH software.

Mahal bilgileri - B-101 Shop 2

Genel Camlar Dış duvar ve çatı İç bölümler İnsanlar Aydınlatma Elektrikli cihazlar Diğer Enfiltrasyon

İşareti Yönü K (Watt/m2K) Ağırlığı Der. L(m)

Uz. (m) Yük. (m) Adet Alan (m2) Saçak H(m) Saçak L(m) Değiştir Yukarı Aşağı

Cam tipi Gölge tipi Gölge f.

İşareti	Yönü	Uz. (m)	Yük. (m)	Adet	Alan (...)	K (Wa...	Ağırlığı	Der. L(...)	Saçak...	Saçak...	Cam tipi	Gölge ...	Gölge f.
Dk 150...	B	1.5	2.3	1	3.45	4.4	Orta	0.1	0	0	NOR...	Gölgeli...	1

Vazgeç

Figure 2.35. Assignment of an external door through the glazing (windows) interface for floor B-101 in the heating load calculation module of MTH software.

Mahal bilgileri - B-101 Shop 2

Genel Camlar Dış duvar ve çatı İç bölümler İnsanlar Aydınlatma Elektrikli cihazlar Diğer Enfiltrasyon

İşaret Yön K (Watt/m2K) Ağırlık Renk

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Gölge Değiştir Yukarı Aşağı

İşaret	Yön	Gen. (m)	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan (...)	K (Wa...	Ağırlık	Renk	Gölge
Dd2	K	0.3	4.66	2.7	1	0	12.58	0.371	Orta	Orta	Hayır
Dd1	B	0.3	6.7	2.7	1	9.48	8.61	0.495	Orta	Orta	Hayır
Dd1	B	0.3	4.91	2.7	1	3	10.26	0.495	Orta	Orta	Hayır
Dd1	D	0.3	2.96	2.7	1	2.25	5.74	0.495	Orta	Orta	Hayır

Vazgeç

Figure 2.36. Definition of external wall and roof elements for floor B-101 in the heating load calculation module of MTH software.

Mahal bilgileri - B-101 Shop 2

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Yaz dt(°C) Kış dt(°C)

Ekle Yeni Sil

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Ağırlık Değiştir Yükan Aşağı

İşaret	Yön	Orta	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (Wa...	Yaz dt...	Kış dt...	Ağırlık
Do_wood		0.58	0	0	1	0	56.5	0.108	5	5	Orta
Do_wood		0.58	0	0	1	0	56.5	0.108	0	0	Orta
İp 100x...	B	0	2	1	1	0	2	2.6	16	16	Orta
İp 310x...	B	0	3.1	2.6	1	0	8.06	2.6	16	16	Orta
İp 170x90	G	0	1.7	0.9	1	0	1.53	2.6	16	16	Orta
İp 210x70	G	0	2.1	0.7	1	0	1.47	2.6	16	16	Orta
İp 150x...	D	0	1.5	1.5	1	0	2.25	2.6	16	16	Orta
İd2	D	0.24	8.43	2.7	1	0	22.76	1.437	10	10	Orta

Vazgeç

Figure 2.37. Definition of internal partitions for floor B-101 in the heating load calculation module of MTH software.

Mahal bilgileri - B-101 Shop 2

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

Aktivite DI (Watt/per) GI (Watt/per) Toplam

Orta derecede aktif mağaza çalışanı 66 110 176

m2/İnsan Toplam insan

3.77 15

Toplam DI (Watt) Toplam GI (Watt) Toplam (Watt) Grafik adı

990 1650 2640 Konut İnsan Yükleri

İemzle Vazgeç

Figure 2.38. Assignment of internal heat gains for floor B-101 by defining fifteen moderately active retail employees in the heating load calculation module of MTH software.

Mahal bilgileri - F-207 Bathroom

Genel Camlar Dış duvar ve çatı İç bölümler İnsanlar Aydınlatma Elektrikli cihazlar Diğer Enfiltrasyon

Mahal no: F-207 Mahal adı: Bathroom Kat: 2 Kot (m): 5.60 Mahal alanı (m²): 5.9 Mahal yüksekliği (m): 2.9

F-207 - Bathroom

- Genel
- Camlar
- Dış duvar ve çatı
- İç bölümler
- İnsanlar
- Aydınlatma
- Elektrikli cihazlar
- Diğer
- Enfiltrasyon

Ekle Yeni Sil Tamam Vazgeç

Figure 2.41. General space information screen for room F-207 in the heating load calculation module of MTH software.

Mahal bilgileri - F-207 Bathroom

Genel Camlar Dış duvar ve çatı İç bölümler İnsanlar Aydınlatma Elektrikli cihazlar Diğer Enfiltrasyon

İşaret Yön K (Watt/m²K) Ağırlık Renk

Uz. (m) Yük. (m) Adet Çıkan (m²) Alan (m²) Gölge

Ekle Yeni Sil Değiştir Yukarı Aşağı

İşaret	Yön	Gen. (m)	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan ...	K (W/a...	Ağırlık	Renk	Gölge
Dd2	B	0.3	1.96	2.9	1	0	5.68	0.371	Orta	Orta	Hayır
Dd1	G	0.3	0.5	2.9	1	0.25	1.2	0.495	Orta	Orta	Hayır

Vazgeç

Figure 2.42. Definition of external wall and roof elements for room F-207 in the heating load calculation module of MTH software.

Mahal bilgileri - F-207 Bathroom

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

İşaret Yön K (Watt/m2K) Yaz dt(°C) Kış dt(°C)

Uz. (m) Yük. (m) Adet Çıkan (m2) Alan (m2) Ağırlık

İşaret	Yön	Orta	Uz. (m)	Yük. (m)	Adet	Çıkan ...	Alan (...)	K (Wa...)	Yaz dt...	Kış dt(...)	Ağırlık
Dö_bath		0.39	0	0	1	0	5.9	0.589	0	0	Orta
Dö_bath		0.39	0	0	1	0	5.9	0.589	0	0	Orta
İK 80x2...	K	0	0.8	2.1	1	0	1.68	2.3	4	4	Orta
İd1	K	0.13	3.34	2.9	1	1.68	8.01	2.101	4	4	Orta
İK 80x2...	D	0	0.8	2.1	1	0	1.68	2.3	8	8	Orta
İd1	D	0.13	1.1	2.9	1	1.68	1.51	2.101	8	8	Orta
İK 80x2...	D	0	0.8	2.1	1	0	1.68	2.3	8	8	Orta
İd1	D	0.13	1.2	2.9	1	1.68	1.8	2.101	8	8	Orta
İp 50x50	G	0	0.5	0.5	1	0	0.25	2.6	16	16	Orta
İd2	G	0.24	1.05	2.9	1	0.25	2.8	1.437	16	16	Orta
İd1	G	0.13	0.84	2.9	1	0	2.44	2.101	8	8	Orta
İp 50x50	G	0	0.5	0.5	1	0	0.25	2.6	22	22	Orta

Figure 2.43. Definition of internal partitions for room F-207 in the heating load calculation module of MTH software.

Mahal bilgileri - F-207 Bathroom

Genel | Camlar | Dış duvar ve çatı | İç bölümler | İnsanlar | Aydınlatma | Elektrikli cihazlar | Diğer | Enfiltrasyon

Aktivite

Vasat bir şahıs istirahatte ayakta

m2/İnsan Toplam insan

Toplam Dİ (Watt) Toplam Gİ (Watt) Toplam (Watt) Grafik adı

Figure 2.44. Assignment of internal heat gains for room F-207 by defining one average person standing at rest in the heating load calculation module of MTH software.

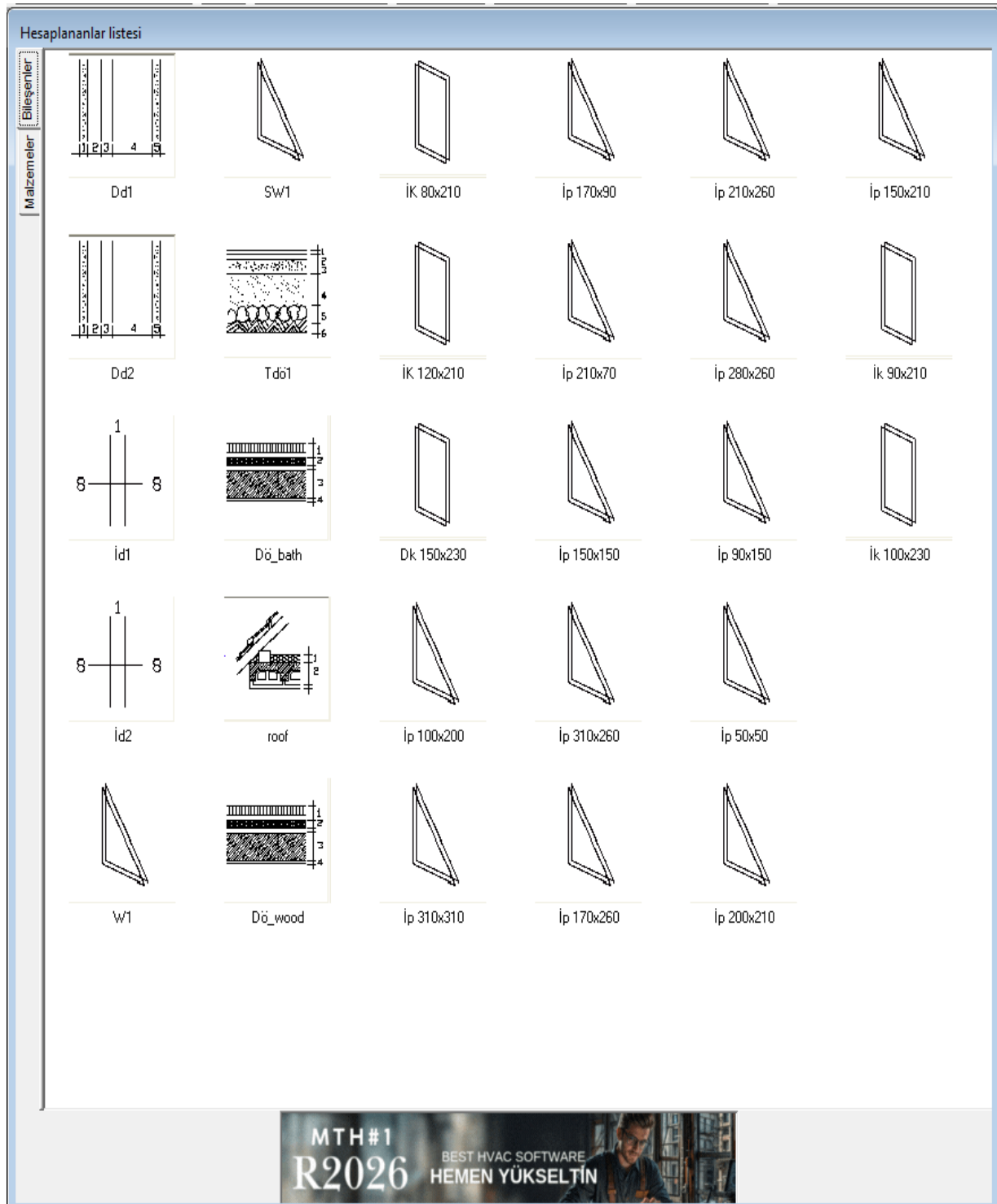


Figure 2.47. List of building envelope components, including walls, windows, doors, and floors, used in the heating load calculation within MTH software.

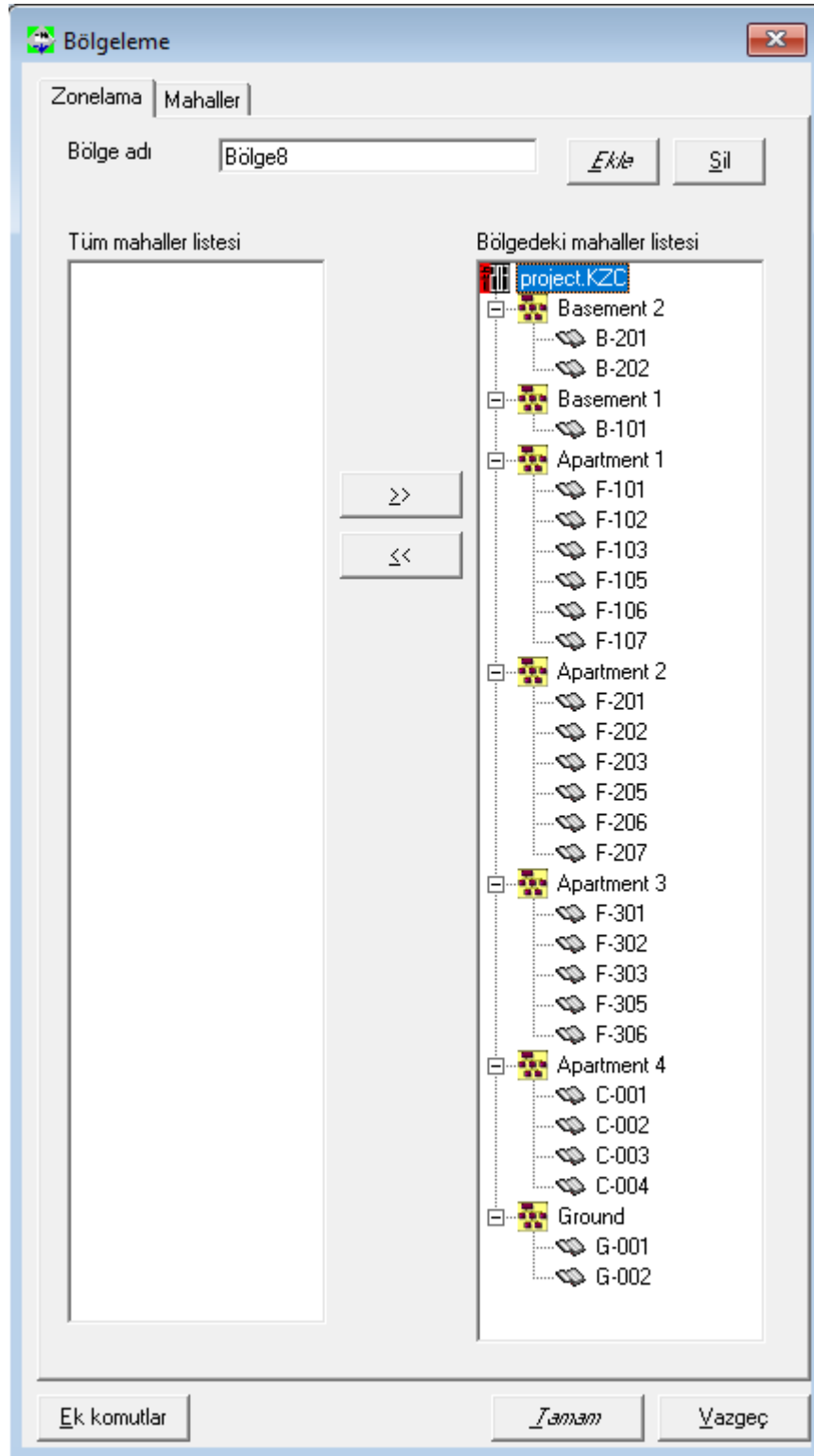


Figure 2.48. Definition of heating zones by grouping spaces according to floors in the heating load calculation module of MTH software.

Tesisat bilgileri

Bölge | Soğutma | Isıtma | Diğer

Bölge Adı:

☐ Bu bölge verilerini tüm bölgelere sakla.

☒ Tüm bölgelere sakla.

☐ Sadece bölge bilgisi verilmeyenlere sakla.

☐ Bu bölge verilerini Otomatik doldur için sakla

Figure 2.49. Zone definition screen in the installation information interface of MTH software.

Tesisat bilgileri

Bölge | Soğutma | Isıtma | Diğer

İç hava uygulama tipi: Genel konfor

İç hava KT (°C): 24

İç hava YT (°C): 17.1

İç hava İN (%): 50

Soğutma kriteri: Sevk havası miktarı (m³/h/m²)

Sevk havası miktarı (m³/h/m²): 1

Gidiş suyu sıcaklığı (°C): 12

Dönüş suyu sıcaklığı (°C): 17

Dış hava uygulama tipi: İkametgah(Vasat)

Dış hava kriteri: Taze hava miktarı (m³/h/m²)

Taze hava miktarı (m³/h/m²): 1

Soğutma emniyet zammı (%): 10

Figure 2.50. Cooling system parameters defined under the cooling tab in the installation information module of MTH software.

Tesisat bilgileri

Bölge Soğutma Isıtma Diğer

İç hava uygulama tipi Genel konfor

İç hava KT (°C) 24

İç hava YT (°C) 17.1

İç hava İN (%) 50

Isıtma kriteri Isıtma suyu gidiş/dönüş farkı (°C)

Isıtma suyu gidiş/dönüş farkı (°C) 20

Gidiş suyu sıcaklığı (°C) 90

Dönüş suyu sıcaklığı (°C) 70

Dış hava uygulama tipi İkametgah(Vasat)

Dış hava kriteri Taze hava miktarı (m³/h/m²)

Taze hava miktarı (m³/h/m²) 1

Isıtma emniyet zammı (%) 10

Vazgeç

Figure 2.51. Definition of heating system parameters under the heating tab in the installation information module of MTH software.

Tesisat bilgileri

Bölge Soğutma Isıtma Diğer

Verici kanal ısı kazancı (%) 0

Verici kanal kaçakları (%) 0

Fan motoru kaybı (%) 5

Dönüş kanalı ısı kazancı (%) 0

Dönüş kanalı kaçakları (%) 0

Soğuk su pompası kaybı (%) 0

Soğutucu borulardan kayıp (%) 0

Serpantin Bypass Faktörü 0.1

Sistem günlük çalışma rejimi 12

Vazgeç

Figure 2.52. Other tab in the installation information interface of MTH software.

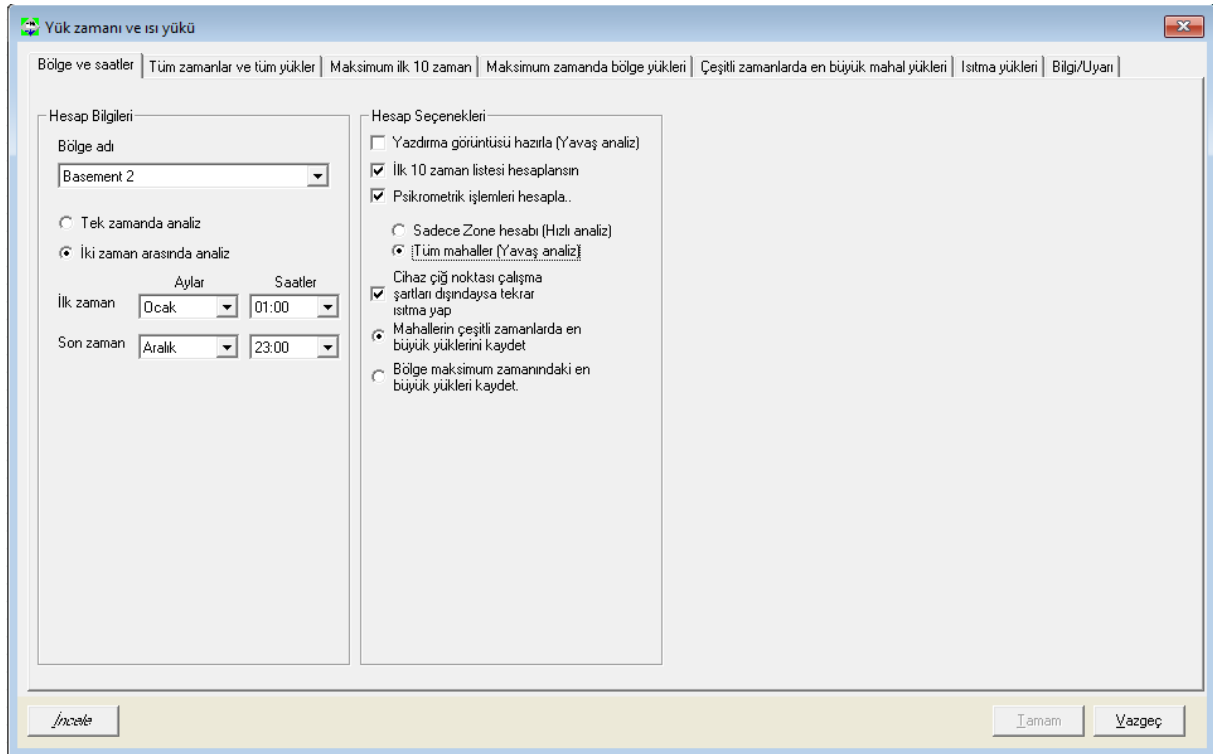


Figure 2.53. Annual heating load analysis screen showing the determination of maximum heating loads for each floor and zone in MTH software.

In this stage of the analysis, heating load calculations are performed over the entire year in order to capture the variation of thermal demand under different outdoor conditions. The software evaluates the heating load continuously throughout the year and identifies the time periods at which peak heating demand occurs.

For each defined zone and each floor, the maximum heating load is determined individually based on the annual load profile. This approach ensures that the heating system is sized according to the most critical operating conditions rather than a single design point. By considering annual load variations, the reliability and adequacy of the heating system are improved, and underestimation of peak loads is avoided.

	Mahal no	Mahal adi	So. DI(Watt)	So. TI(Watt)	Is. TI(Watt)	Kat	Kot (m)
project KZC							
[Basement 2]							
[-2. Kat]							
B-201	B-201	Shop 2 Storage	2239	2711	1383	-2	-6.00
B-202	B-202	WC	496	562	405	-2	-6.00
B-101	B-101	Shop 2	4968	7029	2596	-1	-3.00
B-202	G-001	Shop 1	2623	3314	2120	0	0.00
[Basement 1]							
G-002	G-002	WC	338	404	283	0	0.00
[-1. Kat]							
F-101	F-101	Room	1012	1176	639	1	2.70
B-101	F-102	Kitchen	2309	4327	57	1	2.70
[Ground]							
0. Kat							
G-001	F-103	Living Room	1253	1756	955	1	2.70
G-002	F-105	Room	1287	1480	1013	1	2.70
[Apartment 1]							
1. Kat							
F-101	F-106	WC	168	238	98	1	2.70
F-102	F-107	Bathroom	2676	4584	500	1	2.70
F-103	F-201	Room	1009	1173	636	2	5.60
F-105	F-202	Kitchen	2310	4328	58	2	5.60
F-106	F-203	Living Room	1248	1751	950	2	5.60
F-107	F-205	Room	1277	1470	1003	2	5.60
[Apartment 2]							
2. Kat							
F-201	F-206	WC	153	223	84	2	5.60
F-202	F-207	Bathroom	2676	4584	500	2	5.60
F-203	F-301	Room	851	1015	435	3	8.50
F-205	F-302	Kitchen	2324	4342	71	3	8.50
F-206	F-303	Living Room	1248	1751	950	3	8.50
F-207	F-305	Room	1272	1465	987	3	8.50
[Apartment 3]							
3. Kat							
F-301	F-306	Bathroom WC	2974	4891	882	3	8.50
F-302	C-001	Study Room	819	1200	217	4	10.60
F-303	C-002	Living Area	688	842	706	4	10.60
F-305	C-003	Dressing Room	235	323	260	4	10.60
F-306	C-004	Bathroom WC	2717	4611	504	4	10.60
[Apartment 4]							
4. Kat							
C-001							
C-002							
C-003							
C-004							

Figure 2.54. Summary of heating load calculation results for all floors and spaces obtained from the MTH software. The figure presents the calculated heating loads for each individual space, including transmission heat losses through building envelope elements and internal heat gains. Results are listed on a room-by-room basis and organized according to floors and zones, allowing a detailed evaluation of heating demand distribution throughout the building. The table also provides indoor and outdoor design heat load components, along with elevation data for each space, which are used in subsequent radiator sizing and hydraulic system design. This comprehensive overview ensures that the heating system is sized based on the most critical load conditions for each zone and floor.



Figure 2.55. Transition to the radiator and fan-coil selection module in MTH software, where heating terminals are selected based on the calculated space heating loads. This module is used to match the heating demand of each room with appropriate terminal units by considering standard product capacities and operating conditions.

Dizayn bilgileri

Genel bilgiler | Radyatör seçim bilgileri | Fan-Coil seçim bilgileri | VRF seçim bilgileri

Gidiş suyu sic. (°C) Dönüş suyu sic.

Isıtma tesisatı 90 70

Soğutma tesisatı 7 12

Çabuk seçim ☒ Radyatör ☐ Fan-Coil ☐ Vrf

FC/Vrf seçim kriteri ☒ Soğutma yükü ☐ Isıtma yükü

Isıtma emniyet f(%) 10

Soğutma emniyet f(%) 10

Dosya bağlantısı C:\Users\yusuf\Desktop\MTH Project\Heat Load\project.KZC ...

Tamam Vazgeç

Figure 2.56. Definition of general design parameters in the general information tab of the design information screen within the radiator and fan-coil selection module of MTH software.

B-201 / Shop 2 Storage - Mahal bilgileri ve seçim

Mahal bilgileri | Radyatör seçimi | Fan-Coil seçimi | VFR seçimi

Mahal no	Mahal adı	Katı	Kotu(m)	Mahal alanı (m2)	Mahal yüksekliği
B-201	Shop 2 Storage	-2	-6.00	55	2.7

Oda kış KT (°C)	Qkay (Watt)	Oda yaz KT (°C)	Oda yaz IN (%)	Qduy (Watt)	Qtop (Watt)
15	1521	24	50	2462	2982

B-201

Isısan 0.6/P100/500 (1637 W/att)

Ekle **Yeni** **Sil** **Çabuk seçim** **İmam** **Vazgeç**

Figure 2.57. Space information and heating terminal selection screen for room B-201 in the radiator and fan-coil selection module of MTH software.

B-201 / Shop 2 Storage - Mahal bilgileri ve seçim

Mahal bilgileri | **Radyatör seçimi** | Fan-Coil seçimi | VFR seçimi

Kütüphane: rd_radyator.mdb | Marka: Isısan | Cins: Panel | Tip: P100/500

Seçilebilir Boy/Ad: 0.4 | Brm ver. (Watt): 2729 | Verimi (Watt): 1092 | **Kalan Qkay (Watt): 429**

Ekle **Değiştir** **Sil**

1Marka	2Tip	3Uz(m)/Ad	4Verimi (W...)	5Brm ver. (...)	6Cins	7Kütüphane
Isısan	P100/500	0.4	1092	2729	Panel	rd_radyato...

Seç **İmam** **Vazgeç**

Figure 2.58. Radiator selection screen in MTH software showing the evaluation of a 0.4 m radiator length based on the calculated heating load. Although manual calculations indicated that a 0.4 m radiator would be sufficient, the software-based analysis reveals a heating capacity deficit of approximately 429 W for this configuration. This result demonstrates the discrepancy between preliminary manual sizing and catalog-based performance data used in the software, highlighting the need for revision of the radiator length to ensure adequate heating capacity under design conditions.

Based on the initial heating load calculations, a radiator length of 0.4 m was found to be theoretically sufficient to meet the heating demand of the space. This preliminary selection was carried out using manual calculation methods under the specified design conditions.

However, when the same radiator configuration was evaluated within the MTH software environment, it was observed that a radiator with a length of 0.4 m provided a heating capacity approximately 429 W lower than the calculated space heating requirement. This discrepancy arises from the fact that the software accounts for standardized radiator performance data, operating conditions, and catalog-based capacity values, which may differ from simplified manual estimations.

In order to eliminate this capacity deficit and to ensure that the heating system is capable of meeting the space heating demand under design conditions, the radiator length was revised from 0.4 m to 0.6 m. This revision increases the available heat output and introduces an appropriate safety margin, thereby preventing potential underheating during peak load conditions.

The final radiator selection presented in this study therefore reflects the revised configuration obtained through software-based verification. This approach demonstrates the importance of validating manual calculations using detailed simulation tools and highlights the role of design optimization in achieving reliable and effective heating system performance.

B-201 / Shop 2 Storage - Mahal bilgileri ve seçim

Mahal bilgileri | Radyatör seçimi | Fan-Coil seçimi | VFR seçimi

Kütüphane: rd_radyator.mdb | Marka: Isisan | Cins: Panel | Tip: P100/500

Seçilebilir Boy/Ad: 0.6 | Brm ver. (Watt): 2729 | Verimi (Watt): 1637 | **Kalan Q kay (Watt): 0**

Ekle | Değiştir | Sil

1Marka	2Tip	3Uz(m)/Ad.	4Verimi (W...	5Brm ver. (...)	6Cins	7Kütüphane
Isisan	P100/500	0.6	1637	2729	Panel	rd_radyato...

Seç | Tamam | Vazgeç

Figure 2.59. Radiator selection screen in MTH software after revising the radiator length to 0.6 m. The updated configuration fully satisfies the calculated heating demand of the space, resulting in zero remaining heating capacity deficit. This result confirms that increasing the radiator length from 0.4 m to 0.6 m successfully eliminates the previously identified capacity shortfall and ensures adequate heating performance under design conditions.

B-202 / WC - Mahal bilgileri ve seçim

Mahal bilgileri | **Radyatör seçimi** | Fan-Coil seçimi | VFR seçimi

Kütüphane: rd_havlupan.mdt | Marka: Alarko | Cins: Havlupan | Tip: ABR 50/117 | ...

Seçilebilir Boy/Ad: 1 | Brm ver. (Watt): 489 | Verimi (Watt): 489 | Kalan Qkay (Watt): 0

Ekle | Değiştir | Sil

1Marka	2Tip	3Uz(m)/Ad	4Verimi (W...	5Brm ver. (...	6Cins	7Kütüphane
Alarko	ABR 50/117	1	489	489	Havlupan	rd_havlup...

Seç | Tamam | Vazgeç

Figure 2.60. Radiator selection screen in MTH software showing the placement of a 50/117 (500 × 1170 mm) radiator for space B202. The selected radiator corresponds to the nearest available standard size exceeding the required heating capacity, ensuring that the calculated heating demand is fully satisfied without falling below the design requirement.

Based on the heating load calculations for space B202, a radiator size of 500 × 1000 mm was determined to be required in order to meet the calculated heating demand. However, during the radiator selection process, it was observed that this exact radiator dimension was not available among standard manufacturer sizes used within the MTH software.

The available standard radiator options closest to the required size were 500 × 810 mm and 500 × 1170 mm. Since selecting a radiator with insufficient capacity may lead to underheating under design conditions, the smaller option (500 × 810 mm) was considered inadequate for this space.

To ensure that the heating demand of space B202 is fully satisfied and to avoid any capacity deficit, the larger standard radiator size of 500 × 1170 mm was selected. In the MTH software, this radiator corresponds to the model coded as 50 × 117. This selection provides sufficient heating capacity while maintaining consistency with standard radiator dimensions and design practices.

G-002 / WC - Mahal bilgileri ve seçim

Mahal bilgileri | Radyatör seçimi | Fan-Coil seçimi | VFR seçimi

Kütüphane: rd_havlupan.mdt | Marka: Alarko | Cins: Havlupan | Tip: ABR 50/81 | ...

Seçilebilir Boy/Ad: 1 | Brm ver. (Watt): 337 | Verimi (Watt): 337 | Kalan Qkay (Watt): 0

Ekle | Değiştir | Sil

1Marka	2Tip	3Uz(m)/Ad.	4Verimi (W...)	5Brm ver. (...)	6Cins	7Kütüphane
Alarko	ABR 50/81	1	337	337	Havlupan	rd_havlup...

Seç | Tamam | Vazgeç

Figure 2.61. Radiator selection screen in MTH software showing that the heating demand of space G-002 is fully satisfied using a 500 × 800 mm radiator (ABR 50/81).

Based on the heating load calculations performed for space G-002, a radiator size of 500 × 800 mm was found to be sufficient to meet the calculated heating demand. This preliminary sizing was carried out using manual calculation methods under the defined design conditions.

In order to verify this result, the same radiator configuration was evaluated within the MTH software environment. When a 500 × 800 mm radiator, corresponding to the model coded as **ABR 50/81**, was assigned, the software analysis confirmed that the heating demand of the space was fully satisfied without any remaining capacity deficit.

Since both manual calculations and software-based analysis yielded consistent results, the initially determined radiator size was considered appropriate. Therefore, the radiator selection for space G-002 was retained as 500 × 800 mm (ABR 50/81) without further revision.

Figure 2.62 shows the radiator selection screen in MTH software for space F-107. The interface displays the selected radiator model as ABR 50/117. The 'Kalan Qkay (Watt)' field, which indicates the remaining heating capacity, is highlighted with a red box and shows a value of 61. This suggests that the current selection is insufficient, leading to the revision mentioned in the caption.

1Marka	2Tip	3Uz(m)/Ad.	4Verimi (W...	5Brm ver. (...)	6Cins	7Kütüphane
Alarko	ABR 50/117	1	489	489	Havlupan	rd_havlup...

Figure 2.62. Radiator selection screen in MTH software showing the revision of the radiator model for space F-107 from ABR 50/117 to ABR 60/117 due to insufficient heating capacity.

For space F-107, the radiator size corresponding to the model **ABR 50/117** was initially selected based on heating load calculations. However, when this radiator was evaluated within the MTH software environment, it was observed that the heating capacity provided by the ABR 50/117 model was insufficient to fully satisfy the calculated heating demand of the space.

In order to eliminate the identified capacity deficiency and ensure adequate heating performance under design conditions, the radiator selection was revised. Consequently, the radiator model was upgraded to **ABR 60/117**, which offers a higher heat output due to its increased depth while maintaining the same length.

The revised selection successfully meets the heating requirement of space F-107, demonstrating the importance of verifying preliminary radiator sizing through software-based analysis and adjusting the selection when necessary to ensure reliable system performance.

F-107 / Bathroom - Mahal bilgileri ve seçim

Mahal bilgileri | Radyatör seçimi | Fan-Coil seçimi | VFR seçimi

Kütüphane: rd_havlupan.mdt | Marka: Alarko | Cins: Havlupan | Tip: ABR 60/117 | ...

Seçilebilir Boy/Ad: 1 | Brm ver. (Watt): 572 | Verimi (Watt): 572 | **Kalan Q kay (Watt): 0**

Ekle | Değiştir | Sil

1Marka	2Tip	3Uz(m)/Ad.	4Verimi (W...)	5Brm ver. (...)	6Cins	7Kütüphane
 Alarko	ABR 60/117	1	572	572	Havlupan	rd_havlup...

Seç | Tamam | Vazgeç

Figure 2.63. Radiator selection screen in MTH software showing the updated radiator configuration for space F-107. The initially selected ABR 50/117 radiator was found to be insufficient to meet the calculated heating demand; therefore, the selection was revised to the ABR 60/117 model. The updated radiator fully satisfies the required heating load, ensuring adequate thermal performance under design conditions.

Figure 2.64 shows the towel radiator selection screen in MTH software. The window is titled "F-206 / WC - Mahal bilgileri ve seçim". It features four tabs: "Mahal bilgileri", "Radyator seçimi" (selected), "Fan-Coil seçimi", and "VFR seçimi".

Under the "Radyator seçimi" tab, there are input fields for:

- Kütüphane: rd_havlupan.mdt
- Marka: Alarko
- Cins: Havlupan
- Tip: ABR 50/81

Below these are fields for:

- Seçilebilir Boy/Ad: 1
- Brm ver. (Watt): 412
- Verimi (Watt): 412
- Kalan Qkay (Watt): 0

Buttons for "Ekle", "Değiştir", and "Sil" are present. At the bottom of the window are buttons for "Seç", "Tamam", and "Vazgeç".

1Marka	2Tip	3Uz(m)/Ad	4Verimi (W...	5Brm ver. (...	6Cins	7Kütüphane
Alarko	ABR 50/81	1	412	412	Havlupan	rd_havlup...

Figure 2.64. Towel radiator selection screen in MTH software showing the use of a 500 × 800 mm towel radiator (ABR 50/81) for space F-206 due to the unavailability of the calculated 400 × 700 mm size.

For space F-206, heating load calculations indicated that a towel radiator with dimensions of 400 × 700 mm would be sufficient to meet the required heating demand. This preliminary sizing was determined based on the calculated space heat loss under the specified design conditions.

However, during the selection process within the MTH software, it was observed that a towel radiator with the exact dimensions of 400 × 700 mm was not available among the standard product sizes. In order to avoid selecting a radiator with insufficient capacity, the nearest larger standard size was considered.

Consequently, a towel radiator with dimensions of 500 × 800 mm, corresponding to the model coded as **ABR 50/81**, was selected. This selection ensures that the heating demand of space F-206 is fully satisfied while maintaining consistency with standard manufacturer dimensions and providing an adequate safety margin.

F-306 / Bathroom WC - Mahal bilgileri ve seçim

Mahal bilgileri | Radyatör seçimi | Fan-Coil seçimi | VFR seçimi

Mahal no	Mahal adı	Katı	Kotu(m)	Mahal alanı (m2)	Mahal yüksekliği
F-306	Bathroom WC	3	8.50	8	2.9

Oda kış KT (°C)	Qkay (Watt)	Oda yaz KT (°C)	Oda yaz IN (%)	Qduy (Watt)	Qtop (Watt)
26	970	24	50	3271	5380

F-306

Baymak 1/YH 5/25 (1065 Watt)

Ekle Yeni Sil Çabuk seçim Tamam Vazgeç

Figure 2.65. Towel radiator selection screen in MTH software for the F-306 bathroom, showing the use of a Baymak towel radiator coded as **1/YH 5/25**.

Although a towel radiator with dimensions of 500 × 1800 mm was calculated to be required, this size was not available among the standard products of the initially selected manufacturer, and no higher-capacity model was found within the same brand. Therefore, the Baymak 1/YH 5/25 model was selected as it provides sufficient heating capacity to meet the calculated thermal demand of the space, ensuring adequate heating performance under design conditions.

For space F-306, heating load calculations indicated that a towel radiator with dimensions of 500×1800 mm would be required to meet the calculated heating demand under the specified design conditions. This preliminary sizing was based on the space heat loss and intended to ensure sufficient thermal comfort.

During the selection process within the MTH software, it was observed that a towel radiator with the exact dimensions of 500×1800 mm was not available among the standard product options of the initially selected manufacturer. Furthermore, an evaluation of the available towel radiators within the same brand revealed that no larger or higher-capacity model was available that could meet or exceed the required heating capacity for this space.

In order to avoid underheating and to ensure that the calculated heating demand of space F-306 is fully satisfied, alternative manufacturers were considered. As a result of this evaluation, a towel radiator from the **Baymak** brand, coded as **1/YH 5/25**, was selected. This model provides a higher heat output compared to the available options of the initial manufacturer and successfully meets the required heating capacity determined for space F-306.

The selection of the Baymak 1/YH 5/25 towel radiator ensures that the heating requirement is satisfied without compromising design reliability or system performance. This approach demonstrates the importance of flexibility in product selection when standard dimensions are insufficient and highlights the necessity of considering alternative manufacturers to achieve accurate and effective heating system design.

Similar selection procedures were applied to the remaining spaces following the same methodology, and radiator or towel radiator sizes were determined accordingly based on the calculated heating loads and available standard products.

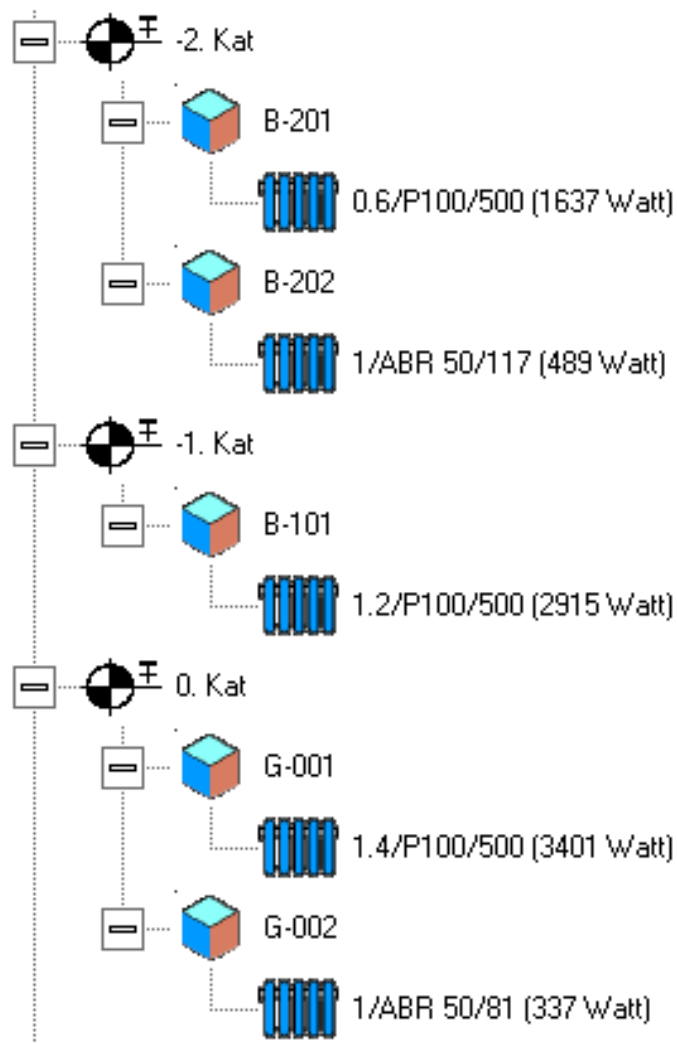


Figure 2.66. Radiator list for the –2nd floor, –1st floor, and ground floor generated in the project calculation module of MTH software. The figure presents the selected radiator types, dimensions, and heat output values for each space on the specified floors, based on the calculated heating loads and finalized radiator selections. This list is used for verification of heating capacity distribution across the building and serves as a reference for installation and system coordination.

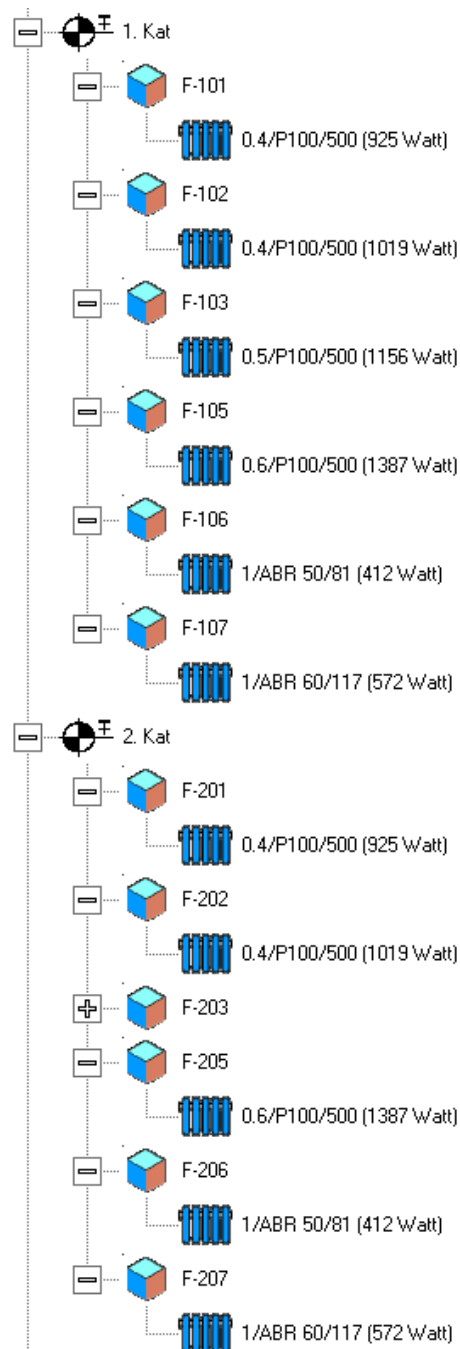


Figure 2.67. Radiator list for the 1st and 2nd floors generated in the project calculation module of MTH software. The list presents the selected radiator types, dimensions, and corresponding heat output values for each space on these floors, determined based on the calculated heating loads and finalized radiator selections. This information is used to verify adequate heat distribution and to support installation planning.

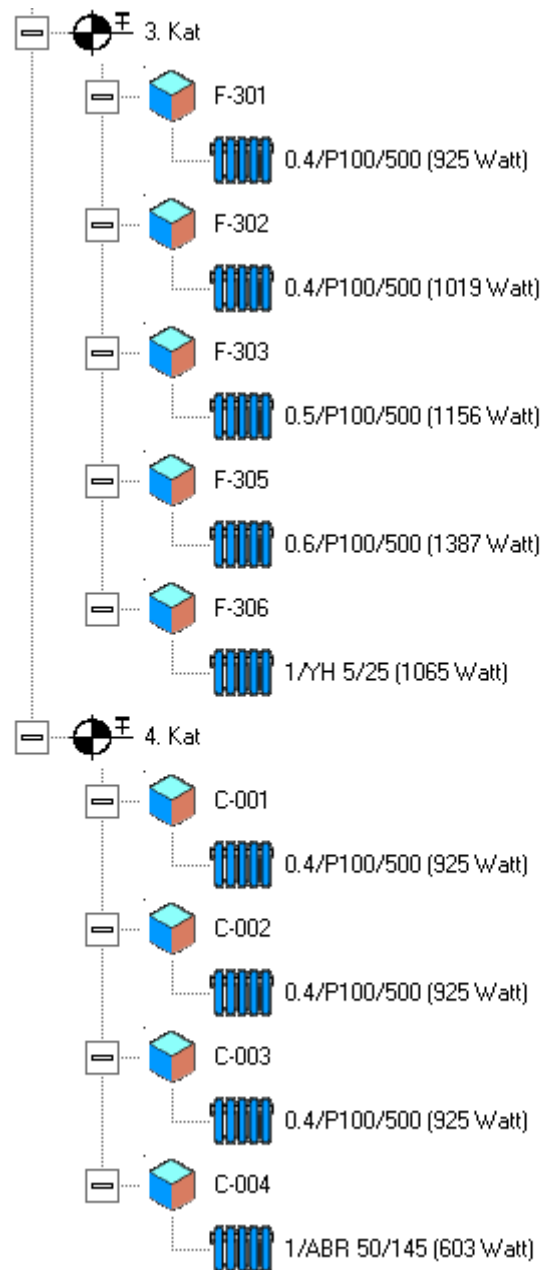


Figure 2.68. Radiator list for the 3rd and 4th floors generated in the project calculation module of MTH software. The list includes the selected radiator types, dimensions, and corresponding heat output values for each space on these floors, determined based on the calculated heating loads and finalized radiator selections. These results are used to ensure adequate heating capacity and consistent heat distribution throughout the upper floors of the building.

RADYATÖR/FAN-COIL METRAJİ					
Proje no :			Sayfa no :	1	
Proje adı :			Revizyon :	0	
Konu :			Tarih :	Cumartesi, 17 Ocak, 2026	
			Hesaplayan :		
			Kontrol :		
Marka	Uz (m)/Ad.	Birimi	Tipi	Cinsi	Grup adedi
Isisan	0.6	M.	P100/500	Panel	4
Alarko	1	Adet	ABR 50/117	Havlupan	1
Isisan	1.2	M.	P100/500	Panel	1
Isisan	1.4	M.	P100/500	Panel	1
Alarko	1	Adet	ABR 50/81	Havlupan	3
Isisan	0.4	M.	P100/500	Panel	9
Isisan	0.5	M.	P100/500	Panel	3
Alarko	1	Adet	ABR 60/117	Havlupan	2
Baymak	1	Adet	YH 5/25	Havlupan	1
Alarko	1	Adet	ABR 50/145	Havlupan	1

Figure 2.69. Quantity take-off screen in the project calculation module of MTH software. The figure presents the calculated quantities of heating system components, including radiators, pipes, and fittings, based on the finalized heating load calculations and system layout. This information is used for cost estimation, material planning, and project implementation.



Figure 2.70. Transition to the project calculation module in MTH software, where the overall system calculations are performed based on the previously defined heating loads, zoning configuration, and selected heating terminals.

Isıtma/Soğutma boru çapı

Genel | Teslat bilgileri | Debi bilgileri | Boru çapı | Özel parçalar

Açıklama
pipediameter

İzole | Sil | Araya | Uygula

Adı	Bağlantısı	Parça yük...	Hat yükü (...	Uz. L(m)	Debi V(m3...	Hız (m/sn)	Boru çapı...	Basınç kay...	Basınç kay...	Özel parça...	Özel parça...	Toplam ka...	Boru tipi
1	1	0	28651	3	1.231	0.26	40 mm - 1 ...	21.41	64.23	0	0	64.23	DIN 2440
2	2	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
3	3	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
4	4	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
5	5	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
6	6	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
7	7	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
8	8	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
9	9	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
10	10	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
11	11	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
12	12	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
13	13	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
14	14	0	28651	0.1	1.231	0.26	40 mm - 1 ...	21.41	2.14	0	0	2.14	DIN 2440
15	15	0	28651	2.15	1.231	0.26	40 mm - 1 ...	21.41	46.03	0	0	46.03	DIN 2440
16	16	0	28651	1.35	1.231	0.26	40 mm - 1 ...	21.41	28.9	0	0	28.9	DIN 2440
17	17	0	28651	1.3	1.231	0.26	40 mm - 1 ...	21.41	27.83	0	0	27.83	DIN 2440
18	18	0	3378	26	0.145	0.21	15 mm - 1/...	49.11	1276.86	0	0	1276.86	DIN 2440

Hesapla | Kritik hat | Metraj

Figure 2.71. General tab of the heating and cooling pipe diameter calculation section in the project calculation module of MTH software. This screen defines the main parameters and calculation settings used for determining appropriate pipe diameters based on system flow rates, heating and cooling loads, and design criteria.

Isıtma/Soğutma boru çapı

Genel | Tesisat bilgileri | Debi bilgileri | Boru çapı | Özel parçalar

Min.hız (m/sn): 0.2 | Max.hız (m/sn): 0.8 | Min.ısı (Watt): 200 | Max.ısı (Watt): 30000 | Gidiş (°C): 90 | Dönüş (°C): 70 | Fark (°C): 20 | Akışkan tipi: Su

Ekle | Sil | Araya | Uygula

Adı	Bağlantısı	Parça yük...	Hat yükü (...)	Uz. L(m)	Debi V(m3...	Hız (m/sn)	Boru çapı ...	Basınç kay...	Basınç kay...	Özel parça...	Özel parça...	Toplam ka...	Boru tipi
1	1	0	28651	3	1.231	0.26	40 mm - 1 ...	21.41	64.23	0	0	64.23	DIN 2440
2	2	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
3	3	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
4	4	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
5	5	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
6	6	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
7	7	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
8	8	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
9	9	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
10	10	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
11	11	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
12	12	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
13	13	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
14	14	0	28651	0.1	1.231	0.26	40 mm - 1 ...	21.41	2.14	0	0	2.14	DIN 2440
15	15	0	28651	2.15	1.231	0.26	40 mm - 1 ...	21.41	46.03	0	0	46.03	DIN 2440
16	16	0	28651	1.35	1.231	0.26	40 mm - 1 ...	21.41	28.9	0	0	28.9	DIN 2440
17	17	0	28651	1.3	1.231	0.26	40 mm - 1 ...	21.41	27.63	0	0	27.63	DIN 2440
18	18	0	3378	26	0.145	0.21	15 mm - 1/2...	49.11	1276.86	0	0	1276.86	DIN 2440

Hesapla | Kritik hat | Metraj

Figure 2.72. Installation information tab in the project calculation module of MTH software. The parameters entered in this section are based on the heating and cooling calculations previously performed and documented in earlier stages of the project. These values are directly transferred from the calculated design data to ensure consistency between analytical calculations and system-level hydraulic evaluations.

Isıtma/Soğutma boru çapı

Genel | Tesiat bilgileri | Debi bilgileri | Boru çapı | Özel parçalar

Adı: 1 Bağlantısı: 1 Parça yükü (Watt): 0 Hat yükü (Watt): 28651 Debisi V(m3/h): 1.231 Uz. L(m): 3 Boru tipi: DIN 2440

Ekle Sil Açaya Uygula

Adı	Bağlantısı	Parça yük...	Hat yükü [...]	Uz. L(m)	Debi V(m3...	Hız (m/sn)	Boru çapı...	Basınç kay...	Basınç kay...	Özel parça...	Özel parça...	Toplam ka...	Boru tipi
1	1	0	28651	3	1.231	0.26	40 mm - 1	21.41	64.23	0	0	64.23	DIN 2440
2	2	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
3	3	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
4	4	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
5	5	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
6	6	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
7	7	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
8	8	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
9	9	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
10	10	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
11	11	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
12	12	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
13	13	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
14	14	0	28651	0.1	1.231	0.26	40 mm - 1 ...	21.41	2.14	0	0	2.14	DIN 2440
15	15	0	28651	2.15	1.231	0.26	40 mm - 1 ...	21.41	46.03	0	0	46.03	DIN 2440
16	16	0	28651	1.35	1.231	0.26	40 mm - 1 ...	21.41	28.9	0	0	28.9	DIN 2440
17	17	0	28651	1.3	1.231	0.26	40 mm - 1 ...	21.41	27.83	0	0	27.83	DIN 2440
18	18	0	3378	26	0.145	0.21	15 mm - 1/...	49.11	1276.86	0	0	1276.86	DIN 2440

Hesapla Kritik hat Metraj

Figure 2.73. Flow rate information tab in the project calculation module of MTH software. The flow rate values defined in this section are derived from the heating and cooling load calculations previously performed and presented in earlier project reports. These values are directly transferred to the software to ensure consistency between analytical calculations and hydraulic system analysis.

Isıtma/Soğutma boru çapı

Genel | Tesiat bilgileri | Debi bilgileri | Boru çapı | Özel parçalar

Adı: 2 Hız (m/sn): 0.24 Boru çapı DN (mm) - (inc): 40 mm - 1 1/2" Basınç kaybı R(Pa/m): 18.69

☒ Hız düşümü ☐ Tablo tavsiyesi

Ekle Sil Araya Uygula

Adı	Bağlantısı	Parça yük...	Hat yükü (...)	Uz. L(m)	Debi V(m3...)	Hız (m/sn)	Boru çapı...	Basınç kay...	Basınç kay...	Özel parça...	Özel parça...	Toplam ka...	Boru tipi
1	1	0	28651	3	1.231	0.26	40 mm - 1 ...	21.41	64.23	0	0	64.23	DIN 2440
2	2	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
3	3	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
4	4	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
5	5	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
6	6	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
7	7	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
8	8	0	3378	2.1	0.145	0.07	25 mm - 1"	3.8	7.98	0	0	7.98	DIN 2440
9	9	0	8930	2.9	0.384	0.19	25 mm - 1"	20.87	60.52	0	0	60.52	DIN 2440
10	10	0	14401	2.9	0.619	0.17	32 mm - 1 ...	12.68	36.77	0	0	36.77	DIN 2440
11	11	0	19872	2.9	0.854	0.24	32 mm - 1 ...	22.28	64.61	0	0	64.61	DIN 2440
12	12	0	23610	2.9	1.014	0.28	32 mm - 1 ...	30.09	87.26	0	0	87.26	DIN 2440
13	13	0	26525	3	1.139	0.24	40 mm - 1 ...	18.69	56.07	0	0	56.07	DIN 2440
14	14	0	28651	0.1	1.231	0.26	40 mm - 1 ...	21.41	2.14	0	0	2.14	DIN 2440
15	15	0	28651	2.15	1.231	0.26	40 mm - 1 ...	21.41	46.03	0	0	46.03	DIN 2440
16	16	0	28651	1.35	1.231	0.26	40 mm - 1 ...	21.41	28.9	0	0	28.9	DIN 2440
17	17	0	28651	1.3	1.231	0.26	40 mm - 1 ...	21.41	27.83	0	0	27.83	DIN 2440
18	18	0	3378	26	0.145	0.21	15 mm - 1/...	49.11	1276.86	0	0	1276.86	DIN 2440

Hesapla Kritik hat Metraj

Figure 2.74. Pipe diameter tab in the project calculation module of MTH software, where pipe diameter values are entered based on the calculations presented in the project report. The defined diameters are derived from previously calculated flow rates and design criteria to ensure consistency between analytical results and hydraulic system design.

Isıtma/Soğutma boru çapı

Genel | Tesiat bilgileri | Debi bilgileri | Boru çapı | Özel parçalar

Bağlantı şekli: ☒ Vidalı ☐ Flanşlı

Adı: 1 Özel parça adı: 90° Dirsek kısa (ksi): 1.20 Adet: 1 Toplam ksi: 1.2 T. BK. PD R(Pa): 39.41

Ekle Sil Araya Uygula ☐ Seçilen özel parçayı tüm hatlara ekle.

Adı	Bağlantı şekli	ksi	Adet	Kayıp k.(ksi)	Toplam basınç...
Tee-Ayrılma	Vidalı	1.6	1	1.6	52.55

Figure 2.75. Special components tab in the project calculation module of MTH software, where additional hydraulic elements such as fittings and special pipeline components are defined. The parameters entered in this section are used to account for local pressure losses and to improve the accuracy of the overall hydraulic system analysis.

Kazan hesabı

Genel Hesap Seçim

Yakıt türü ☒ Sıvı/Gaz ☐ Katı

Isı ihtiyacı Q(Watt) 28651

Dağıtım boruları Sıcak mahalden izoleli

Yakıt tipi Doğal Gaz

Kazan brm.yüz.al.ısı mik. (Watt/m²) 9500

Adet 1

Yedek 0

Kazan kapasitesi (Watt) 31516

Isıtma yüzeyi (m²) 3.17

Kapat

Kazan hesabı

Genel Hesap Seçim

Kütüphane kz_kazan.mdb

Marka Viessmann

Tip Atola 36

Geriye seçim

Kapat

Figure 2.76. Heating boiler calculation tab in the project calculation module of MTH software, where the required boiler capacity is determined based on the total calculated heating load of the building and system design parameters.

DIN 4705 Isıtma bacası hesabı

Genel Bilgiler Hesap Üzel parçalar Sonuçlar

Bağlantı kan. gaz hızı (m/sec) 5.2

Baca gazı hızı (m/sec) 2.5

Baca gazı sıcaklığı (°C) 220

Dış sıcaklık (°C) 0

Kapat

DIN 4705 Isıtma bacası hesabı

Genel Bilgiler Hesap Üzel parçalar Sonuçlar

Kazan kapasitesi Q ☒ kW ☐ Kcal/h 29

Yakıt tipi Doğalgaz

Kazan verimi (%) 90

Brülör tipi Atmosferik

Baca uzunluğu L(m) 16

Baca malzemesi Çelik çekme bo

Bağlantı kan. uz. L(m) 2

Bağ kan. malz. Çelik çekme bo

Kapat

DIN 4705 Isıtma bacası hesabı

Genel Bilgiler Hesap Üzel parçalar Sonuçlar

Üzel parça listesi (ksi) Adet 1

☒ Baca ☐ Bağlantı kan.

(ksi) 0.10 (ksi) 0.40

Keskin dirsek (10°) Saplama (45°)

Kapat

DIN 4705 Isıtma bacası hesabı

Genel Bilgiler Hesap Üzel parçalar Sonuçlar

Kazan kaybı P _w (Pa)	Bağ kan. kaybı (Pa)	Baca kaybı (Pa)	Toplam kayıp P (Pa)
21.25	24.84	21.52	46.36

	Kare a(mm)	Dikd. axb (mm)	Daire. R(mm)
Baca	95	77 × 116	107
Bağlantı kan.	66	54 × 81	74

Hesapla Kapat

Figure 2.77. Heating chimney (flue) calculation tab in the project calculation module of MTH software, where chimney dimensions and performance parameters are evaluated based on the selected boiler capacity, flue gas properties, and system design criteria.

Figure 2.78. Burner calculation tab in the project calculation module of MTH software, where the appropriate burner capacity is determined based on the selected boiler capacity, fuel characteristics, and combustion requirements of the heating system.

Açıklama	Değerler
Markası	Viessmann
Kapasite min. (kWatt)	33
Kapasite max. (kWatt)	40
Yakıt cinsi	Standard
Max. gaz giriş basıncı (mbar)	
Motor gücü (kWatt)	0.23
BBBF No	
Tipi	VEAI-3 3.0
Adı	Unit-V jet brülör
Ağırlığı (Kg)	21
Açıklama	DIN DVGW Onaylı, 230 V, 50 Hz...
Üretici kodu	Viessmann

Figure 2.79. Burner device description used in the heating system. The burner is responsible for supplying the required thermal energy by combusting the selected fuel in accordance with the calculated boiler capacity. The burner selection is based on the heating demand of the system, fuel characteristics, and combustion efficiency requirements to ensure safe, stable, and efficient operation of the heating system.

Figure 2.80. Closed-type expansion tank calculation screen in MTH software used to determine the appropriate expansion tank capacity by considering thermal expansion of the heating system fluid, operating temperature range, and design pressure limits.

Açıklama	Değerler
Markası	Alarko-Carrier
Standartı	
BBBF No	
Tipi	MT 500
Adı	Kapalı genleşme deposu
Su hacmi (Lt)	500
Çalışma basıncı (bar)	
Yüksekliği (mm)	1510
Uzunluğu (mm)	
Genişliği (mm)	750
Ağırlığı (Kg)	
Giriş çapı (")	
Açıklama	Emniyet ventili ile birlikte monte e...
Üretici kodu	Alarko

☐ Üretici bilgisi

Katalog Yazdır Kapat

Figure 2.81. Technical specifications of the selected closed-type expansion tank in MTH software. The figure presents key parameters such as nominal volume, operating pressure limits, pre-charge pressure, and temperature range, which are defined based on the calculated expansion tank requirements to ensure safe and reliable operation of the heating system.

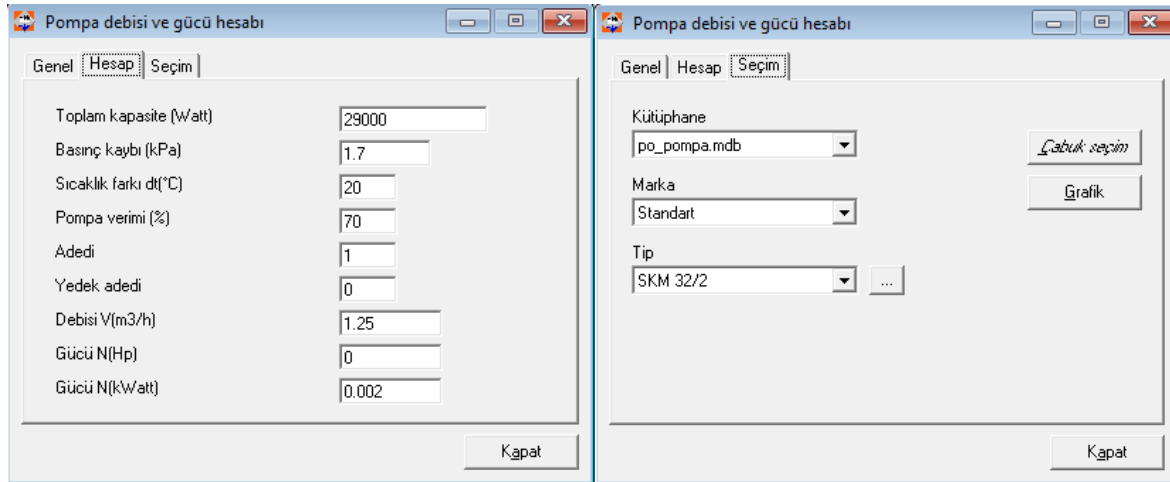
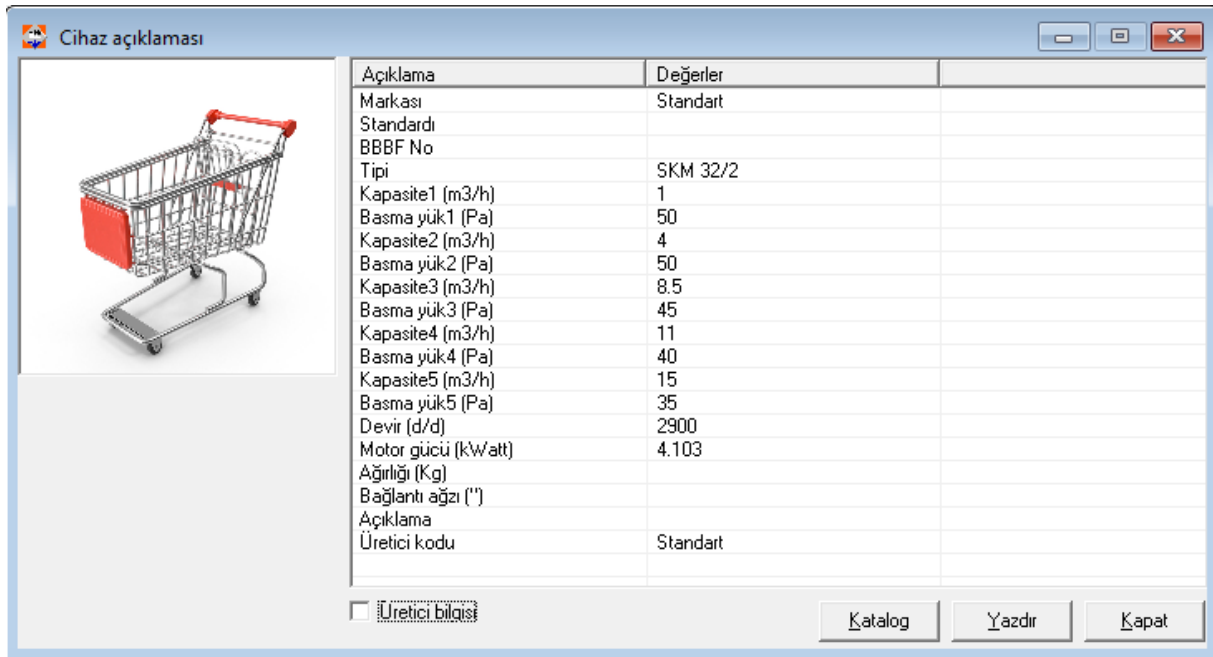


Figure 2.82. Pump flow rate and power calculation screen in the project calculation module of MTH software. The required pump flow rate and motor power are determined based on the total system flow demand, calculated pressure losses, and hydraulic design criteria to ensure sufficient circulation and reliable operation of the heating system.



Açıklama	Değerler
Markası	Standart
Standardı	
BBBF No	
Tipi	SKM 32/2
Kapasite1 (m³/h)	1
Basma yük1 (Pa)	50
Kapasite2 (m³/h)	4
Basma yük2 (Pa)	50
Kapasite3 (m³/h)	8.5
Basma yük3 (Pa)	45
Kapasite4 (m³/h)	11
Basma yük4 (Pa)	40
Kapasite5 (m³/h)	15
Basma yük5 (Pa)	35
Devir (d/d)	2900
Motor gücü (kWatt)	4.103
Ağırlığı (Kg)	
Bağlantı ağzı (")	
Açıklama	
Üretici kodu	Standart

Figure 2.83. Technical specifications of the selected circulation pump in MTH software. The figure presents key pump parameters such as nominal flow rate, head, motor power, and operating characteristics, which are determined based on the calculated pump flow rate and pressure requirements to ensure efficient and reliable operation of the heating system.

Result and Discussion

In this study, the heating system of the building was designed and analyzed by combining manual calculation methods with detailed software-based evaluation using the MTH program. The results obtained from the heating load calculations, radiator and towel radiator selections, and project calculation modules were examined comprehensively to assess the adequacy and reliability of the proposed heating system.

Firstly, space-based heating load calculations were performed considering building envelope properties, indoor design temperatures, outdoor climatic conditions, and internal heat gains arising from occupants, lighting, and electrical equipment. Annual load-time analysis was carried out, and the maximum heating load for each zone and floor was determined. This approach ensured that the system was sized based on peak demand conditions rather than simplified design assumptions.

Based on the calculated heating loads, radiator and towel radiator selections were carried out on a room-by-room basis. In several spaces, the initial radiator sizes obtained from manual calculations were found to be insufficient when evaluated using catalog-based performance data within the MTH software. In such cases, radiator dimensions were revised by selecting larger standard models or deeper radiator types to eliminate capacity deficiencies. This process highlighted the importance of verifying manual calculations with software-based analysis, as standardized product data and operating conditions can lead to differences in actual heat output.

In other spaces, the initially calculated radiator sizes were confirmed to be adequate by the MTH software, and no revision was required. These results demonstrate consistency between analytical calculations and software-based evaluation when standard radiator sizes are available and closely match the calculated requirements. For spaces where the exact calculated radiator dimensions were not available among standard products, the nearest larger standard size was selected to avoid underheating and to ensure sufficient thermal comfort under design conditions.

For towel radiator selections, similar limitations related to standard product dimensions were encountered. In certain cases, the required towel radiator size was not available within the selected manufacturer's product range. When no higher-capacity model could be found within the same brand, alternative

manufacturers were considered. This resulted in the selection of towel radiators from different brands that were capable of meeting the calculated heating demand. This approach demonstrates a flexible and realistic design strategy that prioritizes system performance and thermal comfort over strict adherence to a single manufacturer.

Following the finalization of heating terminal selections, project-level calculations were performed, including pipe diameter determination, flow rate calculations, pressure loss evaluation, and pump selection. Pipe diameters were selected based on flow rates derived from heating load calculations, ensuring acceptable flow velocities and minimizing excessive pressure losses. Special components and fittings were included in the analysis to account for local pressure losses, improving the accuracy of the hydraulic calculations.

The pump flow rate and power were determined based on the total system flow demand and calculated pressure losses. The selected circulation pump was verified to provide sufficient head and flow rate to ensure reliable water circulation throughout the heating network. Additionally, boiler capacity, chimney sizing, burner selection, and closed-type expansion tank calculations were carried out to ensure safe, efficient, and stable operation of the heating system as a whole.

Overall, the results indicate that the designed heating system is capable of meeting the calculated heating demands of all spaces within the building. The combined use of manual calculations and MTH software allowed for systematic verification, identification of capacity deficiencies, and optimization of component selections. The discussion of revised radiator sizes, standard product limitations, and alternative component selections demonstrates a realistic engineering approach consistent with practical design conditions.

Conclusion

In this project, the heating system design of the building was successfully completed by integrating analytical calculation methods with detailed software-based analysis using the MTH program. Heating load calculations were performed for each space by considering building envelope properties, internal heat gains, and climatic design conditions. Maximum heating demands were identified through load-time analysis to ensure that the system was sized for peak operating conditions.

Radiator and towel radiator selections were carried out on a space-by-space basis according to the calculated heating loads. In cases where standard radiator dimensions were insufficient or unavailable, appropriate revisions were made by selecting larger or higher-capacity models. Additionally, when suitable products could not be found within a single manufacturer's range, alternative brands were evaluated to meet the required heating capacity. These revisions ensured that the heating demand of all spaces was fully satisfied while maintaining compliance with standard product availability and practical design constraints.

Following the finalization of heating terminal selections, project-level calculations were performed, including pipe diameter determination, flow rate calculations, pressure loss evaluation, and circulation pump selection. Boiler capacity, chimney sizing, burner selection, and closed-type expansion tank calculations were also completed to ensure safe, efficient, and reliable operation of the heating system. The selected system components were verified to operate compatibly as a complete heating installation.

Overall, the results demonstrate that the designed heating system meets the thermal requirements of the building under design conditions. The combined use of manual calculations and MTH software enabled systematic verification, optimization of component selections, and identification of practical design limitations. This approach reflects a realistic engineering design methodology and provides a reliable basis for the implementation of the proposed heating system.

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